

Exact Standing-Carrier Stratification of the Neutrino Oscillation Realization Space

Complete family moduli, lawful descent, overlap closure, and continuum multiplicity

PCR-5B

Main result

With the oscillation target, PCR-3 comparison class, PCR-4 survivor relation, PCR-5 occupancy result, PCR-5A standing packet, and SCIC identity held fixed, the six principal carrier families assemble into an exact groupoid-valued atlas

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{5,\text{osc}}^{\text{exec}}, \quad \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}.$$

Each family is classified by an independently formed source grammar, an applicability-preserving PCR-4 solution object, its lawful presentation groupoid, the represented seven-channel packet moduli, and a typed complement. Pairwise and higher overlaps are homotopy fibre products over the concrete standing-core groupoid, with framed provenance ensuring well-typed automorphism actions.

A full-rank active-singlet scaling family yields

$$(1, 2) \hookrightarrow Q_{\text{osc}}^{\text{SCIC}}, \quad |Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}.$$

Hence the quotient is neither finite nor countably infinite. The four PCR-5A classes remain pairwise distinct marked points, and the oscillation target does not license a unique carrier realization.

Scope

No additional neutrino phenomenon is introduced. Absolute mass, Majorana character, lepton-number violation, heavy-state observations, nonunitarity, cosmological transport, astrophysical propagation, and $B - L$ remain downstream target extensions. Exactness is asserted at the symbolic source resolution inherited from PCR-3; no finite model list or uniform numerical solver is substituted for that comparison class.

Abstract

PCR-5 fixes one terminal complete-carrier role for the target-frozen neutrino-oscillation interface and proves that its compatible standing-carrier residue is non-singleton. PCR-5A then supplies the complete seven-channel standing packet

$$\Xi_{\text{osc}} = (T^{\text{face}}, R^{\text{stand}}, \Delta^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}}),$$

proves that packet equality is exactly Standing Carrier Identity Closure (SCIC) equality, and exhibits four pairwise distinct classes with law/new-local no-repair deletion pairs $(0, 1), (1, 0), (1, 1), (0, 0)$. What remained open was the complete internal stratification of the LAW, LOCAL, HID, COMP, SOL, and HYB packet images and all duplicate-presentation overlaps.

This paper closes that classification while preserving the inherited target and quotient. The family index is formed before survivor inspection from the finite typed source-incidence grammar of PCR-3. For each source skeleton, the PCR-4 simultaneous relations are written as an explicit applicability-guarded system. Every relation is assigned its pre-evaluation applicability predicate; theoremmatically absent relations retain certified N/A status, whereas active relations evaluate as pass, fail, or fail-closed undefined. The resulting guarded solution object is proved equivalent to the frozen PCR-4 survivor locus, and active failures cannot be reclassified as inapplicable. The raw parameter object therefore decomposes exactly into survivor strata with certified N/A data, active fail/undefined strata, source updates, and target updates.

Each family carries an explicit lawful presentation groupoid generated by source-hypergraph isomorphisms, source automorphisms, fixed-boundary gauge and basis transformations, parameter-chart maps, and the applicable dilation, constituent, profile, gluing, and higher-relation isomorphisms. The packet moduli are represented by a functor of points: a test object carries a family of source solutions together with a solved seven-channel standing record, modulo that lawful action. LAW, LOCAL, HID, COMP, SOL, and HYB receive separate source equations, standing normal forms, provenance fibres, isotropy, and typed frontiers. The HYB domain is source-semantic: it includes active Inflow, Fuse, CoCarry, Project, boundary and edge incidences, arbitrary higher-arity relations, and all surviving TOP, EXT, BDY, and ENV occurrences. The identity-bearing deletion entry remains exactly the PCR-5A kindwise no-repair channel; finer source-subset sufficiency data are retained only as auxiliary source-composition information unless their consequences already factor through the frozen packet.

For every pair of source skeletons, the overlap is the homotopy fibre product of their presentation groupoids over the concrete pointed standing-core groupoid $\mathcal{N}_5$. Over a concrete core, framed provenance fibres carry a literal diagonal automorphism action. All fifteen distinct-family pair types and all forty-two distinct-label higher-overlap types are resolved, while repeated labels are handled by the same finite-tuple construction. Their codescent defines $\mathfrak{A}_{\text{osc}}^{\text{exec}}$, and

$$\pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}.$$

Independently, the active-singlet family $M_\lambda = \lambda M_0$, $Y_\lambda = \sqrt{\lambda} Y_0$, $\lambda \in (1, 2)$, preserves $Y_\lambda M_\lambda^{-1} Y_\lambda^\top$ and therefore the fixed oscillation-facing response, while changing physical threshold and oriented matching data already retained by the standing packet. Distinct values of λ are not related by lawful basis or gauge redescription, giving $|Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}$. The exact oscillation carrier space is consequently a mixed discrete, continuous, functional, and categorical moduli object with at least continuum many SCIC classes. It is not a finite census, and it does not identify a unique realized carrier.

Logical status

Theorems B00–B15 constitute the scientific argument. Supplementary tables and computational checks support scope control, provenance, reproducibility, and consistency, but they do not substitute for the semantic proofs of source coverage, survivor exactness, packet identity, overlap closure, or cardinality.

Keywords. AASC; neutrino oscillation; standing carrier; SCIC; source-incidence grammar; exact residual system; moduli functor; lawful action groupoid; overlap stratification; hybrid carrier; target extension.

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1 Problem, theorem posture, and principal results

1.1 The classification problem after PCR-5A

The physical-carrier sequence is ordered so that no carrier participates in the definition of the target it is later required to realize. PCR-1 fixes the target-relative role complex; PCR-2 fixes the complete carrier-neutral interface; PCR-3 generates the carrier universe source-first; PCR-4 evaluates every generated point under one simultaneous witness system; PCR-5 establishes compatible occupancy and the terminal SCIC quotient; and PCR-5A proves exact packet identity together with a four-class lower bound. PCR-5B addresses the remaining problem entirely within that fixed quotient:

classify every family packet stratum and every duplicate-presentation overlap.

No later neutrino phenomenon enters the calculation.

PCR-5A distinguishes two levels of closure. Exact packet-image representation gives

$$\Xi_{\text{osc}}(x) = \Xi_{\text{osc}}(y) \iff x \sim_{\text{SCIC}} y, \quad Q_{\text{osc}}^{\text{SCIC}} \cong \text{im } \Xi_{\text{osc}}.$$

A complete stratification is stronger: it must determine the source index, residual solution object, applicability structure, lawful quotient action, packet moduli, typed complement, and all overlap incidences. Merely writing $\text{im}(\Xi_{\text{osc}} \circ \iota_F)$, or taking the codescent of an unresolved map, does not supply that internal classification.

Definition 1.1 (Executed symbolic stratum). An executed stratum is a tuple

$$\mathfrak{S}_{F,\sigma} = (\Theta_{F,\sigma}^{\text{raw}}, \mathcal{E}_{F,\sigma}^{\text{src}}, \mathcal{E}_{F,\sigma}^{\text{surv}}, \text{Stat}_{F,\sigma}, G_{F,\sigma} \curvearrowright Z_{F,\sigma}, \mathfrak{M}_{F,\sigma}, \text{Prov}_{F,\sigma}, \partial \text{Fail}_{F,\sigma})$$

for which the skeleton is generated independently, the equations and group action are stated constructively, the packet-moduli functor is represented, and the complement is partitioned by the pre-evaluation applicability mask into certified N/A strata, exact active fail/undefined defect strata, and typed update strata. The stratum may be discrete, continuous, functional, categorical, or mixed.

Definition 1.2 (Executed overlap). For two executed strata, an executed overlap is the represented packet-equality submoduli object together with the pre-evaluation applicability records, theorem-certified N/A dispositions, and the disjoint complement decomposition by the first unequal packet channel, complete active fail/undefined defect pairs, and source/target update. It includes the product action, isotropy, common packet record, provenance multiplicity, and higher-overlap face maps.

The requirement that these objects be fixed at the oscillation target is essential: target-extension fibres can be exhaustive only after their base space has been completely stratified.

1.2 Proof architecture and completeness obligations

The argument separates eight logically independent burdens:

1. form the family source grammar before survivor inspection;
2. state the residual equations, inequalities, existence conditions, applicability predicates, and exact complement for every source skeleton;
3. construct the lawful groupoids by generators, relations, actions, and isotropy;

4. represent the packet moduli through solved standing variables and a universal moduli property;
5. calculate every pairwise skeleton overlap as an exact common-core locus with typed complement;
6. determine every higher distinct-label incidence and the repeated-label construction;
7. identify the complete base-space data required by the target-extension calculus; and
8. keep semantic proof separate from supplementary computational certification.

These obligations are discharged in the theorem chain below and collected in [Section 18](#).

1.3 Exactness at the inherited symbolic scope

PCR-3 permits arbitrary source-defined parameter objects inside one fixed Grothendieck universe and requires only a finite occupancy-bearing incidence presentation. Consequently, exact classification cannot mean a finite list of familiar models, nor can it presume that every moduli problem is a finite-dimensional variety. The common object is a represented groupoid-valued moduli functor. For a test object S , an S -point consists of a source family over S satisfying the displayed equations, together with its solved seven-channel standing record, modulo the displayed lawful action.

Here “symbolic” does not mean “unevaluated.” The source variables, equations, action, packet values, applicability data, and complement strata are all specified. Conversely, exactness does not require a uniform numerical decision procedure for every functional or categorical source object admitted by PCR-3. The theorem determines the moduli object and the status of every point at the declared comparison scope.

1.4 Principal theorem chain

ID	Conclusion
B00	Fix the oscillation target, PCR-3 comparison class, PCR-4 survivor evaluator, PCR-5 occupancy result, PCR-5A packet, SCIC identity, and four marked classes.
B01	Form every family source skeleton independently and associate its source/survivor systems, applicability-aware status stratification, lawful groupoid, represented packet-moduli functor, and canonical standing core.
B02	Classify the LAW family by process type, locality or memory, order, matching, boundary, deletion, and failure data.
B03	Classify the LOCAL family by active graph, representation, rank, multiplicity, threshold, and attachment.
B04	Classify the HID family by minimal active support, dilation, projection, memory, ownership, and failure data.
B05	Determine the complete LOCAL–HID common-core locus, its complement, quotient, provenance, and higher incidence.
B06	Classify the COMP family by constituent support, derivation, binding, matching orientation, boundary incidence, and failure data.
B07	Classify the SOL family by persistent profile, topology, deformation, endability, boundary support, and continuation.
B08	Classify arbitrary finite decorated hybrids, including nontrivial gluing, higher-arity relations, and auxiliary source-composition structure.

ID	Conclusion
B09	Prove lawful parameter descent and determine isotropy and provenance over fixed standing cores.
B10	Resolve all fifteen distinct-family pairwise overlap types and the universal skeleton-product status decomposition.
B11	Resolve all forty-two distinct-label higher-overlap types and prove repeated-label simplicial coherence.
B12	Assemble the resolved chart and overlap objects by codescent and prove $\pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}$.
B13	Determine the mixed moduli endpoint, prove the continuum lower bound, and exclude finite and countably infinite cardinalities.
B14	Preserve $q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}}$ as pairwise distinct marked points.
B15	Supply the complete base-space interface required by the PCR-5C target-extension calculus.

1.5 Global object and endpoint

For each family and source skeleton, let $\mathfrak{M}_{F,\sigma}^{\text{exec}}$ be the represented packet-moduli groupoid defined in Section 4. Set

$$\mathcal{P}_{\text{osc}}^{\text{exec}} = \coprod_{F \in \mathfrak{F}_{\text{osc}}} \coprod_{\sigma \in \Sigma_F^{\text{gram}}} \mathfrak{M}_{F,\sigma}^{\text{exec}}.$$

The complete standing-core map has a kernel-pair object whose distinct-family sectors are the fifteen B10 overlap types and whose repeated-label sectors retain same-family duplicate presentations and isotropy. Iterated kernel products give the B11 higher-overlap objects. Since the chart domains, solution objects, actions, and standing cores are determined before gluing, codescent performs only duplicate-presentation identification:

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} := \text{codesc}_{[k] \in \Delta^{\text{op}}} \mathcal{C}_k^{\text{exec}}, \quad \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}.$$

The cardinal statement is logically independent of this global assembly. The active-singlet scaling family gives

$$(1, 2) \hookrightarrow Q_{\text{osc}}^{\text{SCIC}}, \quad |Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}.$$

Thus the quotient is neither finite nor countably infinite. Its exact description is the mixed moduli atlas itself, not a preferred mechanism and not a count of particle species.

1.6 Organization

Section 2 fixes the dependency packet. Section 3 develops the source-skeleton grammar and proves six-family coverage. Section 4 gives the applicability-aware residual and moduli calculus, while Sections 5 and 6 establish packet/SCIC exactness and lawful descent. Sections 7 to 13 determine the family atlases. Sections 14 and 15 resolve and assemble the overlaps. Section 16 proves the continuum result. Section 17 states the robustness and falsification conditions, Section 18 collects the completeness results, and Section 19 gives the target-extension interface.

2 Frozen target, dependency packet, and no-backflow theorem

2.1 Fixed objects

The paper works entirely inside the oscillation target already fixed by PCR-1 and PCR-2. Write

$$T_{\text{osc}}, \quad A_{\text{osc}}, \quad Q_{\text{osc}}^{\text{SCIC}} = A_{\text{osc}} / \sim_{\text{SCIC}},$$

for the target, the compatible PCR-4/PCR-5 survivor residue at terminal carrier resolution, and the final standing-carrier quotient. The complete packet is

$$\Xi_{\text{osc}}(x) = (T^{\text{face}}(x), R^{\text{stand}}(x), \Delta_{\text{fr}}^{\text{del}}(x), C^{\text{act}}(x), M^{\text{match}}(x), B^{\text{bdy}}(x), F^{\text{same}}(x)). \quad (1)$$

Throughout this manuscript, $\Delta_{\text{fr}}^{\text{del}}$ denotes exactly the deletion channel frozen by PCR-5A: the kindwise no-repair preservation/destruction/fail-closed responses already admitted before PCR-5B, with their behavior under the already licensed continuation monoid carried by the frozen continuation channel. We abbreviate it to Δ^{del} only when no ambiguity is possible. PCR-5B does not enlarge this channel by declaring a newly convenient source-subset intervention to be standing-bearing.

PCR-5A proves

$$\Xi_{\text{osc}}(x) = \Xi_{\text{osc}}(y) \iff x \sim_{\text{SCIC}} y, \quad Q_{\text{osc}}^{\text{SCIC}} \cong \text{im } \Xi_{\text{osc}}. \quad (2)$$

The principal surviving family cover is

$$\mathfrak{F}_{\text{osc}} = \{\text{LAW}, \text{LOCAL}, \text{HID}, \text{COMP}, \text{SOL}, \text{HYB}\}.$$

This is a post-PCR-4 survivor cover, not a claim that the PCR-3 source ontology contains only six primitive source kinds. In particular, PCR-3 also admits active topological, extended/defect, boundary, and environmental carrier-bearing subdiagrams. PCR-4 proves that their pure survivor strata are empty and that every surviving occurrence of those auxiliary kinds lies inside the hybrid locus. The source-to-six-family routing needed by PCR-5B is proved explicitly in [Theorem 3.6](#). For each F , the inherited presentation map is

$$\iota_F : \Theta_F^{\text{surv}} \longrightarrow A_{\text{osc}}.$$

The domains Θ_F^{surv} are the exact PCR-4 preimages of the global semantic model locus, not finite model lists.

2.2 Imported theorem roles

The following dependency table records exactly what is used and what is not imported.

Source	Imported result	Use in PCR-5B
PCR-1 [7]	Target-fixed role necessity and target fingerprint.	Fixes the oscillation target and forbids microscopic carrier coordinates from entering as later selectors.
PCR-2 [2]	Complete carrier-neutral interface, evaluator saturation, shared-witness incidence, lawful redescription.	Supplies the target-facing values, continuation, matching, boundary, failure, and common-cone semantics used by the packet.

Source	Imported result	Use in PCR-5B
PCR-3 [4]	Source-first candidate completeness, Theorem 17.5; arbitrary symbolic domains, Theorem 17.6; fixed universe convention.	Supplies finite active incidence presentations, source-normal-form groupoids, point-wise activity reduction, spectator stripping, and family source semantics.
PCR-4 [3]	Exact family-evaluation theorem, Theorem 15.1; exact hybrid locus, Theorem 18.1; evaluated residue, Theorem 18.3; nonempty proper collapse, Theorem 18.4.	Supplies exact survivor solution loci, relation residuals, typed failure sets, and the one-point simultaneous witness requirement.
PCR-5 [1]	Universal compatible occupancy, carrier-semantic descent, terminal live residue, and over-residue capstone, Theorem 24.6.	Fixes one terminal complete-carrier role and establishes that every PCR-4 survivor remains a live carrier class before SCIC quotienting.
PCR-5A [6]	Packet completeness, Theorem 4.4; family cover and overlap relation, Theorems 5.2–5.5; four-class theorem; global image theorem, Theorem 10.1.	Fixes the identity map to be classified and the four permanent packet classes.
SCIC [5]	Unique same-scope standing-carrier quotient, no hidden classifier, post-quotient finality.	Prevents family labels, provenance, simplicity, or canonical representative choices from refining or coarsening packet identity.

No upstream result gives the family atlas constructed here. Conversely, PCR-5B does not revise any imported theorem.

2.3 Operational target fingerprint

At the resolution needed here, the target retains:

1. the charged-current source and detection boundary;
2. the event-complete propagation instrument and active/null outcome semantics;
3. nontrivial cross-boundary transfer and at least two nuisance-separated visible response-mode classes;
4. causal/support continuation, environment and record transport;
5. gauge, representation, charge, anomaly, vacuum, deformation, matching, and activated-extension data where already admitted by the fixed target;
6. one-completion joint incidence and common-cone data; and
7. the licensed same-target continuation monoid used in SCIC.

The following remain open completion coordinates: primitive field inventory, law-only versus local versus hidden versus composite versus solitonic organization, ultraviolet thresholds, source-incidence architecture, and any other packet-bearing structure not fixed by the target records themselves. Their openness is why a nontrivial atlas exists.

2.4 B00: import lock and target invariance

Theorem 2.1 (PCR-5B import lock and no-backflow). *[B00] Throughout PCR-5B, the objects*

$$T_{\text{osc}}, \quad \Theta_F^{\text{surv}}, \quad A_{\text{osc}}, \quad \Xi_{\text{osc}}, \quad \sim_{\text{SCIC}}, \quad q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}}$$

remain fixed. A construction that changes an empirical record, source/detection boundary, admitted operator category, comparison-class ontology, survivor relation, standing witness family, continuation monoid, or packet coordinate is not a stratum of this atlas. It is an upstream reopening or a successor target.

Proof. The listed objects occur in strict dependency order. PCR-1 fixes the target and role content; PCR-2 fixes the target-facing interface and lawful equivalence; PCR-3 fixes source identity and the candidate universe; PCR-4 fixes survivor membership through a single simultaneous point; PCR-5 fixes terminal occupancy and the one-role carrier resolution; PCR-5A fixes the complete packet and its equality relation. PCR-5B only partitions the image of the already fixed packet on the already fixed survivor domains. If the classification were permitted to change one of those objects after inspecting the desired endpoint, a candidate or cardinality result would help define the test by which it is counted. That reverses the dependency direction and is a target or premise change rather than a classification. The four named classes are packet-separated by admitted deletion witnesses and therefore cannot be merged without contradicting (2). $\square$

2.5 Frozen deletion channel and auxiliary source-sufficiency

A complete hybrid source presentation may support useful no-repair questions finer than the identity-bearing deletion primitives already frozen by PCR-5A. They are retained, but their logical level must be explicit.

Definition 2.2 (Auxiliary source-subset sufficiency). For a survivor x with finite active source-block set K_x , let

$$N_x^{\text{aux}} : 2^{K_x} \longrightarrow \{0, 1, \perp\}$$

record the outcome of retaining exactly a constitutively well-typed subset $S \subseteq K_x$ while holding the parent source point, all retained parameters, the boundary, vacuum branch, matching orientation, continuation data, and target fixed. The value is 1 when the complete role packet is preserved without repair, 0 when a well-typed deletion destroys it, and $\perp$ when the proposed deletion is not a same-source/same-target intervention. No retuning, re-gluing, replacement source, or branch change is permitted.

The full table N_x^{aux} is an exact source-composition diagnostic. It is *not* inserted into $\Delta_{\text{ff}}^{\text{del}}$ by definition.

Theorem 2.3 (Subset-deletion conservativity at the frozen target). *Let D_S be any fixed-target consequence functional extracted from $N_x^{\text{aux}}(S)$. Exactly one of the following applies.*

1. Already standing-bearing. *If D_S is an admitted same-scope standing functional under the frozen*

PCR-5/PCR-5A activation, then there is a target-lawful map F_S on the frozen packet image such that

$$D_S = F_S \circ \Xi_{\text{osc}}.$$

Thus it can separate two base carriers only when the already frozen packet separates them.

2. Standing-inert source information. *If D_S varies inside one Ξ_{osc} -fibre while the target, continuation family, compatibility/failure semantics, and all frozen witnesses remain fixed, that variation is source-composition/provenance data and cannot refine SCIC identity.*
3. Update. *If forming or interpreting D_S requires a new source object, changed target record, changed operator category, changed boundary, new continuation, or new standing witness, it is a source/target update and belongs outside the PCR-5B quotient.*

Consequently the identity-bearing deletion field of every PCR-5B packet normal form is exactly $\Delta_{\text{fr}}^{\text{del}}$. The full source-subset profile may be retained and exported only as an auxiliary atlas annotation unless and until a downstream target extension independently activates one of its distinctions.

Proof. PCR-5A proves that equality of Ξ_{osc} is equality of the complete SCIC standing continuation signature. Therefore every same-scope standing-bearing functional factors through the packet image: otherwise two points with equal complete standing signatures would receive different standing consequences. That would contradict witness completeness and the no-second-classifier consequence of SCIC. This proves the first branch. If the packet remains fixed and the proposed subset value nevertheless differs, the difference has no admitted same-scope standing authority and is therefore provenance/source-composition information at this target, proving the second branch. If the evaluation cannot be formed without changing source or target typing, B00 classifies the move as an update, proving the third. The three cases are disjoint and exhaustive under the fixed target. $\square$

Corollary 2.4 (No retroactive base splitting by source-subset diagnostics). *Two PCR-5B presentations with equal frozen packets remain one base SCIC class even if their auxiliary source-sufficiency tables differ. A later target extension may make such a difference standing-visible and split that base class, but it may not retroactively refine $Q_{\text{osc}}^{\text{SCIC}}$.*

Corollary 2.5 (No phenomenon import). *Absolute-mass records, Majorana/LNV records, heavy-state observations, nonunitarity, charged-lepton-flavour violation, cosmological transport, astrophysical propagation, and local/global/remnant $B - L$ data cannot eliminate, split, or select a PCR-5B stratum. Their lawful use begins only after PCR-5C forms a target-extension map.*

2.6 One-point discipline

Definition 2.6 (One-point family solution). A family parameter point $\theta \in \Theta_F^{\text{surv}}$ is one complete source-coherent object whose channel, response, representation, anomaly, vacuum, matching, continuation, failure, and common-cone data are all evaluated at that same point.

Lemma 2.7 (No marginal witness splicing). *No atlas object can take its channel from one parameter point, anomaly cancellation from another, vacuum branch from a third, or matching map from a fourth.*

Proof. PCR-4 defines every survivor locus as the preimage of one global semantic model object under one family evaluation map. Its relation residuals enforce common source, channel, representation, vacuum, matching, extension, and cone data. A tuple assembled from several points is not in the domain of ι_F and therefore has no packet in A_{osc} . $\square$

2.7 Permanent lower structure

PCR-5A exhibits four points

$$x_W, \quad x_{3N}, \quad x_{\text{red}}, \quad x_{\text{syn}}$$

with law/new-local no-repair deletion pairs

$$(0, 1), \quad (1, 0), \quad (1, 1), \quad (0, 0).$$

Write their SCIC classes as

$$q_W, \quad q_{3N}, \quad q_{\text{red}}, \quad q_{\text{syn}}.$$

These values are standing-bearing because they report preservation or destruction of the complete role packet after fixed source-kind deletion without retuning. Source names alone are provenance; the deletion consequence is target-standing.

Lemma 2.8 (Four-class floor). *Every exact PCR-5B atlas contains four distinct connected components representing $q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}}$.*

Proof. Their deletion-coordinate projections are the four distinct points of $\{0, 1\}^2$. Packet equality implies equality of every coordinate, so no pair can be identified. The atlas is required to cover every PCR-4 survivor and hence contains a presentation of each point. $\square$

2.8 Proof and supplementary representations

The scientific result is the semantic classification established by B01–B15. Supplementary symbolic tables and consistency checks provide an inspectable representation of the source grammar, family relations, overlap incidences, and theorem dependencies, but they carry no independent classificatory authority. Source completeness, packet completeness, family stratification, and overlap closure remain consequences of the mathematical arguments below.

3 Explicit source-skeleton grammar

3.1 Why the skeleton index must be generated before survivor inspection

The PCR-5B atlas cannot define its skeleton set as “the skeletons that occur among survivors.” That would repeat the survivor image in different notation and would leave the candidate-to-stratum bridge unresolved. The skeleton index used below is therefore generated syntactically from the frozen PCR-3 source signature before the PCR-4 equations are imposed. Survivor status is subsequently computed on each generated skeleton.

The source-type tags are inherited from the source-first carrier theory. Write

$$\begin{aligned} \mathbf{N} &= \{\text{Existing, Law, Local, Hidden, Derived, Soliton, Boundary, Extended, Topological, Environment}\}, \\ \mathbf{E} &= \{\text{Couple, Bind, Project, Attach}_{\partial}, \text{Attach}_T, \text{Fuse, Inflow, CoCarry, ActLaw, Compose}\}. \end{aligned}$$

The tag sets are finite. Their source attributes are not: a tag may carry an arbitrary source-defined representation, support, function space, operator family, lattice, moduli category, or other parameter object admitted by PCR-3.

Definition 3.1 (Canonical finite skeleton code). A canonical skeleton code is a tuple

$$\sigma = (n_V, n_E, \nu, \epsilon, s, t, \mathcal{I}, \mathcal{A}, \mathcal{W}, \mathcal{K}, \mathcal{K}^{\text{full}}, \mathcal{G}, \mathcal{H}_{\text{cross}}), \quad (3)$$

with the following data.

1. The vertex and hyperedge sets are the initial segments $V_\sigma = [n_V]$ and $E_\sigma = [n_E]$.
2. $\nu : V_\sigma \rightarrow \mathbf{N}$ and $\epsilon : E_\sigma \rightarrow \mathbf{E}$ are constructor tags.
3. For every hyperedge e , the ordered finite source and target lists $s(e), t(e) \in V_\sigma^{<\omega}$ are specified.
4. $\mathcal{I}$ gives the source sorts of all node and incidence attributes, including representation, support, persistence, derivation, attachment, coupling, profile, matching, and extension sorts.
5. $\mathcal{A}$ gives the type-indexed activity-support maps and their exact null classes.
6. $\mathcal{W}$ gives the finite source-constitutive coherence diagram: representation composition, derivation squares, projection identities, fusion/module cells, boundary functoriality, process axioms, and hybrid gluing cells actually used by the code.
7. $\mathcal{K} \subseteq \{\text{LAW}, \text{LOCAL}, \text{HID}, \text{COMP}, \text{SOL}\}$ is the principal feature mask used by the first five charts.
8. The full activity-reduced carrier-kind mask is

$$\mathcal{K}^{\text{full}} \subseteq \{\text{LAW}, \text{LOCAL}, \text{HID}, \text{COMP}, \text{SOL}, \text{TOP}, \text{EXT}, \text{BDY}, \text{ENV}\}.$$

A kind belongs to $\mathcal{K}^{\text{full}}$ exactly when a source-active subdiagram of that kind lies on, or is source-constitutively required by, a current target-bearing carrier path after spectator stripping. The fixed Existing substrate is not counted merely for being present; the strict BASE branch is handled by the frozen PCR-4 reduction.

9. $\mathcal{G}$ is the finite source-coherent gluing diagram carried by a PCR-3 incidence-reconstruction presentation, and $\mathcal{H}_{\text{cross}}$ records the surviving cross-component incidences and irreducible higher-arity relations. Different lawful reconstruction presentations of the same active carrier trace are related by the HYB source groupoid rather than promoted to standing identity.

Two codes are equal only after the initial-segment convention has fixed the finite combinatorial carrier. Lawful permutations and source isomorphisms are handled by the constructive groupoid, not by silently identifying the codes.

3.2 Formation rules

Let Code be the least class closed under the following productions.

Production	Input/output	Formation guard
Node_a	Introduce one node tagged by $a \in \mathbf{N}$ with a declared source attribute object.	The attribute object is typed by the frozen source ontology; no downstream viability condition is used.
Edge_r	Introduce a finite hyperedge tagged by $r \in \mathbf{E}$ between already formed nodes.	Endpoint sorts match the incidence signature; the required source parameter object is declared.

Production	Input/output	Formation guard
CoherentClose	Adjoin the constitutive cells required by the tags and incidences.	The representation, derivation, projection, fusion, boundary, law, or gluing equations are well typed.
Attach	Add source/detector and target-role attachment edges.	The attachment types are those of the frozen oscillation target.
Compose	Form a finite typed pushout/gluing of two previously formed codes.	Shared supports and incidence maps are declared before any PCR-4 evaluation; all source gluing squares commute.
ActivityReduce	At a source point, delete exactly null incidences and the spectator components disconnected from every active target path.	Activity is determined by the source activity map, with no weak-coupling threshold.
CanonicalRelabel	Relabel the remaining finite vertex and edge sets by initial segments.	Only finite labels change; all source sorts and incidence data are preserved.

Definition 3.2 (Source-semantic hybrid witness). Let σ be source coherent and participating after exact activity reduction. A *hybrid witness* for σ is a finite PCR-3 incidence-reconstruction presentation

$$\omega = (\sigma_1, \dots, \sigma_m; \mathcal{G}, \mathcal{H}_{\text{cross}}), \quad m \geq 2,$$

in which every σ_i is occupancy bearing, the complete active code is the source-coherent gluing of the σ_i , and at least one cross-component gluing/incidence relation survives activity reduction or the constituent diagrams share a nontrivial constitutive gluing cell. Write $\text{HybWit}(\sigma)$ for the groupoid of such witnesses under source isomorphism.

Definition 3.3 (Family grammar predicates). For a source-coherent participating code σ , define

$$\begin{aligned} G_{\text{LAW}}(\sigma) &:\Longleftrightarrow \text{LAW} \in \mathcal{K}_\sigma, \\ G_{\text{LOCAL}}(\sigma) &:\Longleftrightarrow \text{LOCAL} \in \mathcal{K}_\sigma, \\ G_{\text{HID}}(\sigma) &:\Longleftrightarrow \text{HID} \in \mathcal{K}_\sigma, \\ G_{\text{COMP}}(\sigma) &:\Longleftrightarrow \text{COMP} \in \mathcal{K}_\sigma, \\ G_{\text{SOL}}(\sigma) &:\Longleftrightarrow \text{SOL} \in \mathcal{K}_\sigma, \\ G_{\text{HYB}}(\sigma) &:\Longleftrightarrow \text{HybWit}(\sigma) \neq \emptyset. \end{aligned}$$

The first five predicates are feature charts, not mutually exclusive model labels. A mixed LAW–LOCAL point therefore lies in LAW, LOCAL, and HYB simultaneously.

Lemma 3.4 (Complete syntactic witnesses for the HYB predicate). *For an activity-reduced source code, the source-semantic condition $G_{\text{HYB}}(\sigma)$ is witnessed whenever any of the following occurs:*

1. $|\mathcal{K}_\sigma^{\text{full}}| \geq 2$;
 2. a nontrivial cross-component incidence or gluing survives activity reduction, including active Inflow, Fuse, CoCarry, Project, or boundary/edge attachment relations joining constituent subdiagrams;
 3. an irreducible higher-arity cross relation joins independently generated occupancy-bearing components;
- or

4. the PCR-3 gluing diagram $\mathcal{G}_\sigma$ contains at least two occupancy-bearing constituent source subdiagrams.

Conversely every source-semantic hybrid admits a PCR-3 incidence-reconstruction witness satisfying item 4, and therefore the disjunction of the four displayed syntactic witnesses is equivalent to G_{HYB} . No literal **Compose** edge is required to survive canonicalization.

Proof. Each of items 1–3 supplies two or more source-coherent occupancy-bearing constituents together with a retained joint incidence or gluing datum, hence a hybrid witness. For the converse, PCR-3 finite-hybrid closure reconstructs every finitely presented semantic hybrid by generating its occupancy-bearing constituent subdiagrams and adjoining the finite set of cross-component hyperedges and source-coherence data. That reconstruction is exactly item 4. Activity reduction removes source-null relations but retains every nonnull relation without a magnitude cutoff, so a surviving **Inflow**, **Fuse**, **CoCarry**, **Project**, **boundary/edge**, or higher-arity relation remains a valid witness even when no edge is tagged **Compose**. $\square$

A pure principal F code has $\mathcal{K}_\sigma^{\text{full}} = \{F\}$ and $\text{HybWit}(\sigma) = \emptyset$. Pure auxiliary **TOP**, **EXT**, **BDY**, **ENV** and strict **BASE** codes are retained by the pre-routing source grammar for disposition, but they do not become additional terminal family labels; their survivor routing is fixed by PCR-4 and proved below.

For each family set

$$\Sigma_F^{\text{gram}} := \{\sigma \in \text{Code} : G_F(\sigma), \text{SrcCoh}(\sigma), \text{Part}(\sigma)\}. \quad (4)$$

Here **SrcCoh** is the conjunction of the constitutive equations carried by $\mathcal{W}$, and **Part** says that source activity reduction leaves a nonempty target-participating core. Neither predicate mentions a target record, anomaly pass, vacuum pass, matching pass, common cone, or desired carrier endpoint.

3.3 Raw parameter objects and source equations

For $\sigma \in \Sigma_F^{\text{gram}}$, the raw parameter object is the typed product

$$\Theta_{F,\sigma}^{\text{raw}} = \prod_{v \in V_\sigma} P_{\nu(v), \mathcal{I}(v)} \times \prod_{e \in E_\sigma} P_{\epsilon(e), \mathcal{I}(e)} \times P_{\mathcal{W}_\sigma}. \quad (5)$$

The participating source object is the exact subobject cut out by the following named source equations.

ID	Source equation or support condition
S1	Representation-bearing nodes satisfy identity and composition laws and are compatible with every declared source morphism.
S2	Every Derived node has a constituent map and derivation/binding map for which all retained constituent-incidence squares commute.
S3	Every Project edge satisfies its declared embedding, quotient, reduction, or dilation identities.
S4	Every Fuse or module edge satisfies the declared pentagon, triangle, unit, junction, and module equations.
S5	Boundary and target attachments are functorial and preserve the declared source identities.
S6	Every Law node and ActLaw edge satisfies the constitutive axioms of its declared process category.
S7	Every finite hybrid gluing square and shared-support diagram commutes.

ID	Source equation or support condition
A1	A scalar- or operator-valued coupling edge is retained exactly on the nonempty support of its instantiated physical interaction morphism.
A2	A projection/dilation edge is restricted to its source-defined active support; exact rank drop may shrink the diagram but does not invoke a numerical threshold.
A3	Boundary and target attachments count only on their source-non-null support.
A4	Fusion, junction, and module actions retain the non-null action together with its constitutively required cells.
A5	Environmental and hidden co-carriage is retained only when actually connected to an active target path.

Thus

$$\Theta_{F,\sigma}^{\text{part}} = \text{Sol}_{\Theta_{F,\sigma}^{\text{raw}}}(S1:S7, A1:A5). \quad (6)$$

This is a source calculation. The PCR-4 survivor equations enter only in the next section.

3.4 Grammar completeness

Theorem 3.5 (Explicit skeleton-grammar completeness). *Every finite occupancy-bearing source trace in the PCR-3 comparison class determines a code in `Code`, unique up to the constructive source groupoid defined in Section 6. Conversely every point of $\Theta_{F,\sigma}^{\text{part}}$ interprets as a source-coherent activity-reduced carrier presentation of the family features named by G_F . The construction of Σ_F^{gram} is independent of all PCR-4 pass/fail data.*

Proof. PCR-3 reconstructs each finite carrier core from the node and incidence constructor signature and proves source-coherent normal-form soundness. Choose finite labels, apply the initial-segment convention, and record the source sorts, activity maps, and coherence cells in (3). This produces an element of `Code` by the listed formation rules. A different finite labeling or source-equational presentation is generated by graph, basis, chart, or coherence morphisms and is therefore not a new groupoid orbit. Conversely S1–S7 give source objecthood and A1–A5 give the pointwise participating core, so interpretation of a participating skeleton is a PCR-3 candidate. No target evaluator appears in the construction, establishing generation independence. $\square$

Theorem 3.6 (Post-PCR-4 principal-family routing). *Let σ be any PCR-3 activity-reduced participating source code and let θ be a PCR-4 survivor on that code. Then (σ, θ) lies in at least one of the six principal PCR-5B chart domains*

$$\{\text{LAW}, \text{LOCAL}, \text{HID}, \text{COMP}, \text{SOL}, \text{HYB}\}.$$

More precisely, any surviving topological, extended/defect, boundary, or environmental carrier-bearing occurrence is routed to HYB, and the strict BASE survivor branch is empty except through the already frozen overlap reductions.

Proof. If $\mathcal{K}_\sigma$ contains LAW, LOCAL, HID, COMP, or SOL, the corresponding feature predicate holds by construction. If the source architecture is a nontrivial finite composition, Theorem 3.4 places it in HYB even when its only cross incidence is tagged Inflow, Fuse, CoCarry, Project, boundary/edge, or an irreducible higher-arity relation.

It remains to disposition the auxiliary pure source kinds. PCR-4 proves the exact survivor reductions

$$\begin{aligned}\Theta_{\text{TOP}}^{\text{surv}} &= \Theta_{\text{TOP} \cap \text{HYB}}^{\text{surv}}, & \Theta_{\text{EXT}}^{\text{surv}} &= \Theta_{\text{EXT} \cap \text{HYB}}^{\text{surv}}, \\ \Theta_{\text{BDY}}^{\text{surv}} &= \Theta_{\text{BDY} \cap \text{HYB}}^{\text{surv}}, & \Theta_{\text{ENV}}^{\text{surv}} &= \Theta_{\text{ENV} \cap \text{HYB}}^{\text{surv}},\end{aligned}$$

so the pure TOP, EXT, BDY, and ENV survivor strata are empty and every surviving occurrence of those kinds is already part of a complete dynamical hybrid. PCR-4 likewise makes the strict BASE stratum empty; any surviving existing-substrate occurrence appears only in an overlap with a principal surviving construction, and an ENV overlap is again routed through HYB by the preceding equality. Therefore every PCR-4 survivor has at least one principal chart route. These are inherited survivor reductions, not new selection rules. [3] $\square$

Corollary 3.7 (Six-family survivor cover from the source grammar). *The union of the six generated principal chart domains covers every PCR-4 survivor. Thus the PCR-5A family cover is recovered from the independently generated PCR-3 source grammar plus the frozen PCR-4 pure-stratum reductions rather than assumed as a family-name registry.*

Definition 3.8 (Computed survivor-code index). The survivor-code index is

$$\Sigma_F^{\text{surv}} := \{\sigma \in \Sigma_F^{\text{gram}} : \text{Sol}(\mathcal{E}_{F,\sigma}^{\text{surv}}) \neq \emptyset\}, \quad (7)$$

where the finite named system $\mathcal{E}_{F,\sigma}^{\text{surv}}$ is stated explicitly in Section 4 and specialized family by family in Sections 7 to 9 and 11 to 13. Empty codes are retained in the failure register with their exact obstruction; they are not erased from the grammar.

Equation (7) is a determination rule over a pre-existing grammar, not the phrase “all skeletons occurring among survivors.” It supplies the exact index required for the atlas even when the solution object is algebraic, operator-theoretic, functional, categorical, or otherwise not reducible to a finite numerical list.

4 Exact residual, applicability, and packet-moduli calculus

4.1 Universal survivor variables

Fix $F \in \mathfrak{F}_{\text{osc}}$ and $\sigma \in \Sigma_F^{\text{gram}}$. A source point $\theta \in \Theta_{F,\sigma}^{\text{part}}$ determines, by the PCR-3 interpretation and PCR-4 evaluator, one simultaneous tuple

$$\mathcal{V}_F(\theta) = (\mathcal{I}_{\lambda,o}, \rho_\alpha, P_\beta, K_{\beta\alpha}, A^s, A^d, \rho_G, \mathcal{D}_{\text{caus}}, \mathcal{A}, \mathcal{V}, \mathcal{M}, \mathcal{X}, \mathcal{R}, \text{Cone})_\theta. \quad (8)$$

All entries come from the same source point. The instrument includes the active and null/loss outcomes; $K_{\beta\alpha}$ is computed from that instrument and the same source/effect pair; $\mathcal{A}$ is the complete perturbative and activated global anomaly vector; $\mathcal{V}$, $\mathcal{M}$, and Cone are the vacuum, oriented matching, and common-cone solution objects; and $\mathcal{R} = (R_1, \dots, R_9)$ is the cross-signature residual packet.

4.2 The named PCR-4 system

The exact residual solution object is defined by the following finite named system, interpreted in the source category carried by σ .

ID	Equation, inequality, existence condition, or status clause
E00	The source equations S1–S7 and activity conditions A1–A5 of Section 3 hold.
E01	$J_{\lambda,o} \succeq 0$ for every admitted protocol and event outcome.
E02	$\sum_o \mathcal{I}_{\lambda,o}^\dagger(1) = 1$ for every admitted protocol λ .
E03	$K_{\beta\alpha}(\lambda) = \text{Tr}[P_\beta \mathcal{I}_\lambda(\rho_\alpha)]$ using the source/effect and instrument in the same tuple.
E04	$R(K) = R(T_{\text{osc}})$ for every frozen record evaluator $R \in \text{Rec}_T$.
E05	The fixed target visible-mode packet embeds into the candidate packet under the frozen lawful mode equivalence: $\mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_F(\theta))$.
E06	Nonalignment is realized: $\exists \alpha \neq \beta, \lambda$ with $K_{\beta\alpha}(\lambda) > 0$.
E07	Charged-current attachment intertwines the same active representation: $A^{s,d} \rho_G(g) = \rho_{CC}(g) A^{s,d}$ for all admitted g , with operational flavor and residual electromagnetic charge preserved.
E08	The causal/support solution object is nonempty, $\mathcal{D}_{\text{caus}}(\theta) \neq \emptyset$, and in a local representative $\text{supp } \mathcal{I}_{\lambda,o} \subseteq C_\lambda^+$.
E09	The environment action is defined on the frozen environment domain and preserves the charged-current boundary.
E10	The complete anomaly vector, including physically attached inflow terms and every activated global obstruction, vanishes: $\mathcal{A}_F(\theta) = 0$.
E11	The common vacuum/deformation solution object is nonempty: $\mathcal{V}_F(\theta) \neq \emptyset$.
E12	The oriented source-to-target matching solution object is nonempty: $\mathcal{M}_F(\theta) \neq \emptyset$.
E13	For every activated extension relation, all fusion, module, junction, global-form, inflow, and attachment equations hold at the same point; if the relevant operator/extension sort is theoremtically absent, that relation is certified N/A rather than counted as Pass.
E14	The common physical cone is nonempty: $\text{Cone}_F(\theta) \neq \emptyset$.
E15	$R_1 = 0$: source interpretation, active carrier trace, parameter point, and target attachment are restrictions of one source object.
E16	$R_2 = 0$: the bearer assigned to the nested roles is the bearer entering the instrument, kernel, visible-mode map, and boundary attachments.
E17	$R_3 = 0$: every record and visible-mode evaluator is applied to the kernel in E03.
E18	$R_4 = 0$: the continuation/dilation witness reduces to the same instrument and uses the same environment/vacuum state.
E19	$R_5 = 0$: charged-current attachments use the same active representation and residual charge object as the channel.
E20	$R_6 = 0$: the anomaly vector is computed from that representation/charge object, and every inflow term is attached to the obstruction it cancels.
E21	$R_7 = 0$: source equations, carrier nodes, channel, representation, environment, and extension packet are defined on or reduced from one vacuum branch.
E22	$R_8 = 0$: matching commutes with active reduction, interface extraction, record/mode evaluation, anomaly transport, vacuum restriction, and extension transport.
E23	$R_9 = 0$: every active extension attachment or inflow map terminates in the same target/operator fibre as the effect it supports.

Family-specific source equations are appended to E00 and are displayed in the corresponding family section. They do not alter E01–E23.

4.3 Applicability-preserving realization of the PCR-4 evaluator

The named rows E00–E23 and the family equations are schemas for the instantiated PCR-4 relation clauses supported by a particular source skeleton. Let

$$I_{F,\sigma}$$

denote the complete instantiated relation-index set obtained from those rows at (F, σ) . For every $j \in I_{F,\sigma}$, define before evaluating its physical truth value the applicability predicate

$$\begin{aligned} A_j(\theta) &\in \{0, 1\}, \\ A_j(\theta) = 1 &\iff \text{all input witness sorts of relation instance } j \text{ occur} \\ &\quad \text{in the frozen target and PCR-3 activity-reduced trace.} \end{aligned} \tag{9}$$

When $A_j(\theta) = 0$, the point carries a theorem certificate $c_j^{\text{NA}}(\theta)$ exactly when that absence is permitted by the frozen PCR-4 target/operator category. The applicability predicate and its certificate are functions only of the target fingerprint, the PCR-3 active trace, and the relation schema; they are formed before the value of the relation is known. Thus a failed active relation cannot be reclassified as inapplicable.

The row-level source and family equations are interpreted in the same way: each displayed row is the finite/source-indexed family of its typed relation instances. If a row has no instantiated input sort, its row summary is theorem-certified N/A rather than a vacuous physical Pass. In particular, inactive extended/defect, inflow, fusion, module, line/surface, environmental, or dilation relations remain explicitly distinguishable from active relations that fail.

Definition 4.1 (Applicability-aware compiler status). For every instantiated relation clause j , PCR-5B uses the total compiler alphabet

$$\widehat{\text{Val}} = \{\text{NA}, \text{pass}, \text{fail}, \text{undefined}, \text{source update}, \text{target update}\}.$$

Its value $\widehat{e}_j(\theta)$ is assigned in the following dependency order:

1. **target update** if evaluation requires a changed target fibre, primitive target datum, background, boundary semantics, or operator category;
2. **source update** if the target is fixed but the point no longer belongs to the frozen PCR-3 source object/comparison class;
3. **NA** if $A_j(\theta) = 0$ and $c_j^{\text{NA}}(\theta)$ certifies the theorem-permitted absence;
4. **pass** if $A_j(\theta) = 1$ and the complete physical predicate holds;
5. **fail** if $A_j(\theta) = 1$ and the complete physical predicate fails;
6. **undefined** only when an active or purportedly inactive relation has not been assigned one of the preceding scientifically typed outcomes. This is retained as a compiler/evaluation defect and never promoted to Pass or N/A.

The scientific PCR-4 projection is

$$\text{NA} \mapsto \text{N/A}, \quad \text{pass} \mapsto \text{Pass}, \quad \text{fail} \mapsto \text{Fail}, \quad \text{target update} \mapsto \text{Target Update}.$$

The PCR-5B source update branch records departure from the frozen PCR-3 source object before PCR-4 evaluation. `undefined` has no fifth PCR-4 scientific meaning; it is a fail-closed completeness defect that blocks survivor status until resolved.

Theorem 4.2 (No inapplicability laundering). *For every compiled relation instance j , A_j is fixed before $\hat{e}_j$ is evaluated. If $A_j(\theta) = 1$, no subsequent failure, undefinedness, packet value, or desired endpoint can make j N/A while θ , the target, and the PCR-3 activity-reduced source trace are held fixed.*

Proof. This is the PCR-4 no-inapplicability-laundering theorem restricted to the generated source chart. Applicability is the pullback of the frozen PCR-4 activity predicate to the PCR-3 active trace. Its arguments are upstream of the relation value. Changing applicability after a failed evaluation therefore changes the candidate point, source object, or target/operator fibre rather than repairing the same evaluation. $\square$

Definition 4.3 (Exact guarded residual solution object). For a grammar code σ , let $P_j(\theta)$ denote the physical predicate carried by relation instance j . Its guarded clause is

$$\mathcal{G}_j(\theta) :\iff (A_j(\theta) = 0 \wedge c_j^{\text{NA}}(\theta)) \vee (A_j(\theta) = 1 \wedge P_j(\theta)). \quad (10)$$

The exact survivor object is

$$Z_{F,\sigma} := \left\{ \theta \in \Theta_{F,\sigma}^{\text{part}} : \begin{array}{l} \text{no source/target update occurs, every } \mathcal{G}_j(\theta) \text{ holds,} \\ \text{and no active evaluation is undefined} \end{array} \right\}. \quad (11)$$

Equivalently, $Z_{F,\sigma}$ is the applicability-guarded solution object of E00–E23 and the family source equations. An inapplicable relation contributes its theorem-certified N/A record; it is not replaced by a Boolean pass. Nonemptiness of an existential moduli object such as $\mathcal{V}_F$, $\mathcal{M}_F$, or Cone_F remains a displayed active clause wherever its input sorts are present.

Theorem 4.4 (PCR-4 compiler conservativity). *For every generated family code and every $\theta \in \Theta_{F,\sigma}^{\text{part}}$,*

$$\theta \in Z_{F,\sigma} \iff \theta \in \Theta_F^{\text{surv,PCR4}} \text{ with the same PCR-3 source presentation.} \quad (12)$$

Moreover the set of N/A relation instances and their certificates is exactly the restriction of the PCR-4 inactive/theoremically-absent relation set; every active PCR-4 relation has the same Pass/Fail value in the compiled system.

Proof. PCR-4 forms its activity set from the frozen target and the PCR-3 pointwise active carrier trace before evaluating any predicate. The present A_j is that predicate restricted to each named relation instance. When $A_j = 0$, the same upstream absence certificate gives N/A; when $A_j = 1$, the corresponding clause evaluates the same physical relation on the same one-point witness tuple. E00–E23 are the named normal-form presentation of the imported PCR-4 intrinsic, cross-signature, common-cone, and typing relations, while the family equations specialize the already fixed source chart. Hence the guarded conjunction is equivalent to PCR-4’s condition that all active constraints pass, every N/A is theorem-certified, and no target update occurs. Source-update points are outside the frozen PCR-3 candidate

object, and active undefined points are retained as unresolved evaluations rather than admitted as survivors. This proves both directions without changing survivor membership. $\square$

4.4 Total applicability/status decomposition

Define the complete evaluation-status vector

$$\widehat{\text{Stat}}_{F,\sigma}(\theta) = (\widehat{e}_j(\theta))_{j \in I_{F,\sigma}}. \quad (13)$$

For a raw point with no source or target update set

$$\begin{aligned} \mathbf{N}_{F,\sigma}(\theta) &:= \{j : \widehat{e}_j(\theta) = \text{NA}\}, \\ \mathbf{F}_{F,\sigma}(\theta) &:= \{j : \widehat{e}_j(\theta) = \text{fail}\}, \\ \mathbf{U}_{F,\sigma}(\theta) &:= \{j : \widehat{e}_j(\theta) = \text{undefined}\}. \end{aligned}$$

The three sets are pairwise disjoint. Every $j \in \mathbf{N}$ additionally carries its certificate c_j^{NA} . The no-update status record is

$$\mathbf{D}_{F,\sigma}^{\text{app}}(\theta) := (\mathbf{N}_{F,\sigma}(\theta), \mathbf{F}_{F,\sigma}(\theta), \mathbf{U}_{F,\sigma}(\theta), (c_j^{\text{NA}})_{j \in \mathbf{N}}). \quad (14)$$

A point survives exactly when $\mathbf{F} = \mathbf{U} = \emptyset$, all entries of $\mathbf{N}$ are certified, and the guarded family system is satisfied. Thus a survivor may have a nonempty N/A set.

For pairwise-disjoint $N, F, U \subseteq I_{F,\sigma}$ with $F \cup U \neq \emptyset$, let

$$\begin{aligned} \Theta_{F,\sigma}^{\text{def}(N;F,U)} &= \{\theta : \mathbf{N} = N, \mathbf{F} = F, \mathbf{U} = U, \text{ every member of } N \text{ is certified, and no update occurs}\}, \\ Z_{F,\sigma}^N &= \{\theta \in Z_{F,\sigma} : \mathbf{N}_{F,\sigma}(\theta) = N\}, \\ \Theta_{F,\sigma}^{\text{SU}} &= \{\theta : \widehat{\text{Stat}}_{F,\sigma}(\theta) \text{ contains a source update and no target update}\}, \\ \Theta_{F,\sigma}^{\text{TU}} &= \{\theta : \widehat{\text{Stat}}_{F,\sigma}(\theta) \text{ contains a target update}\}. \end{aligned}$$

Target update has priority over source update. The exact raw decomposition is

$$\Theta_{F,\sigma}^{\text{raw}} = \left(\prod_N Z_{F,\sigma}^N \right) \sqcup \coprod_{\substack{N,F,U \text{ pairwise disjoint} \\ F \cup U \neq \emptyset}} \Theta_{F,\sigma}^{\text{def}(N;F,U)} \sqcup \Theta_{F,\sigma}^{\text{SU}} \sqcup \Theta_{F,\sigma}^{\text{TU}}. \quad (15)$$

The N/A set is therefore never erased merely because it is not itself a failure. Failure-only, undefined-only, mixed fail-plus-undefined, and certified-inapplicability patterns are all retained, and an active failure cannot be moved into N by a later choice of interpretation.

4.5 Lawful presentation groupoids

For each source skeleton σ , the source theory supplies automorphism group objects for its typed components. Define

$$G_{F,\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes \left(\prod_{v \in V_\sigma} \text{Aut}_{\text{src}}(v) \times \prod_{e \in E_\sigma} \text{Aut}_{\text{src}}(e) \times \text{Aut}_{\text{chart}}(\Theta_{F,\sigma}^{\text{raw}}) \right)_{\text{fix } T_{\text{osc}}}. \quad (16)$$

The subscript says that the action fixes the target fingerprint, charged-current boundary, source activity class, and the typed orientation of matching and continuation.

Definition 4.5 (Generator system). The action groupoid $\Gamma_{F,\sigma} = G_{F,\sigma} \ltimes Z_{F,\sigma}$ is generated by:

1. finite typed-hypergraph isomorphisms and permutations of repeated isomorphic nodes;
2. source basis transformations in $\text{Aut}_{G_{\text{fixed}}}(V_v)$, together with the induced contragredient action on couplings, mass/kinetic maps, attachments, and anomaly tensors;
3. fixed-boundary gauge transformations and simultaneous weak-basis/rephasing transformations;
4. source-chart isomorphisms and lawful parameter-domain reparameterizations;
5. projection/dilation isometries for HID charts;
6. constituent and derived-object isomorphisms for COMP charts;
7. profile, collective-coordinate, and internal-symmetry reparameterizations for SOL charts; and
8. coherent hypergraph/gluing isomorphisms for HYB charts.

Only the generators applicable to the tags of σ are present.

The relations are: identities and inverses; incidence preservation; the semidirect conjugation relations between graph permutations and component automorphisms; naturality of every source-coherence cell; fixed-boundary covariance; and compatibility with the action on E00–E23. A transformation that changes a physical threshold, retained memory kernel, matching orientation, source responsibility, boundary ownership, deletion result, or failure locus is not a groupoid generator unless the changed value is proven packet-inert.

For $\theta \in Z_{F,\sigma}$, the isotropy group is the explicit stabilizer

$$\text{Iso}_{F,\sigma}(\theta) = \{g \in G_{F,\sigma} : g \cdot \theta = \theta\}. \quad (17)$$

Thus generators, relations, action, and isotropy are fixed constructively rather than by the phrase “all lawful transformations.”

4.6 Standing-normal-form variables

Let

$$\mathbf{s} = (t, r, \delta, c, m, b, f) \in \mathcal{Y}_{F,\sigma}^{\text{NF}} = \mathcal{T}_{F,\sigma} \times \mathcal{R}_{F,\sigma} \times \mathcal{D}_{F,\sigma} \times \mathcal{C}_{F,\sigma} \times \mathcal{M}_{F,\sigma} \times \mathcal{B}_{F,\sigma} \times \mathcal{F}_{F,\sigma} \quad (18)$$

be variables in the seven canonical packet channels. Their domains are fixed by the skeleton types and the PCR-5A packet codomains. In particular, δ is the frozen PCR-5A deletion channel $\Delta_{\text{fr}}^{\text{del}}$; any finer source-subset sufficiency table belongs to the auxiliary annotation of [Theorem 2.2](#) and is not an additional packet coordinate. Define the realization equations

$$\mathcal{R}_{F,\sigma}^{\text{NF}}(\theta, \mathbf{s}) : \begin{cases} t = T^{\text{face}}(\theta), \\ r = R^{\text{stand}}(\theta), \\ \delta = \Delta_{\text{fr}}^{\text{del}}(\theta), \\ c = C^{\text{act}}(\theta), \\ m = M^{\text{match}}(\theta), \\ b = B^{\text{bdy}}(\theta), \\ f = F^{\text{same}}(\theta). \end{cases} \quad (19)$$

Definition 4.6 (Packet-moduli functor of points). Let $\mathbf{Test}_{F,\sigma}$ be the source test category inherited from the parameter sorts of σ . For a test object S , define $\mathfrak{Mod}_{F,\sigma}(S)$ to be the groupoid whose objects are

triples

$$(\theta_S, \mathbf{s}_S, p_S), \quad (20)$$

where $\theta_S : S \rightarrow \Theta_{F,\sigma}^{\text{part}}$ is an S -family satisfying the applicability-guarded E00–E23 and family equations, with every inactive relation carrying its theorem-certified N/A witness, $\mathbf{s}_S : S \rightarrow \mathcal{Y}_{F,\sigma}^{\text{NF}}$ is an S -family of seven-channel records, and p_S is the seven-component equality $\mathbf{s}_S = \Xi_{\text{osc}}(\theta_S)$. Arrows are generated by the base-changed action of $G_{F,\sigma}$. A map $S' \rightarrow S$ acts by pullback of every field, equation, and group arrow.

Theorem 4.7 (Representability and universal property of the family packet moduli). *The functor $\mathfrak{Mod}_{F,\sigma}$ is represented by the action groupoid of the two-sorted solution object in (21). More precisely, for every test object S , there is a natural equivalence between $\mathfrak{Mod}_{F,\sigma}(S)$ and the groupoid of S -points of that quotient presentation. Its map to the standing-record object is representable, and the fibre over an S -record $\mathbf{s}_S$ is the base-changed provenance groupoid (22).*

Proof. An S -point of the displayed two-sorted solution object is exactly a pair $(\theta_S, \mathbf{s}_S)$ satisfying the source/survivor equations and the seven realization equations. The action groupoid identifies precisely the arrows generated in (16); base change is componentwise. This gives the natural equivalence. Fixing $\mathbf{s}_S$ leaves exactly the source-solution fibre on which the subgroup acting trivially on the record operates, which is the provenance groupoid. The statement therefore classifies families and automorphisms over arbitrary test objects; it is stronger than naming a set-theoretic image. $\square$

Definition 4.8 (Executed packet-moduli object). The executed packet-moduli groupoid of (F, σ) is the two-sorted solution quotient

$$\mathcal{M}_{F,\sigma}^{\text{exec}} := \left[\text{Sol}_{Z_{F,\sigma} \times \mathcal{Y}_{F,\sigma}^{\text{NF}}}(\mathcal{R}_{F,\sigma}^{\text{NF}}) / G_{F,\sigma} \right]_{\mathbf{s}}, \quad (21)$$

where $G_{F,\sigma}$ acts on θ by (16) and on $\mathbf{s}$ by the induced packet covariance. The subscript $\mathbf{s}$ means that objects are the solved standing-normal-form records, while the full fibre over one $\mathbf{s}$ is retained as its provenance groupoid.

Equation (21) gives the moduli directly: its variables, equations, group action, stabilizers, typed complement, and provenance fibres are all explicit. The object may be discrete, continuous, functional, categorical, or mixed because those are the source-defined types of the variables in (18).

Definition 4.9 (Provenance fibre). For a solved standing record $\mathbf{s}$, define

$$\text{Prov}_{F,\sigma}(\mathbf{s}) = \left[\{ \theta \in Z_{F,\sigma} : \mathcal{R}_{F,\sigma}^{\text{NF}}(\theta, \mathbf{s}) \} / \ker(G_{F,\sigma} \curvearrowright \mathbf{s}) \right]. \quad (22)$$

It records all source presentations of that standing carrier class without refining the class.

4.7 Universal packet/core groupoid and framed provenance

The solved seven-channel record is a convenient coordinate presentation, but categorical gluing must be performed over concrete standing objects rather than over a set of isomorphism classes. Let

$$\mathcal{N}_5 := \mathcal{R}_5 \quad (23)$$

be the groupoid of concrete pointed reduced standing cores from Section 5. Its objects are the reachable future-separated pointed U_5^{stand} -output systems and its arrows are pointed, output-preserving,

continuation-equivariant isomorphisms. By Theorem 5.5, $\mathcal{N}_5$ is equivalent to the groupoid of concrete solved PCR-5A packet normal forms; we therefore use it as the universal PNF/core groupoid.

For each generated chart, packet covariance gives a functor

$$p_{F,\sigma} : \Gamma_{F,\sigma} = G_{F,\sigma} \ltimes Z_{F,\sigma} \longrightarrow \mathcal{N}_5, \quad \theta \longmapsto \text{Core}_5(\theta). \quad (24)$$

A source-groupoid arrow is sent to the induced pointed standing-core isomorphism. Because the packet/core biconditional is already frozen upstream, this functor neither adds nor removes a standing distinction.

Definition 4.10 (Framed provenance fibre). For a concrete object $n \in \mathcal{N}_5$, define

$$\widetilde{\text{Prov}}_{F,\sigma}(n) := \left[\left\{ (\theta, \phi) : \theta \in Z_{F,\sigma}, \phi : p_{F,\sigma}(\theta) \xrightarrow{\sim} n \right\} / G_{F,\sigma} \right]. \quad (25)$$

The automorphism group $\text{Aut}_{\mathcal{N}_5}(n)$ acts on the right by postcomposition,

$$(\theta, \phi) \cdot \alpha = (\theta, \alpha \circ \phi).$$

The unframed provenance fibre $\text{Prov}_{F,\sigma}(n)$ is the corresponding quotient by this framing action. The frame is auxiliary gluing data; it is not a new carrier identifier.

Lemma 4.11 (Well-typed fixed-core decomposition). *For any two generated charts $a = (F, \sigma)$ and $b = (G, \tau)$, their homotopy fibre product over $\mathcal{N}_5$ admits the fixed-core decomposition*

$$\Gamma_a \times_{\mathcal{N}_5}^h \Gamma_b \simeq \coprod_{[n] \in \pi_0(\mathcal{N}_5)} \left(\widetilde{\text{Prov}}_a(n) \times \widetilde{\text{Prov}}_b(n) \right) // \text{Aut}_{\mathcal{N}_5}(n). \quad (26)$$

Proof. An object of the homotopy fibre product is a triple (x, y, α) with $\alpha : p_a(x) \xrightarrow{\sim} p_b(y)$. Choose a concrete core n in the common isomorphism class and frames $\phi_a : p_a(x) \rightarrow n$, $\phi_b : p_b(y) \rightarrow n$. Replacing the choice of frame acts diagonally by $\text{Aut}(n)$, and every diagonal orbit reconstructs one comparison arrow $\alpha = \phi_b^{-1} \phi_a$. This correspondence is functorial on arrows and independent of the representative of $[n]$ up to canonical equivalence. Thus the displayed fixed-core decomposition is well typed. $\square$

4.8 Exact overlap status for two strata

For two family/skeleton charts (F, σ) and (G, τ) , introduce solved records $\mathbf{s}_F$, $\mathbf{s}_G$, and a comparison arrow

$$\alpha : p_{F,\sigma}(\theta_F) \xrightarrow{\sim} p_{G,\tau}(\theta_G)$$

in the universal PNF/core groupoid. In record coordinates this is exactly an isomorphism of all seven frozen packet channels; when both charts use the same canonical record representative it reduces to literal seven-channel equality. The comparison system is

$$\mathcal{E}_{F\sigma, G\tau}^{\cong} := \mathcal{E}_{F,\sigma}^{\text{surv}} \cup \mathcal{E}_{G,\tau}^{\text{surv}} \cup \mathcal{R}_{F,\sigma}^{\text{NF}} \cup \mathcal{R}_{G,\tau}^{\text{NF}} \cup \{ \alpha : p_{F,\sigma}(\theta_F) \xrightarrow{\sim} p_{G,\tau}(\theta_G) \}. \quad (27)$$

The exact overlap is the homotopy fibre product

$$\mathfrak{D}_{F\sigma, G\tau} := \Gamma_{F,\sigma} \times_{\mathcal{N}_5}^h \Gamma_{G,\tau}. \quad (28)$$

Equivalently, it is the quotient groupoid of triples $(\theta_F, \theta_G, \alpha)$ satisfying (27), with the natural $G_{F,\sigma} \times G_{G,\tau}$ action. By Theorem 4.11, it also has the framed fixed-core presentation (26). No automorphism group

acts on an unframed isomorphism class.

Fix the channel order $T < R < D < C < M < B < F$. On the product of the two raw parameter objects, define the disjoint status regions:

1. **Overlap**: both points satisfy E00–E23 and their concrete PNF/core objects are isomorphic;
2. **Separated_j**: both survive, channels before j agree up to the already fixed channel isomorphisms, and channel j differs;
3. **Defect** $((N_F; F_F, U_F), (N_G; F_G, U_G))$: no update occurs, every N/A entry is certified, and one or both exact fail/undefined pairs have nonempty union;
4. **SourceUpdate** and **TargetUpdate**: a source or target typing change occurs, with target update taking priority over source update.

This gives every skeleton pair a total overlap/separation/applicability/fail-undefined/update classification. The aggregate family overlap can be parameterized even though an individual skeleton-pair comparison groupoid is empty.

4.9 Exact atlas criterion

Theorem 4.12 (Executed atlas criterion). *A family atlas is executed exactly when:*

1. *its skeleton codes are generated by (4), not inferred from survivors;*
2. *every code has the explicit solution system (11) and total status decomposition (15);*
3. *its groupoid is the constructive action groupoid (16) with stabilizers (17);*
4. *its packet moduli are the solved normal-form object (21), not an unspecified image;*
5. *every source provenance fibre is given by (22), while every overlap computation uses the framed fibre (25) over the concrete universal core groupoid;*
6. *every same- and cross-family skeleton pair is dispositioned by the well-typed homotopy fibre product (28) and the total overlap status partition; and*
7. *every deletion datum used to refine standing identity is exactly the frozen field $\Delta_{\text{fr}}^{\text{del}}$ or is proved to factor through it by Theorem 2.3; auxiliary source-subset sufficiency never creates an eighth identity channel.*

Under B00 and the frozen PCR-3/PCR-4/PCR-5A theorems, these conditions are sufficient for the family atlas to classify exactly its contribution to $Q_{\text{osc}}^{\text{SCIC}}$.

Proof. The grammar theorem gives independent source coverage. E00–E23 and the family equations give survivor soundness, survivor completeness, and a complete disposition of the complement. The constructive groupoid removes exactly source-authorized presentation redundancy and records isotropy. The solved seven-channel record is equivalent to the PCR-5A packet, with its deletion entry fixed to $\Delta_{\text{fr}}^{\text{del}}$, so its objects are exactly SCIC identities and its fibres are provenance. The subset-deletion conservativity theorem prevents an auxiliary source-sufficiency diagnostic from over-resolving those fibres. Pairwise and higher gluing is performed over the concrete groupoid $\mathcal{N}_5$; framed provenance supplies the literal automorphism action needed for fixed-core descriptions. The overlap status partition therefore

identifies all and only packet/core-isomorphic records. Hence no source point, survivor point, packet class, applicability/defect face, or duplicate presentation remains unclassified. $\square$

4.10 Supplementary consistency boundary

Supplementary computational representations describe the source grammar, family equations, group actions, pairwise status systems, higher-overlap objects, and the B15 target-extension interface. They can reveal omissions and type mismatches, but their completeness follows only from the source reconstruction theorem, the E00–E23 system, packet completeness, and the proofs below.

5 Canonical standing-behavior core

5.1 Why a behavioral normal form is needed

PCR-5A already proves that the seven-channel packet is complete for SCIC identity. PCR-5B nevertheless requires a uniform normal form that can be attached to every exact source chart, compared across families, and used without assuming that the remaining parameter objects are finite-dimensional or algorithmically enumerable. The normal form below minimizes the *already frozen* standing action. It adds no witness, continuation, deletion, or selector. Its isomorphism type is merely a canonical presentation of the complete PCR-5A continuation signature.

5.2 The frozen total standing action

Let U_5^{stand} be the licensed same-target continuation monoid fixed by PCR-5 and made total and fail closed by PCR-5A. Its action is taken on the standing-faithful normal-form object

$$\mathcal{X}_5^{\text{stand}} = \mathfrak{G}_5^{\text{nf,stand}} \sqcup \mathcal{X}_{5,\perp},$$

where $\mathcal{X}_{5,\perp}$ is the inherited typed failure/undefinedness/update object. The action does not turn a target or source update into a same-target survivor; it only assigns the already prescribed fail-closed value.

Let W_5^{stand} be the target-complete standing witness family. It contains the target-face, role, deletion, continuation, matching, boundary, and same-target failure uses generated upstream. In particular, the PCR-5A kindwise no-repair deletion family is output bearing: at every continued state it records preservation, destruction, undefinedness, or update without retuning any nondeleted datum. The full source-subset sufficiency table of Theorem 2.2 is not freely adjoined to this witness family; its identity-bearing content is governed by Theorem 2.3.

Define the total output object and output map

$$\mathcal{Y}_5^{\text{stand}} = \prod_{w \in W_5^{\text{stand}}} Y_w, \quad \mathbf{O}_5(z) = (w(z))_{w \in W_5^{\text{stand}}}.$$

All products are formed inside the Grothendieck universe fixed by PCR-3. PCR-5A's generator/action factorization identifies the seven packet channels as a standing-faithful finite typed recoding of the family of values $\mathbf{O}_5(u \cdot z)$ for $u \in U_5^{\text{stand}}$.

5.3 Complete standing behavior

Definition 5.1 (Complete standing behavior). For $x \in A_{\text{osc}}$, define

$$\text{Beh}_x : U_5^{\text{stand}} \longrightarrow \mathcal{B}_5^{\text{stand}}, \quad \text{Beh}_x(u) = \mathbf{O}_5(u \cdot x). \quad (29)$$

Thus every survivor is compared on the same continuation domain. Ill-typed, failed, and update-producing continuations have the total values already fixed by the fail-closed action.

Lemma 5.2 (Behavior–packet equivalence). For $x, y \in A_{\text{osc}}$,

$$\text{Beh}_x = \text{Beh}_y \iff \Xi_{\text{osc}}(x) = \Xi_{\text{osc}}(y).$$

Proof. Equality of behaviors is equality of every target-complete standing witness after every licensed continuation. PCR-5A proves that this full continuation signature factors through the seven-channel packet and that the packet reconstructs it. Hence equality of the behavior functions and equality of the complete packets are equivalent. The frozen kindwise deletion coordinates require no extra closure argument: they are members of W_5^{stand} , evaluated after each $u \in U_5^{\text{stand}}$, with the no-repair semantics already fixed upstream. Auxiliary source-subset diagnostics are excluded from this equivalence unless they factor through the frozen packet. $\square$

5.4 Future equivalence and the reduced core

For $u, v \in U_5^{\text{stand}}$, define

$$u \equiv_x v \iff \mathbf{O}_5(au \cdot x) = \mathbf{O}_5(av \cdot x) \text{ for every } a \in U_5^{\text{stand}}. \quad (30)$$

This is a left congruence: if $u \equiv_x v$, then $bu \equiv_x bv$ for every $b \in U_5^{\text{stand}}$.

Definition 5.3 (Pointed reduced standing core). The pointed reduced standing core of x is

$$\text{Core}_5(x) = (S_x, [1]_x, \alpha_x, \omega_x), \quad (31)$$

where

$$\begin{aligned} S_x &= U_5^{\text{stand}} / \equiv_x, \\ \alpha_x(a, [u]_x) &= [au]_x, \\ \omega_x([u]_x) &= \mathbf{O}_5(u \cdot x). \end{aligned}$$

Morphisms are pointed, output-preserving, U_5^{stand} -equivariant isomorphisms.

The state object need not be finite. Finiteness in PCR-3 concerns the occupancy-bearing source-incidence presentation, not the orbit of all licensed continuations. The quotient remains $\mathbb{U}$ -small.

Theorem 5.4 (Reduced standing-core minimization). For every $x \in A_{\text{osc}}$, $\text{Core}_5(x)$ is reachable, future-separated, and minimal among pointed U_5^{stand} -output systems realizing Beh_x . It is unique up to unique pointed output-preserving equivariant isomorphism.

Proof. Every state $[u]_x$ is reached from $[1]_x$ by the action of u , so the core is reachable. If $[u]_x \neq [v]_x$, definition (30) supplies an a for which the future outputs differ; hence the core is future-separated.

Let $M = (S, s_0, \alpha, \omega)$ be any pointed output system realizing Beh_x . On its reachable part set

$$\alpha(u, s_0) \longmapsto [u]_x.$$

If $\alpha(u, s_0) = \alpha(v, s_0)$, then every future action has the same output from those states, so $u \equiv_x v$. The map is therefore well defined, surjective, output preserving, and equivariant. Every realization thus maps onto $\text{Core}_5(x)$, proving minimality. Between two reachable future-separated minimal realizations, the induced maps in both directions are injective by future separation and inverse by reachability. Pointed equivariance determines the map on every reachable state, giving uniqueness. $\square$

Theorem 5.5 (Core–packet–SCIC biconditional). *For $x, y \in A_{\text{osc}}$, the following are equivalent:*

1. $\text{Core}_5(x) \cong \text{Core}_5(y)$;
2. $\text{Beh}_x = \text{Beh}_y$;
3. $\Xi_{\text{osc}}(x) = \Xi_{\text{osc}}(y)$;
4. $x \sim_{\text{SCIC}} y$.

Proof. A pointed output-preserving equivariant isomorphism identifies the output after every continuation, so (1) implies (2). Equal behaviors induce the same future-equivalence relation and therefore canonically isomorphic reduced cores, giving (2) implies (1). Lemma 5.2 gives (2) $\iff$ (3), and the frozen PCR-5A packet theorem gives (3) $\iff$ (4). $\square$

Corollary 5.6 (No second classifier). *The reduced core is a canonical normal form for the unique SCIC identity relation. Every invariant extracted from $\text{Core}_5(x)$ factors through $Q_{\text{osc}}^{\text{SCIC}}$; it cannot refine a packet fibre or merge unequal packets.*

5.5 Family use of the core

For a family survivor groupoid $\mathcal{C}_F^{\text{surv}}$, define

$$\beta_F : \mathcal{C}_F^{\text{surv}} \longrightarrow \mathcal{R}_5, \quad x \longmapsto \text{Core}_5(x),$$

where $\mathcal{R}_5$ is the groupoid of reachable future-separated pointed standing cores. The object fibres of β_F are exactly the SCIC presentation fibres inside the family: every presentation in one fibre has the same reduced-core isomorphism type, although its source-provenance groupoid may have several connected components. Cross-family overlap is equality of reduced-core isomorphism type.

The finite source skeleton developed in the next section and the reduced standing core play complementary roles:

- the source skeleton proves candidate and survivor coverage from the independently generated physical ontology;
- the reduced core proves exact terminal identity and supplies a canonical behavior normal form;
- the residual solution object retains all continuous, functional, representation-valued, or categorical parameters; and
- the provenance fibre records every source presentation of one core without promoting provenance to identity.

Remark 5.7 (Mathematical exactness versus algorithmic decidability). The reduced-core normal form determines equality as an isomorphism problem in the exact continuation/output category. PCR-5B does not claim a uniform finite algorithm for every functional or categorical parameter object admitted by PCR-3. “Decidable” in the atlas means mathematically fixed by the packet/core equivalence, not uniformly Turing-decidable over every symbolic source category.

6 Universal classification theorem and lawful descent

6.1 From source skeletons to solved standing cores

The preceding two sections determine independently:

1. the source grammar Σ_F^{gram} ;
2. the source parameter object $\Theta_{F,\sigma}^{\text{part}}$;
3. the named simultaneous equations E00–E23 and family source equations;
4. the complete applicability vector, theorem-certified N/A data, and fail/undefined/update partition;
5. the action groupoid $G_{F,\sigma} \ltimes Z_{F,\sigma}$; and
6. the solved seven-channel standing-core groupoid $\mathcal{M}_{F,\sigma}^{\text{exec}}$, whose deletion field is the frozen $\Delta_{\text{fr}}^{\text{del}}$.

The reduced standing core of Section 5 is now used only to prove that the solved seven-channel records have exactly SCIC fibres. It is not used to define the source grammar or to hide an uncomputed packet image.

6.2 B01: explicit family normal-form theorem

Theorem 6.1 (Explicit source-skeleton, solution, groupoid, and packet-normal-form theorem). *[B01] For each $F \in \mathfrak{F}_{\text{osc}}$, define*

$$\mathcal{A}_F^{\text{exec}} := \coprod_{\sigma \in \Sigma_F^{\text{gram}}} \mathcal{M}_{F,\sigma}^{\text{exec}}. \quad (32)$$

Then:

1. *every PCR-3 activity-reduced source presentation has a generated finite code σ , and every PCR-4 survivor is routed by Theorem 3.6 into at least one of the six principal charts;*
2. *the survivor objects are exactly the solutions of E00–E23 plus the family source equations;*
3. *every raw point has one complete applicability-aware status vector in (15), with certified N/A retained separately from active failure and undefinedness;*
4. *every lawful presentation morphism is generated by the constructive action groupoid, and every generator is source- and packet-covariant;*
5. *the object space of $\mathcal{M}_{F,\sigma}^{\text{exec}}$ is the explicit solved seven-channel normal-form space, with provenance fibre (22); and*
6. *two objects of $\mathcal{A}_F^{\text{exec}}$ are the same standing carrier exactly when their pointed reduced standing cores are isomorphic.*

Consequently $\mathcal{A}_F^{\text{exec}}$ is an executed exact atlas for the family contribution to $Q_{\text{osc}}^{\text{SCIC}}$, not merely a presentation of an unevaluated image.

Proof. Grammar coverage and source soundness are Theorem 3.5; post-PCR-4 six-family survivor coverage is Theorems 3.6 and 3.7. The explicit residual system is the PCR-4 global family map written as E00–E23, including all intrinsic equations, nonempty moduli requirements, cross-signature residuals, and the common cone. The evaluation-conservativity theorem Theorem 4.4 therefore identifies these guarded solutions exactly with the inherited PCR-4 survivor loci, while Theorem 4.2 preserves PCR-4 applicability and its theorem-certified N/A branch before any failure value is read.

The groupoid generators are the source-authorized graph, basis, gauge, chart, projection, constituent, profile, and gluing transformations supplied by the tags of σ . Their relations are source naturality and fixed-boundary covariance, so every generated arrow is lawful. Conversely PCR-3 source-equational identity and lawful parameter reparameterization factor through those generators; no unnamed class of transformations is needed.

The equations (19) copy the seven PCR-5A packet channels into explicit standing variables, with the deletion component fixed to $\Delta_{\text{fr}}^{\text{del}}$. By Theorem 2.3, any larger source-subset sufficiency profile is auxiliary unless its identity-bearing content already factors through this frozen packet. By the packet-completeness theorem and Theorem 5.5, equality of solved standing records, isomorphism of reduced cores, and SCIC equality are equivalent. The residual θ -fibre at fixed $\mathbf{s}$ is exactly the source-provenance groupoid. Thus the atlas has complete source coverage, explicit routing of every survivor into the six principal charts, complete applicability, survivor, and failure disposition, lawful quotienting, and determined packet moduli. $\square$

Corollary 6.2 (Computed survivor-code index). *A code $\sigma \in \Sigma_F^{\text{gram}}$ is a survivor code exactly when its displayed system $\mathcal{E}_{F,\sigma}^{\text{surv}}$ has a solution. Empty codes remain classified by their applicability-aware status strata, including theorem-certified N/A instances. No skeleton is added or removed by inspection of a preferred model list.*

Corollary 6.3 (No finite-scan assumption). *The code set can be countable or categorical and a solution object can be continuous, functional, representation-valued, or otherwise symbolic. Exactness is supplied by the generated grammar and named equations, not by enumerating parameter values.*

6.3 Four parameter dispositions

Let $\gamma : I \rightarrow \Theta_{F,\sigma}^{\text{raw}}$ be a typed variation. The complete status and standing-normal-form maps give a disjoint primary disposition:

1. *target update*: at least one target anchor, record semantics, operator category, boundary, or comparison regime changes;
2. *source update*: the source ontology or constitutive object changes while the target is fixed;
3. *same-source applicability/defect change*: the source skeleton remains coherent but its fixed applicability set, certified N/A record, or active fail/undefined sets change under an allowed source-point variation; an active failure may not be retyped N/A at the same point;
4. *surviving standing variation*: both points survive and their solved seven-channel records differ;
5. *surviving provenance variation*: both points survive and their solved seven-channel records agree.

The first applicable branch is the primary status. The final two branches are distinguished by exact equality, not by a preference or numerical threshold.

6.4 B09: lawful parameter descent

Theorem 6.4 (Constructive lawful parameter descent). *[B09] For every family and generated code:*

1. *the action of every generator of $G_{F,\sigma}$ preserves E00–E23 and carries the seven-channel standing record covariantly;*
2. *the action therefore descends to $\mathcal{M}_{F,\sigma}^{\text{exec}}$;*
3. *its isotropy is the stabilizer (17);*
4. *a surviving parameter variation defines a new standing class iff at least one solved channel T, R, D, C, M, B, F changes after quotienting the constructive action; and*
5. *a source/provenance variation with all seven channels fixed lies entirely in the provenance fibre and cannot refine SCIC identity.*

No hidden source-history classifier remains beneath the solved packet-normal-form quotient.

Proof. Graph isomorphisms preserve incidence and source sorts. Source basis and gauge transformations act simultaneously on states, effects, couplings, mass/kinetic objects, attachments, anomaly tensors, and matching maps, so E01–E23 are covariant. Chart changes and family-specific isometries preserve the source equations by naturality. The fixed-target restriction excludes changes of threshold, memory, boundary ownership, matching orientation, the frozen deletion value, and failure status unless the complete packet proves them inert. An auxiliary source-subset deletion diagnostic is not a further quotient coordinate: by Theorem 2.3, it either factors through the frozen packet, is standing-inert provenance, or is a target/source/continuation update. Thus the action is well defined on solutions and on solved standing records.

The stabilizer formula is the definition of isotropy in an action groupoid. For two surviving points, the seven-channel records either agree or disagree. Agreement gives one SCIC class by the PCR-5A biconditional and leaves only provenance; disagreement supplies a standing separator and cannot be quotiented without violating packet soundness. These alternatives exhaust the fixed-source survivor branch. $\square$

6.5 Finite separators and complement localization

Proposition 6.5 (Finite packet separator). *Every pair of distinct SCIC classes admits a separator consisting of one first-different channel in the order $T < R < D < C < M < B < F$, together with a finite typed witness/continuation subrecord inside that channel.*

Proof. The seven-channel records are unequal. Choose their first unequal channel. PCR-5A generator factorization expresses every channel as a finite typed recoding of primitive standing witnesses and licensed continuations. At least one primitive or finitely generated subrecord differs. That record is a finite separator; minimality follows by deleting redundant entries. $\square$

Theorem 6.6 (Complete applicability, failure, and undefined localization). *For every generated code and every raw parameter point, exactly one of the applicability-aware status strata in (15) applies. The no-update record contains the theorem-certified N/A set together with the complete failed-equation and*

undefined-equation sets. N/A is permitted only where the pre-evaluation applicability predicate is false and its upstream absence certificate is present; active failure and active undefinedness cannot be laundered into N/A . Source/target updates remain separately typed, and each defect or update carries its upstream reopening locus.

Proof. Each instantiated relation has the applicability-aware six-valued evaluation status of Section 4. Applicability is fixed first. After target- and source-update branches are removed, every relation is either theorem-certified N/A , active Pass, active Fail, or active Undefined. The exact triple (N, F, U) , together with the N/A certificates, therefore determines a survivor stratum when $F = U = \emptyset$ or one exact defect stratum otherwise. Theorem 4.4 identifies this survivor branch exactly with PCR-4, and Theorem 4.2 forbids a failed active relation from moving into N . No point is left without a scientifically typed or fail-closed evaluation status. $\square$

6.6 Consequence for the family theorems

B01 and B09 discharge four obligations that a generic quotient presentation cannot supply: the skeleton domain is generated explicitly; the residual equations are written out; the quotient arrows have generators, relations, actions, and stabilizers; and the packet moduli are represented by solved seven-channel records with explicit provenance fibres. The family sections then specialize the source equations and standing domains without repeating the universal descent argument.

7 The LAW carrier atlas

7.1 Grammar

A LAW skeleton has at least one active **Law** node and one **ActLaw** incidence on the frozen substrate. Its pure branch contains no active **Local**, **Hidden**, **Derived**, or **Soliton** node. Mixed skeletons remain visible in the LAW chart because $LAW \in \mathcal{K}_\sigma$, and they also occur in HYB.

A canonical LAW skeleton has the form

$$\sigma_{LAW} = (n_E, n_L, \mathbf{a}_E, \mathbf{a}_L, I_{\text{ActLaw}}, I_\partial, I_T, \ell, \mathbf{m}, \mathbf{o}, \mathbf{q}, \mathbf{v}, \mathbf{u}), \quad (33)$$

where $n_E, n_L < \infty$, $n_L \geq 1$; the incidence arrays record the source process graph and attachments; ℓ is the support/locality sort; $\mathbf{m}$ is the source memory sort; $\mathbf{o}$ is the process-order or filtration sort; $\mathbf{q}$ is the fixed-boundary gauge-covariance sort; $\mathbf{v}$ is the source vacuum/deformation sort; and $\mathbf{u}$ is the continuation/matching sort. Continuous kernels, coefficients, thresholds, and vacuum values are parameters, not skeleton labels.

The LAW-specific source equations appended to E00 are:

ID	LAW source equation
L1	Every Law node satisfies the axioms of its declared process category and every ActLaw edge is a morphism in that category.
L2	The active process is attached to at least one target-bearing substrate path; a disconnected law object is stripped as spectator.
L3	In the pure LAW branch, no active new primitive local, hidden, derived, or solitonic node is present.

ID	LAW source equation
L4	The process is gauge covariant on the fixed charged-current boundary; because no new chiral node is present, the anomaly contribution is the frozen anomaly-free substrate value.
L5	Any retained memory kernel, nonlocal support, order filtration, or threshold is typed in the continuation or matching source object before target evaluation.

7.2 Residual solution specification

For $\theta \in \Theta_{\text{LAW},\sigma}^{\text{part}}$, write L_θ for the process object and $\mathcal{I}^\theta$ for its induced event-complete instrument. The exact residual object is

$$\begin{aligned}
Z_{\text{LAW},\sigma} = \{ \theta : L1:L5, J_{\lambda,o}^\theta \succeq 0, \sum_o (\mathcal{I}_{\lambda,o}^\theta)^\dagger(1) = 1, \\
R(K^\theta) = R(T_{\text{osc}}) \forall R \in \text{Rec}_T, \mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_\theta), \\
\exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^\theta(\lambda) > 0, A_\theta^{s,d} \rho_\theta(g) = \rho_{CC}(g) A_\theta^{s,d}, \\
\mathcal{D}_{\text{caus}}(\theta) \neq \emptyset, \mathcal{V}_{\text{LAW}}(\theta) \neq \emptyset, \mathcal{M}_{\text{LAW}}(\theta) \neq \emptyset, \\
\mathcal{X}_{\text{LAW}}(\theta) \text{ is inert or coherent, } R_1(\theta) = \dots = R_9(\theta) = 0, \text{Cone}_{\text{LAW}}(\theta) \neq \emptyset \}.
\end{aligned} \tag{34}$$

This is PCR-4 equation (51) expanded into the named variables of E00–E23. The status decomposition has explicit faces $L1:L5$, E01–E23, source update, and target update. In particular, a non-completely-positive process, a record mismatch, a one-mode response, a boundary-changing law, an empty vacuum/matching object, a nonzero cross residual, or an empty cone is a named stratum of the complement rather than an omitted case.

7.3 Constructive LAW groupoid

The group object is

$$G_{\text{LAW},\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes (G_{\text{sub}} \times G_{\text{proc}} \times G_{\text{chart}})_{\text{fix } \partial, T}, \tag{35}$$

where G_{sub} is the simultaneous fixed-substrate basis/gauge group, G_{proc} is the automorphism group object of the declared process category, and G_{chart} is the source parameter-chart groupoid. A generator acts by

$$L_\theta \mapsto U L_\theta U^{-1}, \quad \rho_\alpha \mapsto U \rho_\alpha U^\dagger, \quad P_\beta \mapsto U P_\beta U^\dagger,$$

with the induced action on kernels, attachments, matching maps, and vacuum data. Process-category generators act by natural isomorphism. Relations are graph incidence, naturality, fixed-boundary covariance, and semidirect compatibility. The isotropy group is the stabilizer of the complete tuple $(L_\theta, \mathcal{I}^\theta, A_\theta, \mathcal{V}_\theta, \mathcal{M}_\theta)$.

A rescaling that changes a physical threshold, a memory kernel outside source recharting, an operator-order filtration retained by matching, or a failure boundary is not in $G_{\text{LAW},\sigma}$.

7.4 Solved LAW packet normal form

The LAW standing record is

$$\mathbf{s}_{\text{LAW}} = (t_{\text{osc}}, r_{\text{law}}, \delta_{\text{law}}, c[L_\theta], m[L_\theta : \text{source} \rightarrow \text{target}], b[L_\theta], f[L_\theta]). \tag{36}$$

Here t_{osc} is the fixed target-face record; r_{law} records role ownership/common-cone incidence; $\delta_{\text{law}} := \Delta_{\text{fr}}^{\text{del}}|_{\text{LAW}}$ is the LAW restriction of the frozen PCR-5A kindwise no-repair deletion channel; c is the licensed continuation object, including memory when standing-visible; m is the oriented matching record, including retained order/threshold; b is boundary ownership/transport; and f is the complete same-target fail/undefined frontier.

The corresponding moduli object is therefore the explicit solution quotient

$$\mathcal{M}_{\text{LAW},\sigma}^{\text{exec}} = [\{(\theta, \mathbf{s}_{\text{LAW}}) : \theta \in Z_{\text{LAW},\sigma}, \mathbf{s}_{\text{LAW}} \text{ satisfies (36)}\} / G_{\text{LAW},\sigma}]_{\mathbf{s}_{\text{LAW}}} . \quad (37)$$

The variables, equations, lawful action, stabilizer, and provenance fibre therefore determine the LAW moduli directly.

7.5 B02: LAW classification

Theorem 7.1 (Executed LAW stratum theorem). *[B02] The exact LAW atlas is*

$$\mathcal{A}_{\text{LAW}}^{\text{exec}} = \coprod_{\sigma \in \Sigma_{\text{LAW}}^{\text{gram}}} \mathcal{M}_{\text{LAW},\sigma}^{\text{exec}}, \quad (38)$$

with empty codes retained through their complete status strata. Every LAW survivor occurs in exactly one generated code orbit; every surviving parameter is represented by one solved record (36); two records give one SCIC class exactly when all seven channels agree; and every excluded point has a named LAW/E00–E23 failure or update status.

Proof. The grammar theorem generates every finite LAW source trace before survivor inspection. Equation (34) is the complete LAW specialization of PCR-4’s exact family map. The action (35) is the source-equational and fixed-target redescription groupoid, and B09 proves exact descent. The solved record (36) is the seven-channel PCR-5A packet in family variables. Packet equality is SCIC equality. Hence the coproduct covers every LAW presentation, types its complement, removes exactly lawful presentation redundancy, and retains every standing modulus. $\square$

7.6 Distinguished class and fail/undefined frontiers

The Weinberg-type calibration q_W is one point of (37). Its law/new-local deletion pair is $(0, 1)$. It does not exhaust the LAW atlas. Two LAW solutions with the same effective coefficient remain distinct when their retained memory, continuation, source order, threshold, matching orientation, boundary ownership, frozen deletion value, or same-target fail/undefined frontier differs.

Let

$$\mathcal{I}_{\text{LAW}} = \{L1, \dots, L5\} \cup \{E01, \dots, E23\}.$$

The LAW complement is stratified by the certified inapplicability set $N \subseteq \mathcal{I}_{\text{LAW}}$, the exact active fail/undefined pair (F, U) , and the source- and target-update branches. A survivor may carry certified N/A entries but has $F = U = \emptyset$; every same-target nonsurvivor retains its complete $(N; F, U)$ data rather than a single first-failure label.

8 The LOCAL carrier atlas

8.1 Grammar and active-rank strata

A LOCAL skeleton contains at least one active Local node and at least one active coupling or target attachment from that node to the fixed lepton/Higgs and propagation-role substrate. Its canonical skeleton is

$$\sigma_{\text{LOCAL}} = (n_L, G_{\text{loc}}, \boldsymbol{\rho}, \boldsymbol{\chi}, \boldsymbol{\nu}, I_{\text{cpl}}, I_{\partial}, I_T, \mathbf{a}, \mathbf{q}, \mathbf{v}, \mathbf{m}, \mathbf{u}), \quad (39)$$

where G_{loc} is the finite active local-node graph; $\boldsymbol{\rho}$ and $\boldsymbol{\chi}$ are the representation and chirality/reality sorts; $\boldsymbol{\nu}$ is the multiplicity partition before quotienting repeated-node permutations; and the remaining symbols type attachments, anomaly/global-form data, vacuum, matching, and continuation. The actual matrices and threshold values remain residual variables.

The LOCAL source equations are:

ID	LOCAL source equation
N1	At least one Local node lies on an active source-to-target path after exact activity reduction.
N2	Every active local attachment operator is a well-typed gauge-invariant source morphism linking the fixed lepton/Higgs boundary to the local node graph.
N3	Representation composition, kinetic/mass maps, and coupling incidences are source coherent.
N4	Exact source activity is determined by the support of the instantiated coupling/attachment morphisms; no nonzero coupling is removed by a magnitude cutoff.
N5	The active representation and charge object used by the channel is the one used by the anomaly, vacuum, and matching equations.

For matrix-type attachments define the visible active rank by

$$r_{\text{act}}(\theta) = \text{rank } A_{\text{loc}}^{\text{vis}}(\theta).$$

Its exact rank- r locus is the determinantal stratum on which all $(r+1)$ -minors vanish and at least one r -minor is nonzero. For categorical source objects, rank is replaced by the source-defined essential-image object. This is part of the generated skeleton and status calculus, not a field-count selector.

8.2 Residual solution specification

Let $\mathcal{A}_{\text{loc}}(\theta)$ be the complete active local anomaly vector and $\mathcal{A}_{\text{loc}}^{\text{att}}(\theta)$ the moduli object of gauge-invariant attachment operators. The exact local residual object is

$$\begin{aligned} Z_{\text{LOCAL},\sigma} = \{ \theta : N1:N5, \mathcal{A}_{\text{loc}}^{\text{att}}(\theta) \neq \emptyset, \mathcal{A}_{\text{loc}}(\theta) = 0, \\ J_{\lambda,o}(\theta) \succeq 0, \sum_o \mathcal{I}_{\lambda,o}(\theta)^\dagger(1) = 1, R(K^\theta) = R(T_{\text{osc}}) \forall R, \\ \mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_\theta), \exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^\theta(\lambda) > 0, \\ A_\theta^{s,d} \rho_\theta(g) = \rho_{CC}(g) A_\theta^{s,d}, \mathcal{D}_{\text{caus}}^{\text{LOCAL}}(\theta) \neq \emptyset, \\ \mathcal{V}_{\text{LOCAL}}(\theta) \neq \emptyset, \mathcal{M}_{\text{LOCAL}}(\theta) \neq \emptyset, \mathcal{X}_{\text{LOCAL}}(\theta) \text{ inert or coherent}, \\ R_1(\theta) = \dots = R_9(\theta) = 0, \text{Cone}_{\text{LOCAL}}(\theta) \neq \emptyset \}. \end{aligned} \quad (40)$$

This is PCR-4 equation (52) with every intrinsic and relational term exposed. The complete anomaly vector contains the perturbative $SU(3)^3$, $SU(3)^2U(1)$, $SU(2)^2U(1)$, $U(1)^3$, mixed gravitational, and activated global obstruction components, plus only those inflow terms physically attached in the same witness diagram.

8.3 Constructive LOCAL groupoid

Let V_i be the source representation object of local node i . The group object is

$$G_{\text{LOCAL},\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes \left(\prod_i \text{Aut}_{G_{\text{fixed}}}(V_i) \times \prod_{[i]} S_{\nu_i} \times G_{\text{gauge}}^{\partial} \times G_{\text{chart}} \right). \quad (41)$$

The factor S_{ν_i} permutes repeated isomorphic active nodes. A basis generator U_i acts on a local mass/kinetic object and attachment matrix by the corresponding congruence or similarity action, on Yukawa/coupling tensors contragrediently, and on anomaly tensors through the same representation isomorphism. Gauge transformations fix the charged-current boundary as an operational object. Graph permutations conjugate the component actions, giving the semidirect relations.

The isotropy group is the simultaneous stabilizer of the active graph, representation/charge object, all coupling and mass maps, the vacuum branch, the matching diagram, and the solved standing record. A physical threshold rescaling is not a basis generator merely because the low-energy effective coefficient remains fixed.

8.4 Solved LOCAL packet normal form

The standing record is

$$\mathbf{s}_{\text{LOCAL}} = (t_{\text{osc}}, r[G_{\text{loc}}, r_{\text{act}}, \text{ownership}], \delta_{\text{loc}}, c[G_{\text{loc}}], m[\text{Spec}_{\text{phys}}, \text{threshold}, \text{orientation}], b[A^{s,d}], f_{\text{loc}}). \quad (42)$$

The representation label and microscopic multiplicity are not separate standing coordinates unless they alter one of these solved channels. The physical threshold/spectral object is retained only in the basis-covariant form authorized by the matching channel.

Define

$$\mathcal{M}_{\text{LOCAL},\sigma}^{\text{exec}} = [\{(\theta, \mathbf{s}_{\text{LOCAL}}) : \theta \in Z_{\text{LOCAL},\sigma}, \mathbf{s}_{\text{LOCAL}} \text{ satisfies (42)}\} / G_{\text{LOCAL},\sigma}]_{\mathbf{s}_{\text{LOCAL}}}. \quad (43)$$

Its provenance fibre at fixed $\mathbf{s}_{\text{LOCAL}}$ contains all source-basis, graph, and chart presentations of the same standing record, including their stabilizers.

8.5 B03: LOCAL classification

Theorem 8.1 (Executed LOCAL active-rank and threshold theorem). *[B03] The exact LOCAL atlas is*

$$\mathcal{A}_{\text{LOCAL}}^{\text{exec}} = \coprod_{\sigma \in \Sigma_{\text{LOCAL}}^{\text{gram}}} \mathcal{M}_{\text{LOCAL},\sigma}^{\text{exec}}. \quad (44)$$

The generated code fixes the finite active graph and source sorts; determinantal or essential-image strata fix active rank; (40) fixes the exact simultaneous survivor object; (41) gives generators, relations, action, and isotropy; and (42) gives the evaluated packet moduli. Source-null padding creates no code, while every nonnull active incidence remains represented until the packet quotient determines its standing effect.

Proof. PCR-3 finite-source reconstruction gives the skeleton and N1–N5. PCR-4 equation (52) and E01–E23 give (40). The group action is exactly the source-equational automorphism action on the active graph and its tensor data. B09 proves descent and SCIC finality. Therefore every local point is either a solved standing record carrying its exact theorem-certified N/A set, a complete named active fail/undefined defect record carrying the same applicability data, a source update, or a target update, and no standing-visible threshold, matching, deletion, boundary, continuation, applicability, or failure value is quotiented away. $\square$

8.6 Calibrated active-singlet locus

The PCR-5/PCR-5A point q_{3N} is one solved record with law/new-local deletion pair $(1, 0)$. The continuous threshold-scaling family used in B13 lies in a full-rank local-singlet skeleton. It preserves the effective matrix but changes the physical threshold and matching record, and therefore moves through distinct objects of (43).

After the pre-evaluation applicability mask is fixed, the LOCAL complement is exhaustively partitioned by theorem-certified N/A records together with nonempty active fail/undefined defect pairs of the LOCAL clauses N1–N5 and the universal clauses E01–E23, plus source-update and target-update strata. An anomaly-free label without a common representation witness is not a pass; a channel built at a different parameter point from the anomaly or vacuum object fails the relevant residual equation.

9 The hidden-sector carrier atlas

9.1 Grammar and minimal active hidden support

A HID skeleton contains at least one active Hidden node and one active Project, dilation, or hidden co-carriage incidence whose visible reduction enters the fixed oscillation target. Its canonical skeleton is

$$\sigma_{\text{HID}} = (n_A, n_H, G_{\text{hid}}, I_{\text{proj}}, I_{\text{cpl}}, I_T, \mathbf{p}, \mathbf{h}, \mathbf{q}, \mathbf{v}, \mathbf{m}, \mathbf{u}), \quad (45)$$

where $n_A, n_H < \infty$; G_{hid} records active visible, hidden, and projection incidence; $\mathbf{p}$ is the projection/dilation sort; $\mathbf{h}$ is hidden support, ownership, and memory type; and the remaining symbols type charge/anomaly, vacuum, matching, and continuation.

The HID source equations are:

ID	HID source equation
H1	The visible/hidden embedding, projection, reduction, or dilation maps satisfy the declared source identities; in a Stinespring chart, $V^\dagger V = 1$ on the active input support.
H2	The visible instrument is the declared reduction of the same hidden process, for example $\mathcal{I}_{\lambda, o}(\rho) = \text{Tr}_{\text{hid}}[P_o V_\lambda \rho V_\lambda^\dagger]$ in a Stinespring presentation.
H3	The hidden support is the smallest source-constitutive subobject generated by the active target path under the process and its required adjoints/coherence maps; disconnected ancillary factors are stripped.
H4	Hidden representation, charge, anomaly, vacuum, ownership, memory, and matching data are taken from that minimal active support.

ID	HID source equation
H5	A reduced law with no carrier-detectable active dilation belongs to LAW rather than creating a distinct HID skeleton; an active hidden sector remains HID even if its visible channel agrees with a LAW point.

The minimal support condition is constructive: starting from the visible source/target attachment, iteratively close under the active dilation/process incidences and source-constitutive cells. Finiteness of the incidence skeleton makes this closure terminate. This gives a determined hidden core rather than an unspecified “minimal dilation.”

9.2 Residual solution specification

Let $\mathcal{H}_{\text{hid}}(\theta)$ be the exact moduli object of source-coherent active dilations satisfying H1–H5. The residual solution object is

$$\begin{aligned}
Z_{\text{HID},\sigma} = \{ & \theta : H1:H5, \mathcal{H}_{\text{hid}}(\theta) \neq \emptyset, \mathcal{A}_{\text{HID}}(\theta) = 0, \\
& J_{\lambda,o}(\theta) \succeq 0, \sum_o \mathcal{I}_{\lambda,o}(\theta)^\dagger(1) = 1, R(K^\theta) = R(T_{\text{osc}}) \forall R, \\
& \mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_\theta), \exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^\theta(\lambda) > 0, \\
& A_\theta^{s,d} \rho_\theta(g) = \rho_{CC}(g) A_\theta^{s,d}, \mathcal{D}_{\text{caus}}^{\text{HID}}(\theta) \neq \emptyset, \\
& \mathcal{V}_{\text{HID}}(\theta) \neq \emptyset, \mathcal{M}_{\text{HID}}(\theta) \neq \emptyset, \mathcal{X}_{\text{HID}}(\theta) \text{ inert or coherent}, \\
& R_1(\theta) = \dots = R_9(\theta) = 0, \text{Cone}_{\text{HID}}(\theta) \neq \emptyset \}.
\end{aligned} \tag{46}$$

This is PCR-4 equation (57) expanded. Invisible loss without active cross-boundary transfer fails E06. A visible channel assembled from one dilation and anomaly/vacuum/matching data from another fails H4 and the corresponding residual equations.

9.3 Constructive HID groupoid

The group object is

$$G_{\text{HID},\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes (G_{\text{vis}} \times G_{\text{hid}} \times G_{\text{Kraus}} \times G_{\text{chart}})_{\text{fix } \partial, T}. \tag{47}$$

Here G_{hid} is the source automorphism/isometry group of the minimal active hidden support; G_{Kraus} acts by source-authorized isometric mixing of equivalent Kraus/Stinespring presentations; and G_{vis} is the simultaneous fixed-boundary basis/gauge group. In a Stinespring chart,

$$V \mapsto (U_{\text{vis}} \otimes U_{\text{hid}}) V U_{\text{in}}^{-1}, \quad P_o \mapsto U_{\text{out}} P_o U_{\text{out}}^\dagger,$$

with induced action on hidden charges, memory, vacuum, and matching. Relations are isometry composition, graph incidence, source naturality, fixed-boundary covariance, and semidirect conjugation. The isotropy group is the stabilizer of the minimal active hidden support and the complete source/dilation tuple.

A nonisometric enlargement, a changed hidden ownership relation, a changed physical hidden threshold, or a changed memory/continuation packet is not a groupoid arrow unless all seven standing channels are invariant.

9.4 Solved HID packet normal form

The HID standing record is

$$\begin{aligned} \mathbf{s}_{\text{HID}} = (&t_{\text{osc}}, r[\text{visible/hidden ownership}], \delta_{\text{hid}}, \\ &c[\text{visible projection, memory}], m[\text{hidden support, threshold, orientation}], \\ &b[\text{visible boundary}], f_{\text{hid}}). \end{aligned} \quad (48)$$

The hidden support and ownership enter terminal identity only through the frozen packet uses. A source-distinct ancillary dilation with the same minimal active support and every channel fixed lies in the provenance fibre.

Define

$$\mathcal{M}_{\text{HID},\sigma}^{\text{exec}} = [\{(\theta, \mathbf{s}_{\text{HID}}) : \theta \in Z_{\text{HID},\sigma}, \mathbf{s}_{\text{HID}} \text{ satisfies (48)}\} / G_{\text{HID},\sigma}]_{\mathbf{s}_{\text{HID}}}. \quad (49)$$

9.5 B04: HID classification

Theorem 9.1 (Executed minimal-active-dilation and hidden-support theorem). *[B04] The exact HID atlas is*

$$\mathcal{A}_{\text{HID}}^{\text{exec}} = \coprod_{\sigma \in \Sigma_{\text{HID}}^{\text{gram}}} \mathcal{M}_{\text{HID},\sigma}^{\text{exec}}. \quad (50)$$

The grammar determines every finite active hidden/projection code; H1–H5 determine the minimal active hidden support; (46) determines the exact survivor and status loci; (47) gives the lawful groupoid; and (48) gives the evaluated packet moduli. Every hidden difference is therefore one of: stripped ancillary provenance, a solved standing modulus, a same-target failure, a source update, or a target update.

Proof. PCR-3 supplies every finite hidden/enlarged source trace and pointwise activity reduction. Closure from the active target path gives H3 and removes only spectator support. PCR-4 equation (57) gives the intrinsic and common-witness conditions displayed in (46). The source-authorized isometry/Kraus/chart action is constructive and preserves those equations. B09 gives descent, while packet completeness identifies the solved record with SCIC identity. Hence the atlas is complete at the inherited symbolic scope, including its hidden support and dilation fibres. $\square$

The active-singlet benchmark q_{3N} lies in both LOCAL and HID charts when the singlet sector is retained as an enlarged active state space. That fact supplies a nonempty LOCAL/HID overlap but does not identify all local and hidden solutions.

10 The LOCAL–HID overlap

10.1 Two source resolutions of one standing record

A LOCAL presentation resolves active primitive local nodes and their attachment tensors. A HID presentation resolves an enlarged state space, projection/dilation, hidden support, and visible reduction. The same source architecture can admit both views, as in an active gauge-singlet sector retained explicitly. Equality of the reduced visible channel alone is insufficient: the complete frozen standing packet must agree up to its lawful packet/core isomorphism.

Fix $\sigma \in \Sigma_{\text{LOCAL}}^{\text{gram}}$ and $\tau \in \Sigma_{\text{HID}}^{\text{gram}}$. Introduce points θ_L, θ_H , solved records $\mathbf{s}_L, \mathbf{s}_H$, and a comparison arrow

$$\alpha_{LH} : p_{\text{LOCAL},\sigma}(\theta_L) \xrightarrow{\sim} p_{\text{HID},\tau}(\theta_H)$$

in the universal PNF/core groupoid $\mathcal{N}_5$. The exact LOCAL/HID comparison system consists of:

1. the full LOCAL system N1–N5 and E01–E23 at θ_L ;
2. the full HID system H1–H5 and E01–E23 at θ_H ;
3. the solved frozen packet equations on both sides, with the deletion field fixed to $\Delta_{\text{fr}}^{\text{del}}$; and
4. the standing-core isomorphism α_{LH} , equivalently isomorphism of all seven packet channels.

A typed source-resolution relation Λ_{LH} may additionally identify the LOCAL active node graph with the minimal active HID support when the two presentations claim one PCR-3 source architecture. It is provenance data, not a prerequisite for SCIC overlap: source-distinct presentations can represent one standing carrier class.

$$\begin{aligned} \mathcal{E}_{L,H}^{\cong}(\sigma, \tau) = \{ & \mathcal{E}_{\text{LOCAL},\sigma}^{\text{surv}}, \mathcal{E}_{\text{HID},\tau}^{\text{surv}}, \mathcal{R}_{\text{LOCAL},\sigma}^{\text{NF}}, \mathcal{R}_{\text{HID},\tau}^{\text{NF}}, \\ & \alpha_{LH} : p_{\text{LOCAL},\sigma}(\theta_L) \xrightarrow{\sim} p_{\text{HID},\tau}(\theta_H) \}. \end{aligned} \quad (51)$$

The overlap groupoid is the homotopy fibre product

$$\mathfrak{D}_{\text{LOCAL},\sigma,\text{HID},\tau} = \Gamma_{\text{LOCAL},\sigma} \times_{\mathcal{N}_5}^h \Gamma_{\text{HID},\tau}. \quad (52)$$

Its objects are triples $(\theta_L, \theta_H, \alpha_{LH})$; its arrows are commuting pairs of source-groupoid arrows. For a concrete common core n , the fixed-core presentation is

$$\left(\widetilde{\text{Prov}}_{\text{LOCAL},\sigma}(n) \times \widetilde{\text{Prov}}_{\text{HID},\tau}(n) \right) // \text{Aut}_{\mathcal{N}_5}(n), \quad (53)$$

with the diagonal action on the chosen frames. Thus $\text{Aut}(n)$ acts on framed provenance objects, not on an unframed isomorphism class. The optional source-resolution relation Λ_{LH} records whether a common standing carrier has one dual PCR-3 source presentation, two source-distinct presentations, or a larger source groupoid.

10.2 Total status calculation

For every (θ_L, θ_H) , exactly one of the following applies:

1. **overlap**: both complete systems survive—all active relations pass and every inactive relation is theorem-certified N/A—and the two concrete PNF/core objects are isomorphic;
2. **separated_T, ..., separated_F**: both systems survive with their certified N/A sets retained and the named frozen packet channel is the first non-isomorphic channel;
3. **defect_T($(N_L, F_L, U_L), (N_H, F_H, U_H)$)**: the theorem-certified N/A sets are retained and at least one exact no-update active fail/undefined pair has nonempty union;
4. **source update**: an asserted local/hidden resolution requires a new source object outside the two generated source skeletons;
5. **target update**: boundary, record, operator category, witness family, continuation monoid, or target semantics changes.

Thus every skeleton pair is dispositioned, including empty comparison groupoids. Certified N/A is

retained as applicability data rather than failure; equal visible channels but unequal hidden ownership, threshold, matching orientation, memory, frozen deletion, or active fail/undefined record fall into a specific separation or defect stratum rather than an unresolved overlap question.

10.3 B05: exact LOCAL/HID overlap theorem

Theorem 10.1 (Executed LOCAL/HID overlap theorem). *[B05] The LOCAL/HID overlap is the coproduct of the well-typed fibre products (52) over every generated skeleton pair:*

$$\mathfrak{S}_{\text{LOCAL,HID}}^{\text{exec}} = \coprod_{\sigma, \tau} \left(\Gamma_{\text{LOCAL}, \sigma} \times_{\mathcal{N}_5^h}^h \Gamma_{\text{HID}, \tau} \right). \quad (54)$$

It is nonempty: the active-singlet calibration q_{3N} supplies LOCAL and HID presentations with isomorphic complete standing cores. For arbitrary skeleton pairs, nonemptiness is exactly the existence of an object $(\theta_L, \theta_H, \alpha_{LH})$ satisfying (51); empty pairs carry the complete separation/failure/update status above. The quotient groupoid, common core/packet, framed provenance, source-resolution provenance, and isotropy are all defined without choosing an automorphism group of an isomorphism class.

Proof. B03 and B04 give the two generated grammars, solved residual systems, lawful source groupoids, and frozen packet normal forms. The functors $p_{\text{LOCAL}, \sigma}$ and $p_{\text{HID}, \tau}$ land in the same concrete standing-core groupoid by Equation (24). Their homotopy fibre product therefore contains exactly pairs equipped with a lawful standing-core isomorphism. By Theorem 5.5, this is exactly SCIC equality. The framed decomposition (53) gives a literal $\text{Aut}(n)$ -action and records isotropy without representative ambiguity. The active-singlet construction proves aggregate nonemptiness, and the total status partition closes every complement point without treating reduced-channel equality as complete carrier equality. $\square$

11 The composite-carrier atlas

11.1 Grammar and constituent retention

A COMP skeleton contains a finite nonempty constituent subdiagram, at least one Derived node, and at least one active Bind or derivation incidence. Its canonical skeleton is

$$\sigma_{\text{COMP}} = (n_C, n_D, G_{\text{const}}, I_{\text{bind}}, I_{\text{der}}, I_T, \mathfrak{d}, \mathfrak{a}, \mathfrak{q}, \mathfrak{v}, \mathfrak{m}, \mathfrak{u}), \quad (55)$$

where G_{const} records the finite active constituent and derived-node incidence; $\mathfrak{d}$ types the derivation/binding map; and the remaining entries type attachment, anomaly, vacuum, matching, and continuation.

The COMP source equations are:

ID	COMP source equation
C1	Every derived node has a declared finite constituent diagram and a source-coherent derivation, binding, or emergence map.
C2	Constituent, derived, and retained-incidence squares commute; the derived channel is not selected independently of the constituent source point.
C3	The active constituent closure is the smallest subdiagram containing every constituent and incidence required by the target-bearing derived path.
C4	The source-to-target matching arrow preserves the derived carrier responsibility and the retained constituent/derivation data authorized by the packet.

ID	COMP source equation
C5	Constituent and derived representation, anomaly, vacuum, boundary, and continuation data belong to one witness architecture.

A constituent label is not automatically terminal standing. It is retained in the source skeleton and provenance, and it enters the solved packet only when deletion, continuation, matching, boundary, role ownership, or failure data expose it.

11.2 Residual solution specification

Let D_θ be the derivation/binding map, c_θ the derived carrier, and M_θ the oriented source-to-target matching diagram. Define $\mathcal{C}_{\text{comp}}(\theta)$ as the moduli object of pairs (D_θ, M_θ) satisfying C1–C5. Then

$$\begin{aligned}
Z_{\text{COMP},\sigma} = \{ & \theta : C1:C5, \mathcal{C}_{\text{comp}}(\theta) \neq \emptyset, \mathcal{A}_{\text{COMP}}(\theta) = 0, \\
& J_{\lambda,o}(\theta) \succeq 0, \sum_o \mathcal{I}_{\lambda,o}(\theta)^\dagger(1) = 1, R(K^\theta) = R(T_{\text{osc}}) \forall R, \\
& \mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_\theta), \exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^\theta(\lambda) > 0, \\
& A_\theta^{s,d} \rho_\theta(g) = \rho_{CC}(g) A_\theta^{s,d}, \mathcal{D}_{\text{caus}}^{\text{COMP}}(\theta) \neq \emptyset, \\
& \mathcal{V}_{\text{COMP}}(\theta) \neq \emptyset, \mathcal{M}_{\text{COMP}}(\theta) \neq \emptyset, \mathcal{X}_{\text{COMP}}(\theta) \text{ inert or coherent}, \\
& R_1(\theta) = \dots = R_9(\theta) = 0, \text{Cone}_{\text{COMP}}(\theta) \neq \emptyset \}.
\end{aligned} \tag{56}$$

This is PCR-4 equation (53), including the matching existence condition that is implicit in $\mathcal{C}_{\text{comp}}$ and E12. A derived target channel with constituent anomaly data from another source theory fails C5 and the common-cone/residual equations.

11.3 Constructive COMP groupoid

The group object is

$$G_{\text{COMP},\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes \left(\prod_{v \in V_{\text{const}}} \text{Aut}_{\text{src}}(v) \times \prod_{d \in V_{\text{derived}}} \text{Aut}_{\text{src}}(d) \times G_{\text{bind}} \times G_{\text{chart}} \right)_{\text{fix } \partial, T}. \tag{57}$$

Generators are constituent/derived basis maps, permutations of repeated isomorphic constituents, binding/derivation natural isomorphisms, graph isomorphisms, and source-chart changes. Their relations are commutation with C1–C5, semidirect conjugation by graph permutations, and fixed-boundary covariance. The isotropy group is the simultaneous stabilizer of the constituent diagram, derivation map, derived channel, vacuum, and matching diagram.

A change that erases an active constituent, reverses the matching arrow, or retunes the remaining constituents after a no-repair deletion is not a groupoid arrow.

11.4 Solved COMP packet normal form

The standing record is

$$\mathbf{s}_{\text{COMP}} = (t_{\text{osc}}, r[\text{constituent/derived ownership}], \delta_{\text{const}}, c[D_\theta, c_\theta], m[H_{\text{const}} \xrightarrow{D_\theta} c_\theta \xrightarrow{M_\theta} T_{\text{osc}}], b_\theta, f_{\text{comp}}). \tag{58}$$

The identity-bearing COMP deletion channel is the frozen restriction $\delta_{\text{COMP}} := \Delta_{\text{fr}}^{\text{del}} \downarrow_{\text{COMP}}$. Any finer constituent/block or source-subset deletion table is auxiliary source-composition data unless its consequence is already packet-factorized by Theorem 2.3; it is evaluated without re-binding or retuning. The matching channel retains orientation and every target-authorized constituent trace.

Define

$$\mathcal{M}_{\text{COMP},\sigma}^{\text{exec}} = [\{(\theta, \mathbf{s}_{\text{COMP}}) : \theta \in Z_{\text{COMP},\sigma}, \mathbf{s}_{\text{COMP}} \text{ satisfies (58)}\} / G_{\text{COMP},\sigma}]_{\mathbf{s}_{\text{COMP}}} . \quad (59)$$

11.5 B06: COMP classification

Theorem 11.1 (Executed constituent-retention theorem). *[B06] The exact COMP atlas is*

$$\mathcal{A}_{\text{COMP}}^{\text{exec}} = \coprod_{\sigma \in \Sigma_{\text{COMP}}^{\text{gram}}} \mathcal{M}_{\text{COMP},\sigma}^{\text{exec}} . \quad (60)$$

Every finite constituent/derivation code is generated before viability is known. Equations C1–C5 and E01–E23 determine its exact survivor, failure, and update strata. The constructive groupoid quotients only source isomorphism, while the solved record retains every standing-visible constituent, derivation, deletion, matching, boundary, continuation, and failure coordinate.

Proof. PCR-3 supplies the complete finite derived-carrier grammar and requires constituent/derivation source coherence. PCR-4’s composite evaluation theorem is exactly (56). The automorphisms (57) are the source-equational arrows. B09 gives descent; the seven-channel record gives SCIC identity. Therefore the atlas classifies all composite skeletons, including empty ones, and does not substitute an effective-field label for the constituent calculation. $\square$

Let

$$\mathcal{I}_{\text{COMP}} = \{C1, \dots, C5\} \cup \{E01, \dots, E23\}.$$

The COMP status stratification first fixes the pre-evaluation applicability vector and every theorem-certified N/A entry, then retains every nonempty active fail/undefined pair on $\mathcal{I}_{\text{COMP}}$, together with the source- and target-update branches. A low-energy coefficient shared with a LAW or LOCAL point creates an overlap only when the entire record (58) agrees.

12 The solitonic-carrier atlas

12.1 Grammar and profile strata

A SOL skeleton contains at least one active Soliton node with a source-defined persistent profile object, its supporting nonlinear/topological data, and an active target attachment. Its canonical skeleton is

$$\sigma_{\text{SOL}} = (n_S, G_{\text{prof}}, I_{\text{stab}}, I_{\partial}, I_T, \mathfrak{e}, \mathfrak{q}, \mathfrak{p}, \mathfrak{v}, \mathfrak{m}, \mathfrak{u}), \quad (61)$$

where G_{prof} records the finite incidence support of the profile; $\mathfrak{e}$ is the source equation/energy sort; $\mathfrak{q}$ is the topological or nonlinear charge sort; $\mathfrak{p}$ is persistence and endability type; and the remaining symbols type vacuum/deformation, matching, and continuation.

The SOL source equations are:

ID	SOL source equation
SO1	The profile ϕ_θ solves the declared source field/profile equation $\mathcal{E}_{\text{prof}}(\phi_\theta) = 0$ with the stated boundary conditions.
SO2	The source energy/action or other admitted persistence functional is finite on the declared support, and the source stability/persistence condition holds.
SO3	The declared topological/nonlinear charge and deformation class are well typed; topological labeling alone does not replace SO1–SO2.
SO4	The internal state space and charged-current target attachment are defined on the same persistent profile.
SO5	Endability, decay, annihilation, deformation-to-vacuum, and phase-boundary data are source-defined continuation/failure objects before packet evaluation.

12.2 Residual solution specification

Let $\mathcal{S}_{\text{sol}}(\theta)$ be the moduli object of source-coherent profiles satisfying SO1–SO5. The residual solution object is

$$\begin{aligned}
Z_{\text{SOL},\sigma} = \{ & \theta : \text{SO1}:\text{SO5}, \mathcal{S}_{\text{sol}}(\theta) \neq \emptyset, \mathcal{A}_{\text{SOL}}(\theta) = 0, \\
& J_{\lambda,o}(\theta) \succeq 0, \sum_o \mathcal{I}_{\lambda,o}(\theta)^\dagger(1) = 1, R(K^\theta) = R(T_{\text{osc}}) \forall R, \\
& \mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_\theta), \exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^\theta(\lambda) > 0, \\
& A_\theta^{s,d} \rho_\theta(g) = \rho_{CC}(g) A_\theta^{s,d}, \mathcal{D}_{\text{caus}}^{\text{SOL}}(\theta) \neq \emptyset, \\
& \mathcal{V}_{\text{SOL}}(\theta) \neq \emptyset, \mathcal{M}_{\text{SOL}}(\theta) \neq \emptyset, \mathcal{X}_{\text{SOL}}(\theta) \text{ inert or coherent}, \\
& R_1(\theta) = \dots = R_9(\theta) = 0, \text{Cone}_{\text{SOL}}(\theta) \neq \emptyset \}.
\end{aligned} \tag{62}$$

This is PCR-4 equation (56) expanded. Profile existence without an event-complete target channel fails E01–E06 or E08. A channel whose supporting profile or vacuum belongs to another parameter point fails SO4 and R1/R2/R7.

12.3 Constructive SOL groupoid

The group object is

$$G_{\text{SOL},\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes (G_{\text{int}} \times G_{\text{collective}} \times G_{\text{profile chart}} \times G_{\text{boundary}})_{\text{fix } T_{\text{osc}}} . \tag{63}$$

Generators are internal source symmetries, source-authorized collective-coordinate reparameterizations, profile-chart isomorphisms, finite graph isomorphisms, and boundary-preserving transformations. They act on ϕ , its fluctuation/internal-state operator, charge, attachments, vacuum branch, and matching diagram by pullback or conjugation as appropriate. Relations are source equation covariance, preservation of the persistence functional and charge, fixed-boundary naturality, and semidirect compatibility. The isotropy group is the stabilizer of the profile, its internal target channel, and its complete standing record.

A deformation that changes endability, a phase-boundary crossing, a changed physical profile support, or a changed matching/failure object is not a groupoid arrow merely because a restricted low-energy response agrees.

12.4 Solved SOL packet normal form

The SOL standing record is

$$\begin{aligned} \mathbf{s}_{\text{SOL}} = & (t_{\text{osc}}, r[\phi_\theta, \text{internal ownership}], \delta_{\text{profile}}, \\ & c[\text{profile continuation, endability}], m[\phi_\theta \rightarrow \text{target}], \\ & b[\text{profile attachment}], f_{\text{sol}}). \end{aligned} \quad (64)$$

Topology and profile labels enter terminal identity only through these target-standing consequences. Two source profiles with one solved record lie in the same SCIC class while remaining distinct provenance objects.

Define

$$\mathcal{M}_{\text{SOL},\sigma}^{\text{exec}} = [\{(\theta, \mathbf{s}_{\text{SOL}}) : \theta \in Z_{\text{SOL},\sigma}, \mathbf{s}_{\text{SOL}} \text{ satisfies (64)}\} / G_{\text{SOL},\sigma}]_{\mathbf{s}_{\text{SOL}}}. \quad (65)$$

12.5 B07: SOL classification

Theorem 12.1 (Executed persistence and endability theorem). *[B07] The exact SOL atlas is*

$$\mathcal{A}_{\text{SOL}}^{\text{exec}} = \coprod_{\sigma \in \Sigma_{\text{SOL}}^{\text{gram}}} \mathcal{M}_{\text{SOL},\sigma}^{\text{exec}}. \quad (66)$$

Every finite profile-support code is generated independently; SO1–SO5 and E01–E23 determine its exact profile, survivor, failure, and update strata; (63) gives the lawful action and isotropy; and (64) gives the evaluated packet moduli. No profile is merged by topological label alone when persistence, endability, continuation, matching, deletion, boundary, or failure differs.

Proof. PCR-3 supplies every finite solitonic/localized nonperturbative source trace and its symbolic profile domain. SO1–SO5 state source objecthood and persistence explicitly. PCR-4’s solitonic evaluation theorem is (62). The source symmetry and profile-chart action is constructive and covariant, so B09 applies. Packet completeness identifies the solved record with SCIC identity, and the status vector classifies every complement point. $\square$

Let

$$\mathcal{I}_{\text{SOL}} = \{\text{SO1}, \dots, \text{SO5}\} \cup \{\text{E01}, \dots, \text{E23}\}.$$

The SOL complement is indexed by the theorem-certified inapplicability set $N \subseteq \mathcal{I}_{\text{SOL}}$, the exact active fail/undefined pair (F, U) , and the source- and target-update branches. The atlas does not assume that the resulting profile moduli are finite-dimensional; when the source-defined profile space is functional or categorical, the theorem returns that exact solution object rather than a finite surrogate.

13 The decorated-hybrid carrier atlas

13.1 Grammar of arbitrary lawful finite hybrids

A HYB skeleton is an activity-reduced source skeleton with a nonempty hybrid-witness groupoid $\text{HybWit}(\sigma)$: equivalently, it contains a nontrivial finite source-coherent composition of at least two occupancy-bearing subdiagrams. No literal **Compose** edge is required. The complete witness test is Theorem 3.4; it recognizes multi-kind activity, surviving cross-component incidences, irreducible higher-arity relations, and nontrivial

gluing diagrams. Its canonical presentation is

$$\sigma_{\text{HYB}} = (V, E, \nu, \epsilon, s, t, \mathcal{I}, \mathcal{A}, \mathcal{W}, \mathcal{K}, \mathcal{K}^{\text{full}}, \mathcal{G}, \mathcal{H}_{\text{cross}}), \quad (67)$$

where $(V, E, \nu, \epsilon, s, t)$ is the finite typed hypergraph; $\mathcal{K}^{\text{full}}$ records all active principal and auxiliary carrier-bearing kinds; $\mathcal{G}$ is a finite PCR-3 source-coherent gluing witness with at least two occupancy-bearing constituent subdiagrams; and $\mathcal{H}_{\text{cross}}$ records all surviving cross-component incidences, primitive higher-arity source relations, cross-anomaly terms, cross-vacuum conditions, mixed matching arrows, shared boundary maps, and common-cone incidences created by the gluing. In particular active **Inflow**, **Fuse**, **CoCarry**, **Project**, and boundary/edge relations are retained as hybrid structure whenever they join occupancy-bearing constituents. Hyperedges may have arbitrary finite arity. They are not replaced by pairwise edges unless the source coherence theorem supplies that factorization.

The HYB source equations are:

ID	HYB source equation
Y1	Every occupancy-bearing constituent subdiagram, including TOP/EXT/BDY/ENV components, satisfies its own PCR-3 source-constitutive equations on the restricted source point; principal constituents additionally satisfy their displayed family equations.
Y2	Every overlap map, shared support, and source gluing square in $\mathcal{G}$ commutes.
Y3	Every primitive higher-arity edge satisfies its declared constitutive equation and is retained whenever source-active; no pairwise replacement is imposed without a source theorem.
Y4	The principal mask, full carrier-kind mask, constituent gluing diagram, and source-composition graph are computed after exact activity reduction; source-null incidences are removed and every nonnull cross-component incidence is retained without a magnitude threshold.
Y5	Cross-anomaly, cross-vacuum, mixed matching, extension, boundary, and common-cone data are formed from the complete hybrid source object, not from separately selected constituent ledgers.
Y6	A compensating component may close a constituent obstruction only when the compensating incidence and its own consistency data are part of the previously generated hybrid skeleton.
Y7	The auxiliary no-repair source-sufficiency profile is evaluated on the fixed active hybrid point; deleting a source subset does not permit retuning, re-gluing, or replacing the vacuum/matching branch. Its identity-bearing content is restricted by Theorem 2.3.

13.2 Auxiliary source-sufficiency hypergraph

Let K_θ be the finite active source-block set of a HYB point. Define

$$N_\theta^{\text{aux}}(S) = \begin{cases} 1, & \text{retaining exactly } S \subseteq K_\theta \text{ preserves the complete packet without repair,} \\ 0, & \text{the deletion is well typed and destroys the complete packet,} \\ \perp, & \text{the deletion is fail-closed undefined or changes source/target typing.} \end{cases} \quad (68)$$

The minimal sufficient subsets form a finite auxiliary hypergraph $\mathcal{S}_\theta^{\text{aux}} \subseteq \mathcal{P}(K_\theta)$. Single-route, redundant, synergistic, and partially synergistic source organization are derived from $\mathcal{S}_\theta^{\text{aux}}$, not from family names. By Theorem 2.3, neither N_θ^{aux} nor $\mathcal{S}_\theta^{\text{aux}}$ refines the frozen SCIC quotient unless its relevant consequence already factors through Ξ_{osc} .

13.3 Residual solution specification

Let $\widehat{\Phi}_{\text{HYB}}$ be the universal evaluation on the complete decorated hypergraph. Instead of leaving its inverse image unexpanded, define

$$\begin{aligned} Z_{\text{HYB},\sigma} = \{ & \theta : Y1:Y7, J_{\lambda,o}(\theta) \succeq 0, \sum_o \mathcal{I}_{\lambda,o}(\theta)^\dagger(1) = 1, \\ & R(K^\theta) = R(T_{\text{osc}}) \forall R, \mathfrak{M}^{\text{vis}}(T_{\text{osc}}) \hookrightarrow \mathfrak{M}^{\text{vis}}(X_\theta), \\ & \exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^\theta(\lambda) > 0, A_\theta^{s,d} \rho_\theta(g) = \rho_{CC}(g) A_\theta^{s,d}, \\ & \mathcal{D}_{\text{caus}}^{\text{HYB}}(\theta) \neq \emptyset, \mathcal{A}_{\text{HYB}}(\theta) = 0, \mathcal{V}_{\text{HYB}}(\theta) \neq \emptyset, \\ & \mathcal{M}_{\text{HYB}}(\theta) \neq \emptyset, \mathcal{X}_{\text{HYB}}(\theta) \text{ inert or coherent}, \\ & R_1(\theta) = \dots = R_9(\theta) = 0, \text{Cone}_{\text{HYB}}(\theta) \neq \emptyset \}. \end{aligned} \quad (69)$$

The anomaly, vacuum, matching, extension, residual, and cone objects in (69) include all cross terms in $\mathcal{H}_{\text{cross}}$. Constituentwise survival is neither necessary nor sufficient for membership.

13.4 Constructive HYB groupoid

The group object is

$$G_{\text{HYB},\sigma} = \text{Aut}_{\text{hyp}}(\sigma) \ltimes \left(\prod_{c \in \text{Constit}(\sigma)} G_c^{\text{src}} \times G_{\text{glue}} \times G_{\text{higher}} \right)_{\text{coh, fix } T_{\text{osc}}}. \quad (70)$$

Here $\text{Constit}(\sigma)$ is the finite set of occupancy-bearing constituent subdiagrams in the chosen PCR-3 reconstruction witness. For LAW/LOCAL/HID/COMP/SOL constituents, G_c^{src} is the corresponding displayed source action. For auxiliary TOP, EXT, BDY, and ENV constituents it is the PCR-3 source-constitutive automorphism group preserving, respectively, topological/fusion/inflow data, defect/extended junction data, boundary attachment data, or environmental co-carriage data. G_{glue} consists of natural isomorphisms of shared-support, projection, inflow, co-carriage, fusion, and attachment diagrams, and G_{higher} consists of automorphisms of primitive higher-arity relation objects. The coherence subscript imposes Y1–Y6 and the semidirect relations with hypergraph automorphisms. Isotropy at the source-presentation level is the stabilizer of the entire decorated hypergraph, including auxiliary carrier-bearing components, cross terms, and the auxiliary source-sufficiency profile. This source-level stabilizer does not promote the auxiliary profile into terminal standing identity.

Deleting a higher-arity edge, changing a cross-vacuum branch, replacing a mixed matching map, or retuning a constituent after deletion is not a groupoid arrow.

13.5 Solved HYB packet normal form

The standing record is

$$\begin{aligned} \mathbf{s}_{\text{HYB}} = & (t_{\text{osc}}, r[\text{joint standing ownership}], \delta[\Delta_{\text{fr}}^{\text{del}}(\theta)], \\ & c[\text{hybrid continuation}], m[\text{mixed matching}], \\ & b[\text{shared boundary}], f[\mathcal{H}_{\text{cross}}]). \end{aligned} \quad (71)$$

The identity-bearing deletion entry is exactly the frozen PCR-5A deletion channel. The full auxiliary pair

$$\mathcal{U}_{\text{HYB}}(\theta) = (N_{\theta}^{\text{aux}}, \mathcal{S}_{\theta}^{\text{aux}})$$

is retained beside the standing record as source-composition and provenance data and is supplied to PCR-5C without altering base identity. Higher-arity relations remain standing exactly when they change one of the solved frozen channels; otherwise they remain source structure.

Define

$$\mathcal{M}_{\text{HYB},\sigma}^{\text{exec}} = [\{(\theta, \mathbf{s}_{\text{HYB}}) : \theta \in Z_{\text{HYB},\sigma}, \mathbf{s}_{\text{HYB}} \text{ satisfies (71)}\} / G_{\text{HYB},\sigma}]_{\mathbf{s}_{\text{HYB}}}. \quad (72)$$

13.6 B08: decorated-HYB classification

Theorem 13.1 (Executed decorated-HYB packet theorem). *[B08] The exact HYB atlas is*

$$\mathcal{A}_{\text{HYB}}^{\text{exec}} = \coprod_{\sigma \in \Sigma_{\text{HYB}}^{\text{gram}}} \mathcal{M}_{\text{HYB},\sigma}^{\text{exec}}. \quad (73)$$

It covers every lawful finite source-semantic hybrid generated by PCR-3, including hybrids witnessed only by active Inflow, Fuse, CoCarry, Project, boundary/edge, or arbitrary finite-arity cross relations and including surviving TOP/EXT/BDY/ENV occurrences routed by Theorem 3.6. Equations Y1–Y7 and E01–E23 determine every survivor, failure, source-update, and target-update stratum. The constructive groupoid retains higher coherence and isotropy, and the solved record retains exactly the frozen deletion channel and every standing-visible cross term, while the auxiliary source-sufficiency hypergraph is exported separately without refining SCIC.

Proof. PCR-3’s finite-hybrid closure and incidence reconstruction theorems generate every lawful finite hybrid by constituent subdiagrams plus cross-component hyperedges and do not reduce higher arity to pairwise coupling. Theorem 3.4 makes that inherited semantic criterion explicit in the PCR-5B grammar, while Theorem 3.6 routes every surviving auxiliary TOP/EXT/BDY/ENV occurrence into HYB by the frozen PCR-4 pure-stratum reductions. PCR-4’s hybrid theorem evaluates the complete source object with all cross-anomaly, cross-vacuum, mixed matching, extension, relation, and common-cone data. This is (69). The group action (70) is the source-authorized coherent automorphism action. B09 gives exact descent, Theorem 2.3 prevents auxiliary subset diagnostics from over-resolving the base quotient, and the PCR-5A packet theorem identifies (71) with SCIC identity. Thus no hybrid architecture, higher relation, theorem-certified N/A applicability face, active fail/undefined face, or duplicate presentation remains merely named. $\square$

13.7 Four calibrated deletion points

The redundant and synergistic PCR-5A points lie in LAW–LOCAL–HID–HYB feature charts. Their law/new-local deletion projections are (1, 1) and (0, 0), while q_W and q_{3N} have (0, 1) and (1, 0). These

four two-coordinate projections are marked points of the frozen deletion channel. Their auxiliary source-sufficiency tables may contain additional source-composition information, but no such information refines the four base identities unless it is already visible in the frozen packet.

14 Pairwise overlap atlas

14.1 Skeleton-pair calculation over the universal standing-core groupoid

Let $a = (F, \sigma)$ and $b = (G, \tau)$ be generated family/skeleton charts. Their exact overlap is not formed over a quotient set of packet isomorphism classes. It is the homotopy fibre product of the two constructive source groupoids over the concrete universal PNF/core groupoid:

$$\mathfrak{S}_{a,b}^{\text{exec}} := \Gamma_a \times_{\mathcal{A}_5}^h \Gamma_b. \quad (74)$$

An object is a triple

$$(\theta_a, \theta_b, \alpha), \quad \alpha : p_a(\theta_a) \xrightarrow{\sim} p_b(\theta_b),$$

where both source points satisfy their complete family and E00–E23 systems. In a common canonical packet chart, α is represented by the seven channel equalities; in a noncanonical chart it is the corresponding pointed continuation-equivariant packet/core isomorphism. The deletion comparison is always the frozen field $\Delta_{\text{fr}}^{\text{del}}$. Auxiliary source-subset sufficiency is retained in provenance and never used as an eighth identity channel.

For a concrete common core $n \in \mathcal{A}_5$, Theorem 4.11 gives the fixed-core presentation

$$\mathfrak{S}_{a,b;n}^{\text{exec}} \simeq \left(\widetilde{\text{Prov}}_a(n) \times \widetilde{\text{Prov}}_b(n) \right) // \text{Aut}_{\mathcal{A}_5}(n), \quad (75)$$

with the diagonal postcomposition action on the chosen frames. This is the precise meaning of a common-PNF automorphism in the overlap calculus. No action is assigned to an unframed isomorphism class.

For each skeleton pair, the calculation determines the exact overlap groupoid, the ordered separator strata, and a proof-relevant branch datum. Let $P_{a,b}$ be the raw product parameter object and let $O_{a,b} \subseteq P_{a,b}$ be the all-surviving locus for which an isomorphism α exists. Define

$$\text{Dec}_{a,b} = \text{NE}\left(O_{a,b}, \mathfrak{S}_{a,b}^{\text{exec}}\right) \sqcup \text{E}(\eta_{a,b}), \quad (76)$$

where a nonempty branch contains an object $(\theta_a, \theta_b, \alpha)$, while an empty branch contains a total certificate

$$\eta_{a,b} : P_{a,b} \longrightarrow \{\text{Sep}_T, \text{Sep}_R, \text{Sep}_D, \text{Sep}_C, \text{Sep}_M, \text{Sep}_B, \text{Sep}_F, \text{Defect}, \text{SU}, \text{TU}\},$$

where a **Defect** value carries the exact ordered data

$$\left((\mathbf{N}_a; \mathbf{F}_a, \mathbf{U}_a), (\mathbf{N}_b; \mathbf{F}_b, \mathbf{U}_b) \right),$$

with every member of $\mathbf{N}_a, \mathbf{N}_b$ theorem-certified and at least one nonempty failure/undefined union. The primary decision order is target update, source update, exact applicability/defect record, and only then—when both sides survive—common-core equality or the first unequal frozen packet channel. Thus an empty declaration carries a universal separator/applicability/fail/undefined/update proof rather than the sentence “no example was found.” A nonempty declaration carries an actual core-isomorphism witness and its isotropy.

The complete pairwise datum also contains:

1. the two exact family solution objects and the fibre-product groupoid (74);
2. a concrete common core/packet object or, equivalently, the comparison arrow α ;
3. the framed provenance quotient (75), together with source-presentation isotropy and $\text{Aut}_{\mathcal{N}_5}(n)$;
4. every first-channel separation stratum $\text{Separated}_T, \dots, \text{Separated}_F$;
5. every certified N/A set, exact fail/undefined defect vector, and source/target update reason; and
6. the incidence maps to each higher-overlap object containing the pair.

There is no third global branch between empty and nonempty, while the full parameter product retains the finer overlap/separation/certified-N/A/fail-undefined/update partition.

The comparison objects in PCR-3 are interpreted in ordinary classical, universe-small set/groupoid semantics. Accordingly, for each fixed generated skeleton pair the proposition “the fibre-product groupoid has an object” obeys excluded middle, and nonemptiness is equivalent to the existence of a source-pair and a core-isomorphism witness. This is a mathematical disposition theorem, not a claim that arbitrary functional or categorical source equations admit a finite numerical decision algorithm.

Lemma 14.1 (Total pair-decision record). *For every generated skeleton pair, exactly one branch of (76) is inhabited. In the empty branch, the first-applicable status partition defines the universal certificate $\eta_{a,b}$.*

Proof. The fibre-product groupoid is either empty or nonempty. In the nonempty case, an object supplies the NE witness. In the empty case, every raw product point either carries a target update, carries a source update, has at least one exact no-update applicability/defect record $(N; F, U)$ with $F \cup U \neq \emptyset$, or both points survive but their frozen packet/core objects are nonisomorphic. Packet/core completeness then gives a first unequal frozen channel on the surviving branch. The total disjoint status partition assigns the point uniquely to one first-difference, exact certified-N/A/fail/undefined record, source-update, or target-update stratum, defining η . Disjointness prevents both branches from being inhabited. $\square$

14.2 Fifteen instantiated family-pair comparison systems

The standing-overlap condition for every pair is always the same typed condition: existence of a morphism in $\mathcal{N}_5$. The family-specific material below records how each chart constructs the seven frozen packet channels and, where applicable, how a claimed common PCR-3 source presentation is represented. It is *not* an extra requirement that would exclude source-distinct presentations having the same complete packet.

Pair	Family-specific chart/core comparison and provenance data	Aggregate status and calibrated witness
LAW-LOCAL	Compare the LAW process record with the oriented target-resolution image of the LOCAL source; boundary, threshold/order, frozen deletion, matching, and failure records enter the common core. A direct source-resolution map, when present, is provenance.	Parameterized correspondence; $q_{\text{red}}, q_{\text{syn}}$ give nonempty fibres.

Pair	Family-specific chart/core comparison and provenance data	Aggregate status and calibrated witness
LAW-HID	Compare the LAW process with the HID visible reduction; hidden ownership and memory either appear in the frozen packet/core or remain provenance.	Parameterized correspondence; the mixed singlet calibrations give nonempty fibres.
LAW-COMP	Compare the LAW process with the matched derived-carrier process; retained constituent/derivation data enter standing identity only through matching, deletion, continuation, boundary, or failure channels.	Parameterized/categorical correspondence; nonemptiness is the exact fibre-product status of each skeleton pair.
LAW-SOL	Compare the LAW process with the profile-induced target process; profile persistence/endability enter through continuation/failure when standing-visible.	Parameterized/functional correspondence; each skeleton pair is exactly empty or nonempty by (74).
LAW-HYB	Compare the LAW chart with the LAW-facing restriction of the complete HYB source point; cross terms enter only through frozen packet channels, while auxiliary source-sufficiency remains provenance.	Parameterized correspondence; $q_{\text{red}}, q_{\text{syn}}$ are nonempty.
LOCAL-HID	Compare the active LOCAL graph with the HID visible/minimal-support realization; ownership, threshold, matching, frozen deletion, continuation, and failure enter the common core. A direct graph-resolution map is optional provenance.	Parameterized correspondence; $q_{3N}, q_{\text{red}}, q_{\text{syn}}$ are nonempty.
LOCAL-COMP	Compare the LOCAL channel with the retained derived carrier; constituent traces refine standing only when exposed by the frozen packet/core.	Parameterized/categorical correspondence; exact skeleton-pair status.
LOCAL-SOL	Compare the LOCAL channel/attachment with the profile internal-state channel; profile persistence/endability enter only through frozen standing channels.	Parameterized/functional correspondence; exact skeleton-pair status.
LOCAL-HYB	Compare the LOCAL chart with the LOCAL-facing restriction of the complete hybrid source point; cross-anomaly, cross-vacuum, mixed matching, and frozen deletion data are retained when standing-visible.	Parameterized correspondence; $q_{\text{red}}, q_{\text{syn}}$ are nonempty.
HID-COMP	Compare the HID visible reduction with the derived carrier channel; hidden support and constituent/derivation data enter the common core exactly when standing-visible.	Parameterized/categorical correspondence; exact skeleton-pair status.
HID-SOL	Compare the hidden visible reduction with the profile-induced channel; hidden memory/ownership and profile endability enter through the frozen packet/core when standing-visible.	Parameterized/functional correspondence; exact skeleton-pair status.

Pair	Family-specific chart/core comparison and provenance data	Aggregate status and calibrated witness
HID–HYB	Compare the HID chart with the hidden-facing restriction of the hybrid; cross terms are carried through the frozen core and auxiliary composition data remain provenance unless already packet-visible.	Parameterized correspondence; the mixed singlet calibrations give nonempty fibres.
COMP–SOL	Compare the derived-carrier channel with the profile-induced channel; constituent retention and profile persistence/endability enter exactly through frozen standing channels.	Parameterized/functional-categorical correspondence; exact skeleton-pair status.
COMP–HYB	Compare the COMP chart with the constituent/derived restriction of the hybrid; cross terms and frozen no-repair deletions enter the common core, auxiliary subset profiles do not refine it.	Parameterized/categorical correspondence; exact skeleton-pair status.
SOL–HYB	Compare the SOL chart with the persistent-profile restriction of the hybrid; profile support, endability, cross matching, and failure records enter the common core when standing-visible.	Parameterized/functional correspondence; exact skeleton-pair status.

These fifteen systems exhaust the unordered distinct-family pairs of the six-family cover. Each is an instance of the same universal fibre-product construction and carries its frozen-channel comparison, framed provenance, applicability and complement data, calibrated witnesses, and higher-incidence maps.

14.3 Exact first-difference partition

For a surviving point pair, choose canonical frozen packet representatives after transporting along any already established channel isomorphisms. Define

$$\begin{aligned}
\text{Separated}_T &: \Longleftrightarrow t_F \not\cong t_G, \\
\text{Separated}_R &: \Longleftrightarrow t_F \cong t_G \wedge r_F \not\cong r_G, \\
\text{Separated}_D &: \Longleftrightarrow t_F \cong t_G, r_F \cong r_G \wedge \delta_F \not\cong \delta_G,
\end{aligned}$$

and continue in the fixed order through C, M, B, F . Together with the all-isomorphic locus, these seven regions form a disjoint partition of the surviving product. Adding the complete applicability, failure/undefined, and update strata gives a total partition of the raw product. Thus an empty overlap is accompanied by an exact separator/applicability/failure/update reason, not a bare absence. Surviving overlap points may carry nonempty certified N/A sets; those sets are retained in the comparison record but do not constitute failures.

14.4 Framed provenance multiplicity and isotropy

For a concrete common core n , the overlap provenance is

$$\text{Prov}_{F,G}^{\text{fr}}(n) = \coprod_{\sigma, \tau} \left(\widetilde{\text{Prov}}_{F, \sigma}(n) \times \widetilde{\text{Prov}}_{G, \tau}(n) \right) // \text{Aut}_{\mathcal{N}_5}(n). \quad (77)$$

Its isotropy contains the source-presentation stabilizers compatible with the selected frames and the stabilizer of the common core under the diagonal framing action. A proved source-resolution isomorphism may identify additional provenance arrows, but it does not alter the standing class. This multiplicity is retained for reconstruction and target extension; it does not multiply SCIC classes.

14.5 B10: all pairwise overlaps

Theorem 14.2 (Executed distinct-family overlap atlas). *[B10] For every unordered pair $F < G$ in $\mathfrak{F}_{\text{osc}}$, the exact overlap is*

$$\mathfrak{D}_{F,G}^{\text{exec}} = \prod_{\sigma \in \Sigma_F^{\text{gram}}} \prod_{\tau \in \Sigma_G^{\text{gram}}} \left(\Gamma_{F,\sigma} \times_{\mathcal{N}_5}^h \Gamma_{G,\tau} \right). \quad (78)$$

The fifteen family-pair systems above are exhaustive. Every skeleton-pair fibre has the proof-relevant decision record of Theorem 14.1, an exact well-typed overlap groupoid, a concrete common core/packet or a complete separator, framed provenance and isotropy, complete certified-N/A/fail/undefined/update data, and maps to higher overlaps. No pair remains merely a possible equalizer problem, and no automorphism action is taken on an unframed quotient class.

Proof. There are exactly $\binom{6}{2} = 15$ distinct family pairs. B02–B08 supply generated skeletons, exact systems, constructive source groupoids, and solved frozen packet records for each side. Equation (24) sends every such chart into the same concrete groupoid $\mathcal{N}_5$; hence the homotopy fibre product is automatically well typed and contains every and only pair equipped with a lawful standing-core isomorphism. Theorem 5.5 makes that condition exactly SCIC equality. Theorem 4.11 supplies the fixed-core provenance description with a genuine $\text{Aut}(n)$ -action on frames. The total status partition and Theorem 14.1 supply either a nonempty witness or a universal empty certificate, together with certified applicability/N/A, separation, fail/undefined, and update data for every source-point pair. Hence every distinct-family overlap is resolved at skeleton-pair level. $\square$

15 Higher overlaps, codescent, and exact global assembly

15.1 Higher comparison as an iterated homotopy fibre product

Let

$$\mathcal{I}_1 = \{a = (F, \sigma) : F \in \mathfrak{F}_{\text{osc}}, \sigma \in \Sigma_F^{\text{gram}}\}$$

be the class of generated family-skeleton charts. Each chart carries its constructive source groupoid Γ_a and the functor $p_a : \Gamma_a \rightarrow \mathcal{N}_5$ of Equation (24). For a finite nonempty tuple $\mathbf{a} = (a_0, \dots, a_k)$ with repetitions allowed, define the exact higher overlap

$$\mathcal{C}_{\mathbf{a}} := \Gamma_{a_0} \times_{\mathcal{N}_5}^h \Gamma_{a_1} \times_{\mathcal{N}_5}^h \cdots \times_{\mathcal{N}_5}^h \Gamma_{a_k}. \quad (79)$$

An object is equivalently a tuple of source solutions x_i , one concrete common core $n \in \mathcal{N}_5$, and frames

$$\phi_i : p_{a_i}(x_i) \xrightarrow{\sim} n.$$

Changing the common frame acts diagonally by $\text{Aut}_{\mathcal{N}_5}(n)$. Hence the fixed-core presentation is

$$\mathcal{C}_{\mathbf{a}} \simeq \prod_{[n] \in \pi_0(\mathcal{N}_5)} \left(\prod_{i=0}^k \widetilde{\text{Prov}}_{a_i}(n) \right) // \text{Aut}_{\mathcal{N}_5}(n). \quad (80)$$

This formula is the typed higher analogue of (75). The automorphism group acts on chosen frames, not on an isomorphism class with no representative.

15.2 Direct higher status and emptiness certificate

The scientific status is evaluated on one simultaneous raw source tuple $(\theta_0, \dots, \theta_k)$. First evaluate each complete family system and update status. If all points survive, compare the frozen packet/core of every factor with that of the first factor. Exactly one of the following applies:

1. **CommonCore**: a common concrete core n and frames ϕ_i exist, giving an object of $\mathcal{C}_a$;
2. **Sep_T, ..., Sep_F**: all factors survive but some factor is first separated from the reference factor in the named frozen packet channel;
3. **Defect**: at least one factor has $F_i \cup U_i \neq \emptyset$; the status datum retains the full tuple $((N_i; F_i, U_i))_{i=0}^k$ together with every N/A certificate;
4. **SourceUpdate** or **TargetUpdate**: source or target typing changes.

This gives a total complement certificate

$$\eta_a : \prod_i \Theta_{a_i}^{\text{raw}} \longrightarrow \{\text{CommonCore}, \text{Sep}_T, \dots, \text{Sep}_F, \text{Defect}, \text{SU}, \text{TU}\},$$

where a defect output carries the exact applicability, failed-equation, and undefined-equation sets for every factor, together with the N/A certificates. A higher overlap is therefore not inferred from pairwise nonemptiness. Its nonempty branch requires one simultaneous source tuple and one common core. Its empty branch is certified directly on the same tuple by an exact certified-N/A/fail/undefined vector, update, or a frozen packet separator.

There are

$$\binom{6}{3} + \binom{6}{4} + \binom{6}{5} + \binom{6}{6} = 20 + 15 + 6 + 1 = 42$$

distinct-label family subsets of size at least three. These 42 types exhaust the distinct-label higher incidences of the six-family cover. Repeated labels are governed by the same finite-tuple formula rather than by a separate enumeration.

15.3 Face, degeneracy, and permutation maps

A face map forgets one source solution and its frame. A degeneracy map repeats one source solution with the same frame; in the comparison-arrow presentation this inserts the identity core isomorphism. A permutation reorders source points and frames. These operations commute with the diagonal $\text{Aut}(n)$ -action and therefore descend to (80). The standard identities

$$\begin{aligned} d_i d_j &= d_{j-1} d_i & (i < j), \\ s_i s_j &= s_{j+1} s_i & (i \leq j), \\ d_i s_j &= \begin{cases} s_{j-1} d_i, & i < j, \\ \text{id}, & i = j \text{ or } i = j + 1, \\ s_j d_{i-1}, & i > j + 1 \end{cases} \end{aligned}$$

hold by deletion/repetition of framed factors. The cocycle relation on a triple is composition in the concrete groupoid $\mathcal{N}_5$; all higher cocycles are its associative iterates.

15.4 B11: instantiated higher-overlap coherence

Theorem 15.1 (Instantiated higher-overlap and coherence theorem). *[B11] The 42 distinct-label dispatch records, together with the repeated-label finite-tuple constructor (79), form the complete augmented Čech nerve*

$$\mathcal{C}_\bullet \longrightarrow \mathcal{P}_{\text{osc}}^{\text{exec}} \xrightarrow{p} \mathcal{N}_5,$$

where

$$\mathcal{P}_{\text{osc}}^{\text{exec}} = \coprod_{F \in \mathfrak{F}_{\text{osc}}} \coprod_{\sigma \in \Sigma_F^{\text{gram}}} \Gamma_{F, \sigma}.$$

Every higher object is an iterated homotopy fibre product over the same concrete standing-core groupoid, every fixed-core provenance presentation has a genuine diagonal $\text{Aut}(n)$ -action on frames, and every face, degeneracy, permutation, and cocycle map is well typed. No chain of pairwise coincidences is promoted to a higher point without one common source tuple and one common core.

Proof. B02–B09 give every generated chart and its functor to $\mathcal{N}_5$; the HYB quantifier uses the source-semantic routing domain of Theorems 3.4 and 3.6; B10 gives the degree-one kernel objects. Iterated homotopy fibre products over a groupoid exist in the fixed universe and are represented by the common-core/framing model (80). The diagonal framing action is literal, so the isotropy and quotient are defined without representative ambiguity. Face and degeneracy maps are forget/repeat operations on framed tuples and satisfy the simplicial identities. The direct status map $\eta_{\mathbf{a}}$ prevents pairwise nonemptiness from masquerading as simultaneous higher nonemptiness. The 42 distinct-label subsets exhaust all family subsets of size at least three; repetitions are covered by the universal finite-tuple construction. Thus the complete higher kernel and its coherence are resolved. $\square$

15.5 The resolved standing-core subatlas

Define $\mathcal{N}_{5, \text{osc}}^{\text{exec}}$ to be the full subgroupoid of $\mathcal{N}_5$ on those concrete standing cores n for which there exist a generated chart (F, σ) , a source solution $\theta \in Z_{F, \sigma}$, and an isomorphism

$$p_{F, \sigma}(\theta) \xrightarrow{\sim} n.$$

The definition ranges over the independently generated source grammar and the exact symbolic solution objects of B02–B08. Its objects are exactly the standing cores represented by the family atlases, and its arrows are the standing-core isomorphisms already defined in $\mathcal{N}_5$.

The global atlas is the codescent of the resolved Čech nerve:

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} := \text{codesc}_{[k] \in \Delta^{\text{op}}} \mathcal{C}_k. \quad (81)$$

Every source chart, pairwise comparison, higher comparison, isotropy group, applicability-aware frontier, and auxiliary provenance annotation entering this codescent has already been determined at the source-grammar and solution-object levels.

15.6 B12: exact atlas assembly

Theorem 15.2 (Exact oscillation-atlas assembly). *[B12] There are canonical equivalences*

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{5,\text{osc}}^{\text{exec}}, \quad \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}. \quad (82)$$

The first equivalence is the executed categorical atlas of concrete standing cores and their lawful isomorphisms; the second identifies its connected components with the frozen SCIC quotient. Source presentation, provenance, auxiliary source-sufficiency data, certified-N/A/fail/undefined faces, and isotropy remain available in the atlas but cannot refine a connected component unless already represented by the frozen packet/core.

Proof. B01–B09 give complete generated source coverage, including the explicit post-PCR-4 six-family routing theorem, explicit survivor/status systems, constructive source groupoids, frozen packet records, and chart functors into $\mathcal{N}_5$. B10 computes the complete degree-one homotopy fibre product, and B11 supplies all higher kernel objects and their coherence. Standard codescent of the Čech nerve of a functor to a groupoid reconstructs its full essential subgroupoid; here that subgroupoid is exactly $\mathcal{N}_{5,\text{osc}}^{\text{exec}}$ by the displayed membership condition. Because the source charts and all their status strata were explicitly classified before codescent, codescent performs only duplicate-presentation gluing.

By Theorem 5.5, two survivor presentations lie in the same connected component of $\mathcal{N}_5$ exactly when their frozen PCR-5A packets are equal, equivalently when they are SCIC-equivalent. B00 and Theorem 2.3 prevent auxiliary source-subset diagnostics from enlarging that identity relation. Therefore $\pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}$, with no merger of unequal packet classes and no refinement of a packet fibre. $\square$

15.7 Why codescent does not replace classification

For an arbitrary unclassified map $X \rightarrow Y$, the codescent of its kernel pair can reconstruct an image without resolving its internal structure. Here the classification precedes codescent:

1. source skeletons are generated by the explicit PCR-3 source grammar;
2. each skeleton has named source equations, E00–E23, family equations, and a complete applicability/N/A/fail/undefined/update stratification;
3. quotient arrows have explicit generators, relations, actions, and stabilizers;
4. packet moduli are solved frozen seven-channel records with canonical reduced standing cores;
5. the full source-subset sufficiency table is separated from the identity-bearing frozen deletion channel by Theorem 2.3;
6. every pairwise overlap is a well-typed homotopy fibre product over the concrete groupoid $\mathcal{N}_5$, with framed provenance and a genuine $\text{Aut}(n)$ -action;
7. all 42 distinct-label higher types and every repeated-label tuple are built by the same concrete fibre-product construction; and
8. B15 supplies the resulting core frames, overlap incidences, typed frontiers, and auxiliary source-composition annotations to the target-extension calculus without permitting downstream backflow.

Codescent is therefore the final gluing theorem, not a substitute for an unresolved family or overlap classification.

16 Exact moduli type, continuum lower bound, and four-class permanence

16.1 Discrete deletion grading

The frozen PCR-5A deletion channel induces a functor from the global atlas to the discrete groupoid of admitted kindwise no-repair profiles. Restricting to the LAW/new-local projection gives

$$\delta_{W \oplus N} : \mathfrak{A}_{\text{osc}}^{\text{exec}} \longrightarrow \{0, 1, \perp\}^2.$$

The four calibrated classes occupy the four finite values in $\{0, 1\}^2$. The atlas therefore has at least four nonempty deletion sectors, each of which can contain further discrete, continuous, functional, or categorical moduli.

For HYB source analysis one may additionally record the auxiliary finite source-subset sufficiency profile

$$N_x^{\text{aux}} : 2^{K(x)} \longrightarrow \{0, 1, \perp\}.$$

By Theorem 2.3, this auxiliary object is not an additional standing coordinate: a component that is already packet-factorized reports frozen standing content, a component that varies inside one frozen packet fibre is provenance/source-composition data, and a component requiring a new witness, continuation, or target primitive belongs to an update. The active block set varies with the finite source-incidence skeleton, so infinitely many finite combinatorial source-composition profiles can occur without creating extra base SCIC classes by label alone.

16.2 A uniform active-singlet survivor family

The cardinality result uses the constructive full-rank active-singlet lift already established in PCR-5 and represented by the standard heavy-singlet matching class [8]. Fix a full-rank target-admissible symmetric coefficient

$$\kappa_* = UDU^\top, \quad D = \text{diag}(d_1, d_2, d_3), \quad d_i > 0,$$

and choose a positive nonsingular diagonal singlet matrix M_0 . Set

$$Y_0 = UD^{1/2}M_0^{1/2}.$$

Then every column of Y_0 is active and

$$Y_0 M_0^{-1} Y_0^\top = \kappa_*. \quad (83)$$

PCR-5 proves that this construction, together with the inherited PCR-4 common vacuum, channel, response, matching, failure, and common-cone packet, gives an active LOCAL/HID survivor; the construction is uniform for every positive nonsingular diagonal choice of the heavy matrix [1]. No new empirical heavy-state or absolute-mass record is used.

For $\lambda > 0$, define

$$M_\lambda = \lambda M_0, \quad Y_\lambda = \sqrt{\lambda} Y_0. \quad (84)$$

Then

$$Y_\lambda M_\lambda^{-1} Y_\lambda^\top = Y_0 M_0^{-1} Y_0^\top = \kappa_*. \quad (85)$$

The active source graph, rank, singlet representation, anomaly contribution, target attachment, deletion type, and low-energy target-facing matrix are unchanged; only the physical threshold and the associated oriented matching/continuation data vary.

Lemma 16.1 (Uniform survivor interval). *Let $I_* = (1, 2) \subset \mathbb{R}_{>0}$. For every $\lambda \in I_*$, the point*

$$x_\lambda = (Y_\lambda, M_\lambda)$$

is a source-coherent, full-rank, active-singlet PCR-4 survivor in the same finite source-incidence skeleton.

Proof. For every $\lambda \in I_*$, M_λ is positive and nonsingular, Y_λ is full rank, and every singlet remains source active. Equation (85) preserves the exact target-admissible coefficient used by the PCR-5 constructive lift. Gauge representation, anomaly status, source graph, charged-current attachment, and source-constitutive incidence are unchanged. The imported PCR-5 construction is valid for every positive nonsingular diagonal heavy matrix and supplies the same PCR-4 common vacuum, target channel, visible response, oriented matching existence, failure disposition, and common-cone witness at the corresponding point. Thus each x_λ is a survivor. Since no node or incidence becomes source null, all points have the same finite source-incidence skeleton. $\square$

This is a direct parameter-uniform specialization of an upstream survivor construction, not an appeal to genericity, numerical sampling, or topological openness of an unspecified survivor locus.

16.3 Threshold separation

Lemma 16.2 (Physical scale is not lawful redescription). *If $\lambda \neq \mu$, the source points (Y_λ, M_λ) and (Y_μ, M_μ) are not related by a lawful LOCAL/HID basis, gauge, graph, or parameter-chart transformation preserving the physical matching threshold.*

Proof. A lawful singlet basis change acts by congruence on M and the corresponding right action on Y . It preserves the basis-invariant singular-value spectrum of the heavy threshold. Graph relabeling and parameter recharting preserve the same invariant, and gauge redescription is trivial on a singlet mass scale. Multiplying every nonzero threshold singular value by $\lambda/\mu \neq 1$ changes a physical source-to-effective matching coordinate relative to the fixed target scale. It is therefore not an identity-preserving redescription. $\square$

Lemma 16.3 (Packet separation along the scale family). *For $\lambda, \mu \in I_*$ with $\lambda \neq \mu$,*

$$\Xi_{\text{osc}}(x_\lambda) \neq \Xi_{\text{osc}}(x_\mu).$$

Proof. The target-face and low-energy response coordinates agree by (85). The oriented matching channel M^{match} , however, retains threshold and retained-resolution data by the frozen PCR-5A packet construction [6]. The basis-invariant threshold spectra of M_λ and M_μ differ by Lemma 16.2. Hence the matching channels, and therefore the complete packets, are unequal. $\square$

Corollary 16.4 (Continuum injection). *The map*

$$I_* \longrightarrow Q_{\text{osc}}^{\text{SCIC}}, \quad \lambda \longmapsto [x_\lambda]_{\text{SCIC}}$$

is injective. Consequently

$$|Q_{\text{osc}}^{\text{SCIC}}| \geq |I_*| = \mathfrak{c}.$$

Proof. Every x_λ is a survivor by Lemma 16.1. Distinct parameters have unequal packets by Lemma 16.3, hence unequal SCIC classes by the packet biconditional. Every nondegenerate real interval has cardinality $\mathfrak{c}$. $\square$

16.4 Why no absolute upper cardinal is a target theorem

PCR-3 permits arbitrary $\mathbb{U}$ -small continuous, discrete, representation-valued, functional, and categorical parameter objects. It imposes neither second countability nor finite-dimensionality, algebraicity, separability, or a real/complex-only parameter convention. Adding one of those restrictions in PCR-5B would narrow the already frozen comparison class.

Accordingly, the target licenses the exact groupoid-valued moduli atlas and the lower bound $|Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}$, but not a comparison-class-independent assertion $|Q_{\text{osc}}^{\text{SCIC}}| = \mathfrak{c}$. Under a separately declared standard-analytic restriction in which the skeleton index is countable and each residual parameter object has cardinality at most $\mathfrak{c}$, equality would follow; that restriction is not part of the present target and is not used.

16.5 B13: exact cardinality/moduli theorem

The global classification object has already been calculated in B12 as the codescent of the resolved chart groupoids over the concrete universal standing-core groupoid:

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} = \text{codesc}_{[k] \in \Delta^{\text{op}}} \mathcal{C}_k^{\text{exec}} \simeq \mathcal{N}_{5, \text{osc}}^{\text{exec}}, \quad \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}.$$

Its objects are solved seven-channel standing cores reached by the displayed family systems; its arrows are concrete pointed continuation-equivariant core isomorphisms induced from the constructive source groupoids. Pairwise and higher gluing is performed by homotopy fibre products over $\mathcal{N}_5$, with framed provenance supplying the literal common-core automorphism action. No unspecified image functor and no automorphism action on an unframed isomorphism class occurs in the endpoint.

Theorem 16.5 (Exact categorical moduli and cardinality theorem). *[B13] The complete oscillation standing-carrier space is the coarse class space of the executed atlas*

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} = \text{codesc}_{[k] \in \Delta^{\text{op}}} \mathcal{C}_k^{\text{exec}} \simeq \mathcal{N}_{5, \text{osc}}^{\text{exec}}, \quad Q_{\text{osc}}^{\text{SCIC}} \cong \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}). \quad (86)$$

For every family/skeleton code the contributing moduli object is the explicit solution quotient of E00–E23, its family equations, the seven frozen standing-record equations, and the constructive source action group. The chart-to-core functor lands in $\mathcal{N}_5$. Every pairwise skeleton fibre and every higher family tuple is the corresponding homotopy fibre product over $\mathcal{N}_5$, with framed provenance, exact empty/nonempty and first-separator status, failure/update data, isotropy, and incidence as calculated in B10–B11. The deletion field used by these equality tests is exactly $\Delta_{\text{fr}}^{\text{del}}$; any larger source-subset profile remains auxiliary subject to the conservativity theorem.

The object is $\mathbb{U}$ -small and contains the injective interval family of Corollary 16.4. Therefore

$$|Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c},$$

and exact four, every finite endpoint, and every countably infinite endpoint are false. Equation (86), rather than a finite list or an unsupported absolute upper cardinal, is the exact parameterized/categorical endpoint.

Proof. B01 gives generated source-skeleton coverage, the explicit post-PCR-4 six-family routing theorem, explicit residual/status systems, constructive source groupoids, and chart functors to the concrete standing-core groupoid. B02–B08 instantiate those objects for every family while Theorem 2.3 prevents auxiliary subset diagnostics from refining the frozen quotient. B10 calculates all fifteen family-pair

systems as homotopy fibre products over $\mathcal{M}_5$; B11 instantiates all 42 distinct-label higher-overlap types and the universal repeated-label finite-tuple construction with common-core frames. B12 proves that their codescent is the resolved standing-core groupoid and that its connected components are SCIC classes. The source universe convention is preserved by the finite products, solution subobjects, action groupoids, U-small coproducts, and codescent used in the construction. The continuum injection gives the cardinal lower bound. No upper cardinal follows without adding a size restriction absent from the frozen comparison class. $\square$

16.6 B14: four-class permanence

Theorem 16.6 (Four-class permanence). *[B14] The executed global atlas contains solved packet/core points representing*

$$q_W, \quad q_{3N}, \quad q_{\text{red}}, \quad q_{\text{syn}},$$

and they lie in four distinct fibres of the frozen LAW/new-local projection $\delta_{W \oplus N}$ of $\Delta_{\text{fr}}^{\text{del}}$. No self-overlap, cross-family overlap, higher coherence map, or categorical quotient identifies any pair. The four classes remain distinguished inside the continuum-sized quotient.

Proof. Atlas coverage includes the four PCR-5A survivor points. Their frozen kindwise deletion projections are the four distinct values $(0, 1), (1, 0), (1, 1), (0, 0)$. Every generating morphism of the codescent object preserves the complete packet and therefore those frozen deletion values. No overlap chain can connect two different pairs, and B12 sends them to four distinct connected components. $\square$

16.7 Strengthened realization diagnosis

PCR-5 established a terminal over-residue from at least two classes, and PCR-5A strengthened the lower bound to four. B13 shows that the fixed oscillation target admits at least continuum many standing carrier classes. The terminal role set remains the one complete-carrier role fixed upstream. No target-internal realized-carrier trace or same-scope impossibility theorem selects one class.

Corollary 16.7 (Oscillation-target realization status). *At the fixed oscillation target,*

$$\neg \text{RealizationLicensed}(T_{\text{osc}}),$$

and the realization-level endpoint is irreducible structural underdetermination with at least continuum standing carrier classes.

This is not a statement that continuum many particles, neutrino species, or propagation eigenstates exist. It is a classification of standing-distinct complete carrier realizations of one target-relative terminal role.

17 Robustness, falsifiability, and reopening conditions

17.1 Admissible challenges

A substantive objection must preserve the fixed target and do at least one of the following: construct a faithful same-regime counterexample; defeat a named implication; expose candidate-domain incompleteness; identify an omitted standing-bearing coordinate; show that a purported stratum changes the target; exhibit an ill-typed quotient, overlap, or codescent construction; or show that an asserted classification has not been semantically resolved. Preferences based on familiarity, field count, simplicity, or naturalness do not address the theorem.

17.2 Load-bearing chain

The principal implications are

PCR-3 finite source completeness $\implies$ finite active source-incidence skeletons
 $\implies$ exact skeleton-indexed PCR-4 solution objects
 $\implies$ lawful quotient groupoids and family atlases
 $\implies$ standing-core self, pairwise, and higher overlaps
 $\implies \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}$
 $\implies$ exact categorical moduli endpoint,
 active-singlet scale family $\implies I_* \hookrightarrow Q_{\text{osc}}^{\text{SCIC}}$.

Any successful challenge must identify the implication it defeats.

17.3 Adversarial map

Challenge	Required construction	Consequence
Target replacement	Show that a B02–B14 construction changes an inherited record, boundary, continuation, or comparison coordinate while being treated as a base stratum.	The object belongs to a successor target; B00 excludes it from PCR-5B.
Candidate omission	Exhibit a finitely presented same-target active carrier architecture absent from every PCR-3 source skeleton and HYB incidence class.	PCR-3 and B01 reopen.
Packet under-resolution	Exhibit two survivors with equal atlas packet objects but unequal admitted continuation signatures.	PCR-5A packet completeness or B09 fails.
Packet over-resolution	Exhibit an atlas separation invisible to every admitted packet evaluator after every licensed continuation.	The distinction must collapse under B09/SCIC.
Auxiliary-deletion promotion	Exhibit two points with equal frozen packet that are separated solely by an auxiliary source-subset sufficiency profile.	Theorem 2.3 fails, reopening B00/B08/B09.
Survivor splicing	Assemble one atlas point from marginally successful values belonging to different source points.	The object is excluded by Lemma 2.7.
Finite sampling	Replace a symbolic family solution object by finitely many named examples.	The relevant family theorem remains unproved.
Hidden parameter collapse	Erase a continuous, functional, representation-valued, or categorical packet modulus.	The corresponding family atlas and B13 reopen.
Family-name counting	Treat LAW, LOCAL, HID, COMP, SOL, and HYB as six carrier identities.	Rejected: the families are overlapping charts.

Challenge	Required construction	Consequence
Overlap undercount	Exhibit equal complete packets in two family charts absent from $\mathfrak{D}_{F,G}$.	B10 fails.
Overlap overcount	Exhibit an overlap object whose complete packets differ.	B10 soundness fails.
Unframed automorphism action	Exhibit an overlap or higher object that invokes $\text{Aut}(n)$ without a concrete core and chosen frame.	The framed decomposition and B05/B10–B12 reopen.
Higher cocycle failure	Exhibit a triple overlap whose packet isomorphisms do not compose in $\mathcal{N}_5$.	B11 fails.
Higher-arity omission	Exhibit a standing-bearing finite hyperedge or coherence cell removed by HYB reduction.	B08, and possibly PCR-3 incidence completeness, reopen.
Dilation overcollapse	Identify two HID points with equal visible channels but different retained hidden continuation, ownership, or failure packets that B04 merges.	B04/B05 fail.
Dilation undercollapse	Retain two source-authorized isometric dilations with equal complete active trace and packet as distinct SCIC classes.	B04/B09 require collapse.
Matching backflow	Infer unique local, hidden, or composite source identity from equality of a low-energy coefficient.	Rejected: matching is oriented and overlap requires the full packet.
Threshold-scaling objection	Show that the family M_λ, Y_λ fails the fixed target on every interval or that threshold variation is absent from the standing packet.	Lemma 16.1 or 16.3 fails; B13 reopens.
Scale-as-basis objection	Construct a lawful source or gauge transformation rescaling all physical threshold singular values by $\lambda/\mu \neq 1$ while preserving the fixed scale reference.	Lemma 16.2 fails.
Naturalness selection	Prefer one stratum because it has fewer fields, smaller couplings, lower operator order, or more familiar dynamics.	Outside the standing target.
Certification substitution	Treat supplementary tables, hashes, or computational encodings as the completeness proof.	Rejected: B01–B13 remain the semantic basis.
Downstream backflow	Allow PCR-5C/D or PCR-6 terminology to alter base membership or SCIC equivalence.	Rejected by B00 and B15.

17.4 Direct falsifiers

A same-target PCR-4 survivor whose occupancy-bearing active trace has no finite source-incidence skeleton, or a packet-bearing distinction removed by every proposed source-skeleton, core, and groupoid reduction.

A survivor in the relevant family whose source semantics, packet-bearing discrete type, residual solution object, applicability or active-defect frontier, or lawful morphism is absent from the stated atlas.

A duplicate packet presentation missing from the overlap nerve, an overlap connecting unequal packets, a comparison arrow outside $\mathcal{N}_5$, an ill-defined automorphism action on an unframed fibre, or an SCIC class absent from or duplicated in the coarse codescent object.

An auxiliary source-subset deletion diagnostic that changes terminal carrier identity while the frozen PCR-5A packet remains equal and no target, source, or continuation update occurs.

Failure of the scale family to contain a nonempty survivor interval, lawful equivalence of distinct threshold scales, or omission of the threshold and matching difference from the inherited standing packet.

A lawful packet isomorphism between two of the four distinct frozen deletion values.

17.5 Representative stress tests

The following variations isolate common failure modes without replacing the proofs:

1. add a source-null spectator and verify that no new carrier stratum appears;
2. perturb a packet-bearing threshold and verify a change in the matching packet;
3. perform a lawful local-basis relabeling and verify quotient invariance;
4. replace a minimal HID dilation by an isometric presentation and verify collapse;
5. change hidden ownership while holding the visible channel fixed and verify separation;
6. splice channel and anomaly data from different points and verify survivor rejection;
7. remove a higher-arity HYB relation that supports the common cone and verify failure;
8. duplicate one packet across family charts and verify one global connected component;

9. force unequal frozen deletion profiles into one overlap and verify rejection;
10. add a Majorana or cosmological datum and verify classification as a target extension rather than a base-atlas modification;
11. replace an exact solution object by sampled representatives and verify failure of the exact-classification criterion;
12. compare a lawful basis congruence, which preserves the packet, with a physical threshold rescaling, which does not;
13. vary only an auxiliary source-subset sufficiency entry at fixed seven-channel packet and verify that no base SCIC split occurs; and
14. remove the frame from a fixed-core overlap, which destroys the literal automorphism action, then restore it and recover the well-typed postcomposition action.

17.6 Reopening map

A defect is assigned to the narrowest load-bearing locus:

Observed defect	Required reopening
Missing source sort or incidence	PCR-3 source ontology and comparison class, followed by B01 onward.
Missing simultaneous relation or common-point constraint	PCR-4 survival theory, followed by recomputation of the affected solution objects and overlaps.
Missing standing evaluator or continuation	PCR-5/SCIC activation and PCR-5A packet completeness.
Auxiliary deletion asserted as base identity	Apply Theorem 2.3; the datum must factor through the frozen packet, remain provenance, or define an update.
Incorrect source reduction or groupoid	B01/B09 and the affected family atlas.
Missing pairwise or higher overlap	B10/B11/B12, unless the omission originates in family incompleteness.
Unframed or representative-dependent overlap action	B05/B10/B11/B12; replace the coarse-class action by the concrete $\mathcal{N}_5$ fibre product with framed provenance.
Invalid continuum family	B13 only; the categorical atlas may remain valid while the cardinal lower bound changes.
New neutrino phenomenon	PCR-5C target extension, not PCR-5B.

17.7 Supplementary verification boundary

Supplementary stress tests can confirm selected consequences and expose regressions in the encoded data. They cannot establish source completeness, standing identity, overlap exactness, or the continuum theorem. Those conclusions remain consequences of the manuscript's mathematical arguments.

18 Completeness of the classification

The exact-atlas conclusion depends on several distinctions that cannot be supplied by codescent alone. Source formation must precede survivor inspection; applicability must precede predicate evaluation; source equivalence must remain distinct from terminal SCIC identity; auxiliary source-composition data must not refine the frozen packet; and overlap groupoids must be formed over concrete standing cores. This section collects the resulting completeness statements and identifies the theorem supporting each one.

18.1 Source, survivor, and family completeness

Obligation	Mathematical resolution	Primary result
Independent source index	The finite source-skeleton grammar is formed from PCR-3 node, incidence, activity, attachment, and coherence data before any PCR-4 predicate is evaluated.	Theorem 3.5
Six-family survivor coverage	LAW, LOCAL, HID, COMP, and SOL are feature charts; source-semantic HYB includes non-trivial gluing, auxiliary carrier-bearing sectors, and arbitrary higher-arity relations. PCR-4's pure TOP/EXT/BDY/ENV reductions route every surviving auxiliary occurrence through HYB.	Theorem 3.6
Exact PCR-4 survivor locus	Every relation instance receives its applicability predicate before its truth value. Certified N/A is retained, active pass/fail is preserved, and an undefined active evaluation is fail closed. The guarded solution object is equivalent to the inherited PCR-4 survivor object.	Theorems 4.2 and 4.4
Complete complement	Every raw point lies in a survivor stratum with certified N/A , an active fail/undefined stratum, a source-update stratum, or a target-update stratum. Mixed fail-plus-undefined patterns are retained rather than reduced to a first-failure label.	(15)
Family-specific resolution	The universal relations are specialized to explicit LAW, LOCAL, HID, COMP, SOL, and HYB source equations, standing normal forms, and typed frontiers.	B02–B08

18.2 Identity, descent, and overlap completeness

Obligation	Mathematical resolution	Primary result
Frozen carrier identity	The seven-channel PCR-5A packet remains the identity invariant. The identity-bearing deletion entry is exactly $\Delta_{\text{fr}}^{\text{del}}$; finer source-subset sufficiency data either factor through the frozen packet, remain provenance, or require an update.	Theorem 2.3
Constructive lawful descent	Each source chart has an action groupoid generated by the appropriate source-hypergraph isomorphisms, source automorphisms, fixed-boundary gauge and basis transformations, parameter reparameterizations, and family-specific isomorphisms. Isotropy is the stabilizer of the full solution.	B09
Represented packet moduli	The family moduli are defined by their functor of points: source solutions together with solved seven-channel standing data, modulo the lawful action. The fibre over one standing core is provenance rather than a second identity classifier.	Theorems 4.6 and 4.7
Well-typed pairwise overlap	Every chart maps to the concrete standing-core groupoid $\mathcal{N}_5$; overlap is a homotopy fibre product over that groupoid. Framed provenance gives the literal $\text{Aut}_{\mathcal{N}_5}(n)$ -action required by fixed-core decompositions.	Theorems 4.11 and 14.2
Higher-overlap coherence	Every nonempty higher overlap contains one simultaneous tuple and one common concrete core. Face, degeneracy, permutation, and co-cycle maps are induced by projection, identity, permutation, and composition in $\mathcal{N}_5$.	Theorem 15.1
Global assembly	Codescent is applied only after the chart domains, residual systems, actions, standing cores, and overlaps are resolved. It therefore identifies duplicate presentations rather than standing in for an omitted classification.	Theorem 15.2

18.3 Cardinality, permanence, and downstream finality

The active-singlet scaling family proves a continuum injection independently of the atlas assembly, while the four PCR-5A classes remain separated by their frozen deletion projections. These results establish

$$|Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}, \quad q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}} \text{ pairwise distinct.}$$

B15 then supplies the complete base-space object required by the target-extension calculus. No downstream characterization may alter base membership or SCIC equivalence; a richer neutrino phenomenon can only eliminate, preserve, or split the already classified base classes through an explicit target extension.

18.4 Semantic proof and supplementary certification

The preceding conclusions are mathematical statements about the source grammar, guarded survivor equations, lawful groupoids, standing-core functors, and their overlaps. Supplementary tables and computational checks may expose omissions, inconsistent identifiers, or broken incidence relations, but they do not establish semantic source completeness, packet completeness, or uniqueness of the standing quotient. Those burdens are discharged only by B00–B15 and their imported hypotheses.

19 Interface to the PCR-5C target-extension calculus

19.1 Why a universal base-stratum datum is required

PCR-5C forms target-extension restrictions and lift fibres over exact base classes and moduli strata. Because the PCR-5B source grammar ranges over arbitrarily many finite source skeletons, the required interface cannot be a finite enumeration of all σ . It must specify a uniform datum for every admissible source skeleton, together with the distinguished base points and the finite family-label overlap types that organize the global atlas.

Five features are retained explicitly. First, the identity-bearing deletion field is the frozen PCR-5A channel $\Delta_{\text{fr}}^{\text{del}}$; any full source-subset sufficiency profile remains a separate source-composition datum. Second, overlap provenance is framed over a concrete object of the universal standing-core groupoid $\mathcal{N}_5$. Third, every relation retains its pre-evaluation applicability predicate and, when theoretically absent, its certified N/A status. Fourth, every no-update point carries the exact triple $(\mathbf{N}_{\text{NA}}, \mathbf{F}_{\text{fail}}, \mathbf{U}_{\text{undef}})$. Fifth, HYB ranges over the complete source-semantic hybrid domain, including surviving TOP, EXT, BDY, and ENV components and non-Compose cross incidences.

19.2 Universal base-stratum datum

For every $F \in \mathfrak{F}_{\text{osc}}$ and $\sigma \in \Sigma_F^{\text{gram}}$, define

$$\mathbf{H}_{F,\sigma} = \left(\begin{array}{l} \text{id}_{F,\sigma}, F, \sigma, \Theta_{F,\sigma}^{\text{raw}}, \mathcal{E}_{F,\sigma}^{\text{surv}}, \mathbf{A}_{F,\sigma}, \widehat{\text{Stat}}_{F,\sigma}, \\ \Gamma_{F,\sigma}, p_{F,\sigma}, \text{Iso}_{F,\sigma}, \mathcal{M}_{F,\sigma}^{\text{exec}}, \widetilde{\text{Prov}}_{F,\sigma}, \\ \Delta_{\text{fr}}^{\text{del}}, \mathcal{U}_{F,\sigma}^{\text{aux}}, \partial \text{Fail}_{F,\sigma}, \text{Sep}_{F,\sigma}, \\ \text{Ov}_{F,\sigma}^{(2)}, \text{Ov}_{F,\sigma}^{(\geq 3)}, \text{reopen}_{F,\sigma} \end{array} \right), \quad (87)$$

where $\mathcal{U}_{F,\sigma}^{\text{aux}}$ is empty unless the source chart carries a nontrivial auxiliary source-composition profile; in HYB it contains $(N^{\text{aux}}, \mathcal{S}^{\text{aux}})$.

Carrier identity is the connected component of the concrete standing core in $\mathcal{N}_5$, not an auxiliary annotation or source label. The datum $\mathbf{H}_{F,\sigma}$ contains:

1. the finite source skeleton and raw source parameter object;
2. the complete E00–E23 system together with the family equations;
3. the applicability vector, certified N/A data, and the exact survivor/active-defect/source-update/target-update stratification;
4. the lawful source groupoid, its action and isotropy, and the chart functor $p_{F,\sigma} : \Gamma_{F,\sigma} \rightarrow \mathcal{N}_5$;
5. the solved seven-channel standing core and its framed provenance fibres;

6. the frozen deletion channel together with any separately typed auxiliary source-composition profile;
7. the complete applicability/fail/undefined frontier and the theorem that governs each reopening;
8. a finite first-difference separator for every adjacent or compared standing core; and
9. every pairwise and higher homotopy-fibre-product incidence containing the stratum.

This datum is determined functorially from the source skeleton and the fixed theorem chain; it does not depend on a preferred endpoint.

19.3 Distinguished finite incidence families

In addition to the universal family datum, PCR-5C receives:

1. the six family-indexed strata $H_{F,\sigma}$, quantified over every $\sigma \in \Sigma_F^{\text{gram}}$;
2. the four marked points $q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}}$, with their frozen deletion values and overlap incidences;
3. the fifteen distinct-family pairwise incidence types supplied by B10; and
4. the forty-two distinct-label higher-incidence types supplied by B11.

Repeated labels are governed by the finite-tuple construction (79); no finite list is substituted for the unbounded source-skeleton index.

19.4 B15: exact base-space interface theorem

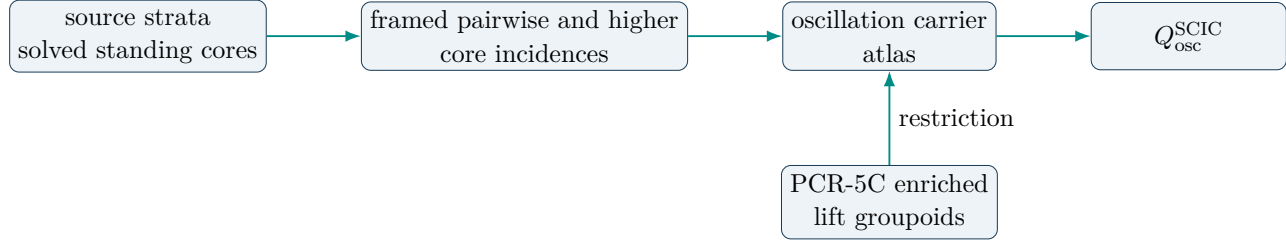
Theorem 19.1 (Exact populated base-space handoff). *[B15] The record constructor (87), the six family record families, the four marked-point records, the fifteen pairwise records, the forty-two distinct-label higher records, and the repeated-label finite-tuple constructor reconstruct $\mathfrak{A}_{\text{osc}}^{\text{exec}}$ and therefore $Q_{\text{osc}}^{\text{SCIC}}$. Every PCR-4 survivor has a proved principal-family route and every generated chart code has a determined residual system, exact applicability-preserving six-valued compiler partition whose scientific projection preserves the PCR-4 four-valued disposition, lawful source groupoid, chart-to-core functor, solved frozen packet record, framed provenance fibre, certified-N/A/fail/undefined frontier, separator policy, overlap incidence, and auxiliary source-composition annotation where applicable. PCR-5C may use these records to form restriction and lift fibres but may not mutate their base packet membership or SCIC equivalence.*

Proof. For each source skeleton, B01 supplies the grammar, applicability-guarded residual system, source groupoid, solved frozen packet, and chart functor. [Theorem 4.4](#) identifies the guarded survivor object with the inherited PCR-4 survivor locus, including the same certified N/A and active Pass/Fail dispositions, while [Theorem 4.2](#) forbids active failures from being reclassified as inapplicable. [Theorem 3.6](#) proves that every PCR-4 survivor enters at least one of the six family charts. B02–B08 determine the family equations and source annotations, and [Theorem 2.3](#) separates the frozen identity channel from auxiliary source-subset information. B09 supplies lawful descent and isotropy. B10 supplies all fifteen pairwise homotopy fibre products with framed provenance; B11 supplies all forty-two distinct-label higher-incidence types and the repeated-label finite-tuple construction. B12 proves that their codescent is the global atlas and that its connected components are $Q_{\text{osc}}^{\text{SCIC}}$. B14 supplies the four permanent marked points.

A lawful target extension retains the old witness algebra, so restriction of an enriched standing core lands in one object of the frozen base atlas. Changing a base packet value, promoting an auxiliary source-subset distinction into a retroactive base separator, or identifying separator-distinct base cores would reinterpret the original target and would therefore be a target replacement rather than an extension. $\square$

19.5 Restriction-ready structure

The relation among the objects is summarized by



Restriction is formed on source-presentation and concrete standing-core groupoids before descending to the coarse quotient.

19.6 Base selection and enriched singleton selection

The interface preserves the distinction

$$|\mathrm{im} \rho_j| = 1 \not\iff |Q(T_1)| = 1.$$

A target extension may select one base standing-core sector while its enriched fibre remains finite, continuous, functional, or categorical. A singleton enriched quotient requires both a singleton base image and a singleton lift fibre.

19.7 No-merger consequence

Every pair of distinct base cores has a finite first-difference separator in the frozen packet. A genuine target extension retains that witness and therefore cannot merge the corresponding base classes. It may eliminate a base class, preserve it, or split it into enriched classes. A source-subset distinction that was standing-inert at the base target may become standing-bearing only in the richer target; when it does, it produces a downstream split rather than a retroactive refinement of the base quotient. Coarsening the base quotient requires changed target semantics and a new certificate.

20 Conclusion

PCR-5A fixes the identity object classified in this paper: the seven-channel standing packet whose equality is exactly SCIC equality, together with four permanent marked classes. PCR-5B neither changes that quotient relation nor enlarges the oscillation target. In particular, its identity-bearing deletion field remains the frozen PCR-5A kindwise no-repair coordinate $\Delta_{\mathrm{fr}}^{\mathrm{del}}$. Finer source-subset sufficiency data are retained only as auxiliary source-composition information unless their consequences already factor through the frozen packet.

The classification begins before survivor inspection. Every finite occupancy-bearing source architecture in the PCR-3 comparison class determines a finite source skeleton formed from the node, incidence, activity, attachment, and coherence structure. For each skeleton, the source equations and E00–E23 define the exact simultaneous solution object. Pre-evaluation applicability separates theorem-certified N/A from active pass, fail, or undefined evaluation; the resulting guarded survivor object is exactly the PCR-4 survivor locus, and no active failure can be reclassified as inapplicable. The raw parameter space is therefore partitioned into survivor strata with certified N/A data, exact active fail/undefined strata, source updates, and target updates.

Each family carries an explicit lawful action. Source-hypergraph isomorphisms, source automorphisms, fixed-boundary gauge and basis changes, parameter reparameterizations, and the applicable dilation, constituent, profile, gluing, and higher-relation isomorphisms generate the groupoid; naturality, incidence preservation, fixed-boundary covariance, and semidirect compatibility supply its relations. Isotropy is the stabilizer of the complete source solution and standing core. Transformations that alter retained threshold, memory, matching orientation, boundary ownership, deletion response, or failure data are not treated as presentation equivalences.

The packet moduli are represented by their functor of points. An S -point is an S -family of source solutions together with solved T, R, D, C, M, B, F data, modulo the lawful action. The fibre over one standing core is the provenance groupoid. LAW, LOCAL, HID, COMP, SOL, and HYB specialize this universal construction to process, active-rank, dilation/support, constituent/derivation, profile/endability, and finite decorated-hypergraph variables. The HYB domain is source-semantic and includes auxiliary carrier-bearing kinds, non-Compose cross incidences, and irreducible higher-arity relations; the family-routing theorem proves that these six charts cover every PCR-4 survivor.

Overlap closure is both parameter-resolved and categorically well typed. Every chart a carries a functor $p_a : \Gamma_a \rightarrow \mathcal{N}_5$ to the concrete pointed standing-core groupoid, and each pairwise overlap is

$$\Gamma_a \times_{\mathcal{N}_5}^h \Gamma_b.$$

Its objects contain the actual isomorphism between concrete standing cores. Over a chosen core n , framed provenance makes the $\text{Aut}_{\mathcal{N}_5}(n)$ -action literal. All fifteen distinct-family pairwise types and all forty-two distinct-label higher types are resolved, while repeated labels are handled by the same finite-tuple construction. Every nonempty higher overlap contains one simultaneous source tuple, one common concrete core, and compatible frames, so pairwise compatibility cannot be spliced into a false higher intersection.

The global atlas is assembled only after its chart and overlap objects have been determined:

$$\mathfrak{A}_{\text{osc}}^{\text{exec}} = \text{codesc}_{[k] \in \Delta^{\text{op}}} \mathcal{C}_k^{\text{exec}}, \quad \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}.$$

Codescent therefore performs duplicate-presentation gluing rather than standing in for an unresolved family image. B15 supplies this exact base space to the target-extension calculus while forbidding any downstream change to its membership or equivalence relation.

The cardinal result is independent of the atlas assembly. The full-rank active-singlet family

$$M_\lambda = \lambda M_0, \quad Y_\lambda = \sqrt{\lambda} Y_0, \quad \lambda \in (1, 2),$$

preserves the fixed low-energy coefficient while changing the physical threshold and oriented matching data already retained by the standing packet. Distinct values of λ therefore define distinct SCIC classes:

$$(1, 2) \hookrightarrow Q_{\text{osc}}^{\text{SCIC}}, \quad |Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}.$$

Exact four, every other finite cardinality, and countable infinity are excluded. Because PCR-3 admits arbitrary universe-small source parameter objects, no comparison-class-independent absolute upper cardinal follows. The exact answer is the represented mixed moduli atlas together with its continuum lower bound.

The four PCR-5A classes remain distinct because their frozen LAW/new-local deletion projections are $(0, 1), (1, 0), (1, 1), (0, 0)$. They are four marked points in a much larger carrier space, not four particles

or four neutrino species. The oscillation target fixes one terminal complete-carrier role while leaving at least continuum many standing-distinct complete realizations. No target-internal realized-carrier trace or same-scope impossibility theorem selects one of them. The realization endpoint is therefore continuum-strength structural underdetermination at the fixed oscillation target.

PCR-5B capstone

$$\begin{aligned} \pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) &\cong Q_{\text{osc}}^{\text{SCIC}}, & |Q_{\text{osc}}^{\text{SCIC}}| &\geq \mathfrak{c}, \\ q_W, q_{3N}, q_{\text{red}}, q_{\text{syn}} &\text{ remain pairwise distinct,} \\ &\neg \text{RealizationLicensed}(T_{\text{osc}}). \end{aligned}$$

The exact oscillation standing-carrier space is a mixed parameterized and categorical moduli atlas. Later neutrino phenomena may eliminate, preserve, or split its classes only through explicit target extensions.

A Notation and type hierarchy

Symbol	Meaning
T_{osc}	Fixed neutrino-oscillation target inherited from PCR-1/PCR-2.
A_{osc}	PCR-4/PCR-5 compatible carrier-survivor family at terminal carrier resolution.
$Q_{\text{osc}}^{\text{SCIC}}$	Final SCIC standing-carrier quotient $A_{\text{osc}}/\sim_{\text{SCIC}}$.
Ξ_{osc}	Complete seven-channel packet (T, R, D, C, M, B, F) , with $D = \Delta_{\text{fr}}^{\text{del}}$ fixed by PCR-5A.
$\mathfrak{F}_{\text{osc}}$	Six principal family charts $\{\text{LAW}, \text{LOCAL}, \text{HID}, \text{COMP}, \text{SOL}, \text{HYB}\}$.
U_5^{stand}	Licensed total fail-closed same-target continuation monoid.
W_5^{stand}	Target-complete standing witness family.
Beh_x	Complete continuation/output behavior of survivor x .
$\text{Core}_5(x)$	Canonical reachable future-separated pointed reduced standing core.
$\mathcal{R}_5$	Groupoid of reduced standing cores and pointed output-preserving continuation-equivariant isomorphisms.
$\mathcal{N}_5$	The same concrete groupoid, used as the universal PNF/core gluing base for all overlap fibre products.
$\mathcal{V}_{\Xi}$	Groupoid of complete packet objects and packet-preserving isomorphisms.
Θ_F^{part}	Source-coherent active parameter object before PCR-4 survival.
Θ_F^{surv}	Exact PCR-4 simultaneous survivor object.
$\mathcal{C}_F^{\text{surv}}$	Groupoid of exact family survivor presentations and lawful morphisms.
σ	Finite reduced source-incidence skeleton.
Σ_F^{gram}	Independently generated PCR-3-derived source skeletons for family F , before survivor evaluation.

Symbol	Meaning
$Z_{F,\sigma}$	Exact residual PCR-4 survivor solution object for skeleton σ .
$\Gamma_{F,\sigma}$	Full lawful reparameterization/presentation groupoid.
$[Z/\Gamma]$	Action or quotient groupoid, retaining isotropy.
$\beta_{F,\sigma}$	Reduced-core functor on one source-skeleton chart.
$\mathcal{P}_F$	Family presentation atlas $\coprod_{\sigma}[Z_{F,\sigma}/\Gamma_{F,\sigma}]$.
$\mathcal{P}_{\text{osc}}$	Complete presentation groupoid $\coprod_{F \in \mathfrak{F}_{\text{osc}}} \mathcal{P}_F$.
β	Global reduced-core functor $\mathcal{P}_{\text{osc}} \rightarrow \mathcal{R}_5$.
$\mathfrak{D}_{F,G}$	Exact well-typed core/packet-isomorphism fibre product $\mathcal{P}_F \times_{\mathcal{N}_5}^h \mathcal{P}_G$.
$\mathcal{C}_k$	$(k+1)$ -fold self/cross kernel object of β .
$\mathfrak{A}_{\text{osc}}$	Codescent groupoid of the resolved Čech nerve over $\mathcal{N}_5$.
π_0	Connected-component/coarse-class functor π_0 .
Prov	Unframed source-presentation provenance above a reduced core.
$\widetilde{\text{Prov}}$	Framed provenance fibre $(x, \phi : p(x) \xrightarrow{\sim} n)$; $\text{Aut}(n)$ acts on the chosen frame by postcomposition.
∂Fail	Typed same-target fail/undefined frontier object; source/target updates are separately typed.
N_x^{aux}	Full no-repair source-subset sufficiency profile of a finite hybrid, retained as auxiliary source-composition data unless its consequence already factors through the frozen packet.
$\text{MinSupp}^{\text{aux}}(x)$	Antichain of minimal sufficient source subsets in the auxiliary source-composition profile.
I_*	Explicit nondegenerate survivor interval $(1, 2)$ in the active-singlet scaling family.
$\mathbb{U}$	Fixed Grothendieck universe of the PCR-3 comparison class.
$\mathfrak{c}$	Cardinality of a nonempty real interval.

A.1 Identity hierarchy

The paper uses the strict hierarchy

$$\begin{aligned} \text{source presentation} &\longrightarrow \text{source-equational instance} \longrightarrow \text{PCR-4 survivor} \\ &\longrightarrow \text{reduced standing core/packet} \longrightarrow \text{SCIC class.} \end{aligned}$$

Source-presentation equality is finer than candidate identity; candidate identity is finer than interface equality; reduced-core isomorphism and packet equality are exactly SCIC identity at the terminal oscillation target. The atlas retains the finer levels as provenance and groupoid structure without allowing them to refine the connected-component standing quotient.

A.2 Status codomain

Every instantiated PCR-4 relation instance first carries the frozen pre-evaluation applicability predicate and then uses the applicability-aware fail-closed evaluation codomain

$$\widehat{\text{Val}} = \{\text{NA}, \text{pass}, \text{fail}, \text{undefined}, \text{source update}, \text{target update}\}.$$

Here NA is permitted only when the corresponding relation is theoremtically absent under the frozen PCR-4 activity predicate and carries its upstream N/A certificate. On active relation instances, `pass` and `fail` reproduce the PCR-4 scientific Pass/Fail disposition; `undefined` is a PCR-5B undefined evaluation rather than a scientific N/A. Source and target updates leave the fixed source or target fibre respectively.

A status is refined by a named evaluator or residual, an exact parameter locus, the applicability certificate, and the theorem locus that would have to reopen for the status to change. On the no-update branch, the complete status object is the ordered triple

$$\widehat{D}(\theta) = (N(\theta), F(\theta), U(\theta)),$$

with

$$N = \{i : \widehat{e}_i = \text{NA}\}, \quad F = \{i : \widehat{e}_i = \text{fail}\}, \quad U = \{i : \widehat{e}_i = \text{undefined}\}.$$

The active sets F and U are disjoint, while N records certified inapplicability rather than failure. A no-update point is a survivor exactly when $F = U = \emptyset$, every member of N has its theorem-certified N/A witness, and the guarded source/family equations hold. Thus a survivor may carry a nonempty N ; every no-update point with $F \cup U \neq \emptyset$ is a typed nonsurvivor stratum.

B Supplementary proofs for the source-skeleton theorem

B.1 Finite support and finite skeleton size

Lemma B.1 (Finite active support). *For every PCR-3 candidate instance in the comparison class, the occupancy-bearing active carrier core has finitely many source nodes, active incidences, target attachments, and source-constitutive coherence cells.*

Proof. This is the finite occupancy-bearing presentation condition of PCR-3. Parameter domains and multiplicity variables can be infinite as family objects, but each instantiated candidate has a finite carrier-incidence description at target-relative resolution. Pointwise activity reduction deletes exactly null incidences before the core is formed. $\square$

Lemma B.2 (Finite packet-bearing type support). *For one active candidate instance, only finitely many discrete type and incidence entries are required to specify the reduced source-incidence skeleton.*

Proof. The active source core and witness diagram are finite. Each node and hyperedge carries a finite tuple of source and packet types. The complete continuation value can be infinite as a function object, but its source/codomain and finite incidence type belong to the skeleton while the actual function lies in the residual moduli. The same separation applies to representation, matrix, kernel, profile, and categorical values. $\square$

B.2 Reduction independence

The reduction relation is generated by:

1. source-null deletion;
2. target-inert spectator removal;
3. finite graph/hypergraph isomorphism;
4. source-equational coherence replacement;

5. lawful basis, gauge, and chart transformation; and
6. proved source-equational or source-presentation equivalence within the complete PCR-3 carrier trace.

Proposition B.3 (Weak confluence up to lawful isomorphism). *Any two finite reduction sequences starting from one survivor presentation terminate in reduced skeletons connected by a lawful PCR-3 source-presentation isomorphism. Packet equality is not itself a source-reduction rule.*

Proof. Source-null and spectator removal are fixed by the source-active carrier core and therefore commute up to graph isomorphism. Source-equational and basis/gauge/chart moves are morphisms in the inherited PCR-3 source groupoid and preserve the complete activity-reduced carrier trace. A difference that is packet-inert but PCR-3-source-distinct is *not* a source reduction: it remains in the provenance fibre and can collapse only later at terminal SCIC resolution if the frozen standing packet agrees. This forbids downstream target/packet equivalence from flowing backward into candidate identity. Finiteness of the active graph prevents an infinite descending sequence of node/edge-removing reductions. The theorem requires source-lawful equivalence, not a unique normal-form representative. $\square$

B.3 Residual moduli completeness

Lemma B.4 (No value loss). *Every parameter value removed from the finite skeleton remains represented either in the exact residual solution object or in the lawful groupoid action.*

Proof. The skeleton retains only finite discrete/categorical type and incidence. Actual source parameters, matrices, kernels, profiles, representation objects, thresholds, continuation values, and failure loci remain coordinates of $\Theta_{F,\sigma}^{\text{part}}$, hence of $Z_{F,\sigma}$ when surviving. Values changed by a lawful source chart transformation are related by $\Gamma_{F,\sigma}$. Source-level deletion is allowed only for PCR-3 source-null incidence, target-inert spectator structure, or proved source-equational/presentational equivalence in the complete carrier trace. A source-distinct value that is merely packet-inert is retained in provenance until terminal SCIC descent; it is not erased by source normalization. $\square$

B.4 U-smallness

Proposition B.5 (Universe control). *Every skeleton groupoid, residual solution object, quotient groupoid, overlap fibre product, and global atlas lies in the fixed universe $\mathbb{U}$.*

Proof. PCR-3 fixes all source groupoids and parameter families inside $\mathbb{U}$. Finite typed diagrams formed from $\mathbb{U}$ -small data are $\mathbb{U}$ -small. Exact preimages, finite products, action groupoids, finite-family fibre products, coproducts indexed by $\mathbb{U}$ -small skeleton classes, and the finite-family Čech construction remain in $\mathbb{U}$. The global codescent object is formed inside the same universe. $\square$

B.5 Packet functor totality

Lemma B.6 (Fail-closed total packet action). *For every atlas object and every licensed continuation, the complete packet has a defined value in the fail-closed codomain.*

Proof. PCR-5A reconstructs the continuation monoid as a total fail-closed action on the standing-faithful normal form. Undefined internal operations are mapped to typed failure values rather than omitted. Restricting to a skeleton or quotient groupoid preserves this totality. $\square$

B.6 Exactness is not a tautology

The family atlas is not defined as “the set of packet values.” Its objects are independently sourced finite skeletons, independently formed PCR-4 solution objects, and independently fixed lawful groupoids. Packet equality is applied only after those constructions. Soundness and coverage therefore depend on PCR-3 reconstruction and PCR-4 semantic evaluation, not on a circular definition of the desired quotient.

C Summary of the family atlases

C.1 Source data, defining relations, lawful actions, and standing moduli

Family	Generated skeleton data	Family relations	Lawful action	Solved standing data
LAW	Existing/law nodes, process incidences, locality or memory, order, and attachments.	L1–L5 together with E00–E23.	Hypergraph isomorphisms, fixed-substrate basis and gauge maps, process natural isomorphisms, and source-chart changes.	Target face, role ownership, frozen deletion, continuation or memory, oriented matching and order, boundary, and same-target frontier.
LOCAL	Active local graph, representations, chirality or reality, multiplicity, and attachments.	N1–N5, rank minors, and E00–E23.	Hypergraph isomorphisms, repeated-node permutations, representation automorphisms, fixed-boundary gauge maps, and chart changes.	Active graph and rank, frozen local deletion, continuation, threshold and matching, boundary, and same-target frontier.
HID	Visible/hidden graph, active support, projection or dilation, memory, and ownership.	H1–H5, dilation identities, and E00–E23.	Hypergraph isomorphisms, visible-basis maps, hidden isometries, Kraus mixing, and chart changes.	Visible/hidden ownership, frozen hidden deletion, projection or memory, hidden support and threshold, boundary, and same-target frontier.

Family	Generated skeleton data	Family relations	Lawful action	Solved standing data
COMP	Constituent graph, derived nodes, derivation or binding, and residual incidence.	C1–C5 together with E00–E23.	Hypergraph isomorphisms, constituent permutations, constituent and derived isomorphisms, binding naturality, and chart changes.	Constituent and derived ownership, frozen constituent deletion, derived continuation, oriented derivation and matching, boundary, and same-target frontier.
SOL	Profile support, topology, persistence, deformation or endability, and attachments.	SO1–SO5, profile existence and stability, and E00–E23.	Hypergraph isomorphisms, internal symmetries, collective-coordinate maps, profile-chart changes, and boundary-preserving maps.	Profile ownership, frozen profile deletion, persistence and endability, profile matching, boundary, and same-target frontier.
HYB	The full activity-reduced decorated hypergraph, principal and full carrier-kind masks, source-semantic gluing, non-Compose cross incidences, and higher-arity relations.	Y1–Y7, all constituent and cross relations, and E00–E23.	Hypergraph isomorphisms, all constituent source actions including auxiliary TOP, EXT, BDY, and ENV sectors, cross-incidence and gluing naturality, and higher-relation automorphisms.	Joint standing ownership, frozen kindwise deletion, hybrid continuation, mixed matching, shared boundary, and cross-sector frontier; the full subset-sufficiency hypergraph remains auxiliary source-composition data.

C.2 Universal residual relations

Every family residual system contains the following universal requirements:

1. source coherence and pointwise activity reduction;
2. Choi positivity and trace preservation of the event-complete instrument;
3. the source/effect/channel kernel equation and every frozen record equality;

4. embedding of the two target-visible response classes and boundary nonalignment;
5. charged-current attachment, causal support, and environment typing;
6. the complete perturbative and activated-global anomaly equation;
7. nonempty common vacuum/deformation, oriented matching, and common-cone objects;
8. active extension coherence; and
9. the nine cross-signature residual equations enforcing one source point and one physical witness architecture.

The identifiers E00–E23 provide a common notation for these relations; the family sections state their specializations and prove the corresponding solution theorems.

C.3 Applicability and the complete complement

For a generated source skeleton let $I_{F,\sigma}$ be the instantiated relation-index set. For a no-update point define the pairwise-disjoint sets

$$N = \{j : \hat{e}_j = \text{NA}\}, \quad F = \{j : \hat{e}_j = \text{fail}\}, \quad U = \{j : \hat{e}_j = \text{undefined}\},$$

with a theorem certificate c_j^{NA} for every $j \in N$. The exact decomposition is therefore

$$\Theta_{F,\sigma}^{\text{raw}} = \left(\coprod_N Z_{F,\sigma}^N \right) \sqcup \coprod_{\substack{N,F,U \text{ pairwise disjoint} \\ F \cup U \neq \emptyset}} \Theta_{F,\sigma}^{\text{def}(N;F,U)} \sqcup \Theta_{F,\sigma}^{\text{SU}} \sqcup \Theta_{F,\sigma}^{\text{TU}}.$$

A survivor may have a nonempty certified inapplicability set N , but has $F = U = \emptyset$. A same-target nonsurvivor retains the same applicability information together with its exact active fail/undefined pair. Failure-only, undefined-only, and mixed fail-plus-undefined cases are therefore distinguished, and an active failed relation cannot be reclassified as inapplicable.

C.4 Provenance and isotropy

At fixed standing data $\mathbf{s}$, the source provenance fibre is the quotient of all source solutions mapping to $\mathbf{s}$ by the subgroup acting trivially on that standing record. Overlap calculations use instead the framed fibre $\widetilde{\text{Prov}}_{F,\sigma}(n)$, whose objects include a chosen core isomorphism to a concrete $n \in \mathcal{N}_5$. Only the framed fibre carries the literal postcomposition action of $\text{Aut}_{\mathcal{N}_5}(n)$. Connected components retain source presentations and isotropy records their lawful automorphisms; neither refines the SCIC class. A target extension may use provenance or auxiliary source-sufficiency data when constructing lifts, but it cannot retroactively promote them to base identity coordinates.

D Overlap objects and coherence details

D.1 Concrete gluing base

All overlaps are formed over the universal pointed standing-core groupoid

$$\mathcal{N}_5 = \mathcal{B}_5,$$

not over a set of packet isomorphism classes. For each generated chart $a = (F, \sigma)$, the lawful source groupoid carries a functor

$$p_a : \Gamma_a \longrightarrow \mathcal{N}_5.$$

An overlap object therefore contains its comparison isomorphism as part of the data; it does not merely assert that two unframed classes admit isomorphic representatives.

D.2 Pairwise overlap form

For generated charts $a = (F, \sigma)$ and $b = (G, \tau)$, define

$$\mathcal{O}_{a,b} := \Gamma_a \times_{\mathcal{N}_5}^h \Gamma_b.$$

An object is a triple

$$(\theta_a, \theta_b, \alpha), \quad \alpha : p_a(\theta_a) \xrightarrow{\sim} p_b(\theta_b).$$

In a common canonical packet chart this condition is the simultaneous equality of the seven frozen channels T, R, D, C, M, B, F , with $D = \Delta_{\text{fr}}^{\text{del}}$. The first-difference loci require agreement in every preceding channel and nonisomorphism at the named channel. Complement strata retain the certified inapplicability sets, the complete active fail/undefined pairs on both sides, and separate source- and target-update branches.

For a concrete core $n \in \mathcal{N}_5$, the framed provenance fibre is

$$\widetilde{\text{Prov}}_a(n) = \{(\theta, \phi) : \phi : p_a(\theta) \xrightarrow{\sim} n\} // G_a.$$

The group $\text{Aut}_{\mathcal{N}_5}(n)$ acts by postcomposition on the chosen frame, yielding the fixed-core presentation

$$\mathcal{O}_{a,b} \simeq \coprod_{[n] \in \pi_0(\mathcal{N}_5)} (\widetilde{\text{Prov}}_a(n) \times \widetilde{\text{Prov}}_b(n)) // \text{Aut}_{\mathcal{N}_5}(n).$$

No automorphism action on an unframed isomorphism class is used. The fifteen unordered family pairs are precisely the $\binom{6}{2} = 15$ distinct-label sectors of this universal construction; family-specific source-resolution data remain provenance rather than additional SCIC overlap conditions.

D.3 Higher overlaps

For a finite distinct-label tuple $\mathbf{a} = (a_0, \dots, a_k)$, $k \geq 2$, set

$$\mathcal{C}_{\mathbf{a}} = \Gamma_{a_0} \times_{\mathcal{N}_5}^h \cdots \times_{\mathcal{N}_5}^h \Gamma_{a_k}.$$

Equivalently, an object consists of one simultaneous source tuple, one concrete core n , and frames

$$\phi_i : p_{a_i}(\theta_i) \xrightarrow{\sim} n,$$

modulo the source actions and the diagonal postcomposition action of $\text{Aut}(n)$. There are

$$\binom{6}{3} + \binom{6}{4} + \binom{6}{5} + \binom{6}{6} = 42$$

distinct-label family-subset types. Each inherits its pairwise faces, common-core condition, framing convention, applicability and defect data, isotropy, and first-difference information from the universal fibre-product construction.

A higher overlap is not inferred from pairwise nonemptiness. Its nonempty branch contains one simultaneous source tuple and one common concrete core. Its complement records a source or target update, an exact applicability/fail/undefined vector, or the first frozen packet channel at which a common core becomes impossible.

D.4 Repeated labels and simplicial structure

For a finite string $\mathbf{a} = ((F_0, \sigma_0), \dots, (F_k, \sigma_k))$, repetitions allowed, the same iterated homotopy fibre product is used. A face forgets one source solution and its frame; a degeneracy repeats one source solution with the same frame, equivalently inserting the identity core isomorphism; and a permutation reorders source points and frames. These operations commute with the diagonal $\text{Aut}(n)$ -action.

Consequently the standard identities

$$d_i d_j = d_{j-1} d_i \quad (i < j), \quad s_i s_j = s_{j+1} s_i \quad (i \leq j),$$

and the mixed identities hold in the framed fibre-product model. Cocycle closure is composition in the concrete groupoid $\mathcal{N}_5$, so no representative-choice ambiguity remains.

D.5 Codescent boundary

At coarse resolution,

$$Q_{\text{osc}}^{\text{SCIC}} \cong \text{coeq}(\pi_0(\mathcal{C}_1^{\text{exec}}) \rightrightarrows \pi_0(\mathcal{C}_0^{\text{exec}})),$$

with higher levels enforcing path coherence. This coequalizer is taken only after the family charts, pairwise fibre products, higher framed objects, and all applicability and complement strata have been determined. It identifies duplicate source presentations of one standing core; it does not replace the family classification.

E Detailed active-singlet continuum witness

E.1 Source object and imported survivor construction

Fix a complex symmetric full-rank coefficient

$$\kappa_* = U D U^\top, \quad D = \text{diag}(d_1, d_2, d_3), \quad d_i > 0,$$

whose induced target channel and visible response satisfy the frozen oscillation target. Choose any positive nonsingular diagonal singlet matrix M_0 and set

$$Y_0 = U D^{1/2} M_0^{1/2}.$$

Then

$$Y_0 M_0^{-1} Y_0^\top = \kappa_*.$$

Every column of Y_0 is nonzero, all three singlets are source active, and the singlet representation contributes no Standard-Model gauge anomaly. PCR-5's constructive three-singlet lift states that any such positive nonsingular choice, together with the inherited PCR-4 common vacuum, channel, response, matching, failure, and common-cone data, gives a survivor point. The present appendix specializes that already proved family; it does not add an empirical heavy-state target.

E.2 Scaling family

For $\lambda \in I_* = (1, 2)$, set

$$M_\lambda = \lambda M_0, \quad Y_\lambda = \sqrt{\lambda} Y_0.$$

The active source graph, multiplicity, representation, attachment, rank, gauge/anomaly data, and deletion type are fixed. Moreover,

$$Y_\lambda M_\lambda^{-1} Y_\lambda^\top = \kappa_*$$

for every λ . By the parameter-uniform PCR-5 construction, each point is an exact PCR-4 survivor in the same finite source-incidence skeleton.

E.3 Why the scaling path is not a lawful groupoid orbit

A lawful singlet basis transformation V acts, in the relevant convention, by

$$Y \mapsto YV, \quad M \mapsto V^\top MV.$$

For unitary or orthogonal source redescription it preserves the basis-invariant threshold singular-value spectrum. Graph isomorphism, gauge redescription, and parameter recharting preserve the same physical spectrum. The transformation

$$(Y_0, M_0) \mapsto (\sqrt{\lambda} Y_0, \lambda M_0)$$

changes every nonzero threshold singular value by the factor λ ; it is a path in the source parameter object, not a morphism of the lawful presentation groupoid unless $\lambda = 1$.

E.4 Equal low-energy output, unequal complete packets

The target-face and low-energy response coordinates induced by κ_* are constant along the family. The physical threshold and retained-resolution validity data are not. They belong to the frozen oriented matching channel M^{match} , and can also change the fail-closed continuation boundary. Therefore

$$\lambda \neq \mu \implies \Xi_{\text{osc}}(x_\lambda) \neq \Xi_{\text{osc}}(x_\mu).$$

Packet completeness then gives pairwise SCIC inequality. Since $|I_*| = \mathfrak{c}$, the family injects a continuum into $Q_{\text{osc}}^{\text{SCIC}}$.

E.5 Disposition of standard objections

Perturbativity or finite-coupling bounds. The witness uses the bounded interval $(1, 2)$ around one finite benchmark. More importantly, its survivor status is imported from the parameter-uniform PCR-5 construction rather than inferred from an unproved genericity claim. A different source theory with an explicit bound would require choosing a nondegenerate subinterval inside that declared source domain.

Absolute-mass or heavy-state import. No threshold measurement is added. The threshold is an already admitted source-to-effective matching coordinate of the oscillation carrier presentation. Using its standing value to distinguish carriers does not turn it into a new empirical target record.

Low-energy equivalence. Equality of κ_* establishes equality of the relevant low-energy projection. It does not invert the heavy-to-light matching arrow or erase threshold, continuation, source-responsibility, and failure data retained by the complete packet.

Field-count selector. Multiplicity is fixed throughout the family. The separation is produced by the matching packet, not by field count, simplicity, or naturalness.

Comparison-class change. The source graph, operator category, charged-current boundary, target records, and background remain fixed. Only an already open source parameter varies.

F Adversarial consistency checks and falsification criteria

The following perturbations isolate the principal ways in which the classification could fail. They are consequences of the stated theorems, not substitutes for them.

ID	Perturbation	Required disposition	Controlling result
T01	Add a disconnected source-null local or hidden copy.	The skeleton and packet are unchanged after activity and spectator reduction.	B01, B03, B04
T02	Add a nonzero active copy with arbitrarily small coupling.	Retain the source block and evaluate its packet; no magnitude threshold authorizes deletion.	B01, B03, B08
T03	Apply a lawful local basis transformation.	The quotient object is unchanged.	B09
T04	Rescale a physical heavy threshold while preserving the effective matrix.	The matching channel changes, producing a distinct packet class.	B13
T05	Replace a HID dilation by a source-authorized isometric presentation.	The HID quotient object is unchanged.	B04
T06	Change hidden ownership while holding the visible channel fixed.	Separate the packets when ownership is standing-bearing; otherwise collapse the difference or identify a target extension.	B04, B05
T07	Combine channel and anomaly data from different parameter points.	Reject the spliced object as a nonsurvivor.	B00, Lemma 2.7
T08	Remove a higher-arity HYB relation required by the common cone.	Enter the corresponding relation or cone frontier.	B08
T09	Present one packet in both LAW and LOCAL charts.	Obtain one global connected component with two provenance maps.	B10–B12
T10	Force unequal frozen deletion profiles into one overlap.	The common-core condition fails.	B10, B14
T11	Replace an exact symbolic solution object by finitely many samples.	The family classification remains unproved.	B01–B08

ID	Perturbation	Required disposition	Controlling result
T12	Add a Majorana or cosmological record after the base classification.	Treat the datum as a target extension; the base atlas is unchanged.	B00, B15
T13	Delete an active COMP constituent and retune the remainder.	Treat the result as a new candidate, not as the original no-repair deletion.	B06
T14	Identify two SOL profiles by topological label alone although their endability differs.	Reject the identification by the continuation or frontier separator.	B07
T15	Infer source identity from equality of an effective coefficient.	Reject the reversal of the oriented matching map.	B02–B06
T16	Count the six family names as six carrier classes.	Reject family-name counting and use the packet-overlap codescent.	B10–B12
T17	Promote a packet-inert provenance label to carrier identity.	Collapse the distinction under B09 and SCIC.	B09
T18	Remove a standing-visible functional continuation modulus.	Reopen the affected family atlas and the moduli theorem.	B01, B09, B13
T19	Choose pairwise comparison arrows that do not compose on a triple overlap.	Reject the triple; B11 coherence would otherwise fail.	B11
T20	Use naturalness to select one point of the continuum family.	No standing consequence follows; the preference lies outside the target.	B00, SCIC
T21	Describe a pair locus only as “nonempty when a solution exists.”	Reject the description unless the common-core object and its typed complement are determined.	B10
T22	Omit a higher family subset while retaining all of its faces.	The higher-overlap completeness theorem fails.	B11
T23	Omit a required field from the PCR-5C base-stratum datum.	The B15 interface theorem is not established.	B15
T24	Change a base SCIC equality while preparing PCR-5C.	Reject the change as target replacement or backflow.	B15
T25	Change only an auxiliary source-subset sufficiency value while the frozen seven-channel packet is fixed.	Retain one base SCIC class; keep the difference as provenance or lift data unless it already packet-factorizes.	B00, B08, B09

ID	Perturbation	Required disposition	Controlling result
T26	Remove the frame from a fixed-core overlap and apply $\text{Aut}(n)$ to an unframed class.	Reject the action as ill typed; restore a concrete core and framed provenance.	B05, B10–B12
T27	Give a no-update raw point an undefined active relation and no failed relation.	Place it in the exact stratum ($F = \emptyset, U \neq \emptyset$).	B01, B10, B11, B15
T28	Give a no-update raw point both failed and undefined active relations.	Retain both exact index sets; do not reduce them to a first defect.	B01, B10, B11, B15
T29	Join an active LOCAL carrier to a target-bearing topological sector only through an Inflow hyperedge.	Route the activity-reduced source object to HYB even without a literal Compose edge.	B01, B08
T30	Add an active environmental co-carrier through CoCarry while the principal mask remains singleton.	Route to HYB because the full carrier-kind and gluing data are nontrivial.	B01, B08
T31	Declare a pure TOP, EXT, BDY, or ENV source object to survive outside HYB.	Reject it by the frozen PCR-4 pure-stratum reductions.	B01, B08
T32	Glue two occupancy-bearing subdiagrams of the same principal kind by an irreducible higher-arity relation.	Route to HYB from the nontrivial gluing witness; the kind-mask cardinality alone is not decisive.	B01, B08
T33	Remove an optional extended or defect input sort while preserving the target and the rest of the active trace.	Place the corresponding relation in N with its theorem-certified N/A witness, not in F or U .	B01
T34	Reactivate that input sort and force the physical relation to fail.	Keep the relation active and place it in F ; it cannot be reclassified as N/A without changing the source point or target.	Theorem 4.2

Together these tests distinguish lawful redescription from physical variation, certified inapplicability from active failure, source provenance from SCIC identity, and pairwise coincidence from coherent higher overlap. Any failed disposition identifies the theorem that must be reopened.

G Theorem dependency and responsibility matrix

ID	Primary dependencies	Output	Used by
B00	PCR-1–PCR-5A; SCIC	Frozen classification domain and no-backflow rule	All later theorems
B01	PCR-3 17.5–17.6; PCR-4 15.1; PCR-5A 4.4	Finite skeleton plus residual exact moduli	B02–B13
B02	LAW source/evaluation maps; B01/B09	Exact LAW atlas	B10–B15
B03	LOCAL source/evaluation maps; B01/B09	Exact LOCAL atlas	B05, B10–B15
B04	HID source/evaluation maps; Stine-spring/Kraus; B01/B09	Exact HID atlas	B05, B10–B15
B05	B03, B04, packet completeness	Exact LOCAL/HID overlap	B11–B15
B06	COMP source/evaluation maps; B01/B09	Exact COMP atlas	B10–B15
B07	SOL source/evaluation maps; B01/B09	Exact SOL atlas	B10–B15
B08	PCR-3 finite hybrids; PCR-4 18.1; B01/B09	Exact decorated-HYB atlas	B10–B15
B09	PCR-5A packet completeness; SCIC	Lawful parameter descent	Every atlas/overlap
B10	B02–B09	Fifteen pairwise overlap groupoids	B11/B12/B15
B11	B10; packet groupoid composition	Coherent higher-overlap nerve	B12/B15
B12	B02–B11; PCR-5A family cover	$\pi_0(\mathfrak{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}}$	B13–B15
B13	B12; active-singlet scale family	Exact categorical endpoint and $ Q_{\text{osc}}^{\text{SCIC}} \geq \mathfrak{c}$	Final interpretation
B14	PCR-5A four packets; B12	Four-class permanence	Final interpretation/B15
B15	B02–B14	PCR-5C base-space interface	Target-extension mini-ARC

G.1 Claim-source classification

Every principal statement belongs to one of four categories:

1. **imported theorem:** an upstream AASC/PCR result used at its proved scope;
2. **local mathematical theorem:** proved in this manuscript;

3. **representative physical family**: used to establish nonemptiness or a cardinal lower bound, not candidate completeness; or
4. **supplementary certification**: scope, provenance, and reproducibility support with no independent semantic authority.

The active-singlet scaling family is category 3 inside the already complete family domain. It proves a continuum lower bound; it does not define the LOCAL universe.

H Nonclaims and exact scope boundary

1. The paper does not add any neutrino phenomenon beyond the frozen oscillation target.
2. It does not derive the absolute neutrino mass scale, mass ordering, PMNS parameters, a Majorana/Dirac choice, or a neutrinoless-double-beta-decay rate.
3. It does not assert that a standing carrier class is a particle, species, propagation eigenstate, mass gap, or particle–antiparticle pair.
4. It does not assert that Nature instantiates every object of the mathematical carrier atlas.
5. It does not select a preferred carrier by naturalness, simplicity, field count, operator dimension, familiarity, ultraviolet narrative, or model popularity.
6. It does not identify equal low-energy coefficients with equal ultraviolet sources.
7. It does not collapse hidden dilations solely because their visible reduced channels agree.
8. It does not treat source provenance as terminal carrier identity when the complete packet is unchanged.
9. It does not treat the four PCR-5A classes as exhaustive; B13 proves that they are not exhaustive.
10. It does not claim $|Q_{\text{osc}}^{\text{SCIC}}| = \mathfrak{c}$. It proves $|Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}$ and gives the exact $\mathbb{U}$ -small categorical moduli object.
11. It does not infer a universal upper cardinal from a target that permits arbitrary $\mathbb{U}$ -small symbolic parameter domains.
12. It does not use a finite enumeration, computational check, theorem table, or hash as the proof of semantic classification.
13. It does not permit later target-extension or PCR-6 characterization labels to change the completed base quotient.
14. It does not infer joint compatibility of future phenomenon lifts from their separate nonemptiness.
15. It does not deny that a richer target can eliminate or split current base classes. Such a target belongs to PCR-5C/D.

H.1 Exact reopening conditions

The endpoint reopens only if a named load-bearing theorem fails. Examples include a faithful same-target carrier outside PCR-3 finite source completeness, an omitted PCR-4 simultaneous relation, an incomplete SCIC witness/continuation interface, an incorrect packet biconditional, a missing source-incidence skeleton coordinate, an overlap inconsistency, or failure of the continuum witness. New empirical data alone do not retroactively reopen the base target; they define a new target revision.

I Supplementary materials and reproducibility

I.1 Supplementary source and data

The supplementary materials contain the complete $\text{\LaTeX}$ source and bibliography, together with machine-readable dictionaries for the source-skeleton grammar, residual relations, family atlases, lawful groupoid actions, standing-core maps, overlap incidences, the four distinguished classes, the continuum witness, and the PCR-5C interface. The upstream dependencies are identified by title, version or status, and theorem role in the bibliography and dependency matrix.

These materials make the symbolic classification independently inspectable. They record, in a second representation, the correspondence among source skeletons, instantiated relation families, standing moduli, pairwise and higher overlaps, and the target-extension handoff. The mathematical objects and their semantic interpretation remain those defined in the manuscript.

I.2 Reproducibility and identity

The supplied source reproduces the publication PDF with its bibliography and internal cross-references resolved. The accompanying checks verify the declared theorem and equation indices, family and overlap incidence counts, required data fields, internal references, and file identity. In addition, the immutable distributed v1.2.3 project archive is hash-verified and its accepted source is extracted directly for byte comparison with every theorem-like formal environment and every labeled equation in the publication source. Cryptographic hashes permit verification that a local copy agrees with the distributed materials.

I.3 Scope of machine-readable verification

Machine-readable checks can expose missing identifiers, inconsistent family attachments, malformed incidence data, incomplete pairwise or higher-overlap indexing, broken cross-references, and typesetting regressions. They do not prove that

- every same-target source architecture has a source-skeleton presentation;
- E00–E23 exhaust the simultaneous semantic relations;
- every family specialization retains all standing-bearing coordinates;
- the displayed actions are exactly the lawful source groupoids;
- the represented moduli have precisely the SCIC fibres;
- every pairwise and higher overlap is physically correct; or
- the active-singlet scaling family survives and remains packet-distinct.

Those are theorem-level conclusions. The supplementary materials support inspection, reproducibility, and internal consistency; they do not replace the semantic arguments of B00–B15.

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